

STAR Robot Control System

Comprehensive Technical Documentation

Complete Reference for Software Engineers

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Locked Inc.

Contents

1	Nanopb (Protobuf) Protocol with ARQ and CRC for SPI Communication	9
1.1	Executive Summary	9
1.2	Key Design Decisions	9
1.2.1	Protocol Stack	9
1.2.2	Byte Order and Standards Compliance	9
1.2.3	Memory Budget	10
1.3	Implementation Requirements	11
1.3.1	Protocol Stack Architecture	11
1.3.2	ESP32 (SPI Peripheral) Implementation	12
1.3.3	RPi5 (SPI Controller) Implementation	13
1.3.4	ARQ Symmetry and State Management	13
1.3.5	Example Transaction: RPi5 Sends Velocity Command	14
1.4	ARQ Communication Flow	15
1.4.1	Normal Operation (No Retransmission)	15
1.4.2	Error Recovery Scenarios	15
1.5	Error Handling Strategies	17
2	Protocol Buffer Schema Architecture	18
2.1	Architecture Overview	18
2.2	Service Specifications	18
2.2.1	MotorControlService	18
2.2.2	TelemetryService	19
2.2.3	BatteryManagementService	20
2.2.4	ConfigurationService	20
2.2.5	FirmwareUpdateService	21
2.3	common.proto – Shared Types	21
2.4	Code Generation and Integration	22
3	Hardware Pin Connections	23
3.1	ESP32-S3-WROOM-1-N16 Pinout	23
3.1.1	Main GPIO Assignments	23
3.1.2	Pin Function Categories	24
3.2	74HC238 Decoder (Motor Driver Chip Select)	24
3.3	74HC4052 Multiplexer (Encoder Channel Selection)	25
3.4	Raspberry Pi 5 Pinout	26
3.4.1	RPi5 Pin Function Categories	26
3.5	Additional Peripheral Connections	27
3.5.1	IMU (Inertial Measurement Unit)	27
3.5.2	Stereo Cameras	28
3.5.3	Temperature Sensor	28
4	Protocol Buffer Style Guide	29
4.1	Terminology Standard	29
4.2	Protocol Buffer Naming Conventions (Boston Dynamics Style)	29
4.3	Unit Suffixes (MKS System)	29
4.4	General Documentation Standards	30

5	ESP32 Firmware Style Guide	31
5.1	General Guidelines for ASM and Inline ASM	31
5.2	When to Use ASM and Inline ASM	32
5.3	Guidelines	32
5.4	ESP32-Specific Assertions	32
5.5	General Rules	33
5.6	Control Structures	33
5.6.1	For Loops	34
5.6.2	While Loops	34
5.7	Functions	34
5.8	Structs and Enums	35
5.9	Initializer Lists	36
5.10	Closing Brace Placement	37
5.11	Summary Table	37
5.12	General Rules	37
5.13	Basic Switch Structure	38
5.14	Case Block Braces	38
5.15	Break Statement Requirements	39
5.16	Default Case Handling	40
5.17	Switch on Enum Types	41
5.18	Indentation	42
5.19	Summary	42
5.20	General Commenting Guidelines	43
5.21	File Header Comments	43
5.22	Source File Header	44
5.23	Doxygen Documentation	44
5.23.1	Function Documentation	44
5.23.2	Parameter Direction Markers	45
5.23.3	Code Examples with @code/@endcode	46
5.23.4	Struct Member Documentation	46
5.24	Special Comment Keywords	47
5.24.1	Keyword Definitions	47
5.25	Section Comments	47
5.26	Aligned Comments	48
5.27	Implementation Comments	48
5.28	Private Function Documentation	49
5.29	Summary	50
5.30	RTOS Types, Mutexes, Semaphores, and Atomic Operations	50
5.31	General Rules	50
5.32	Basic Enum Structure	51
5.33	Explicit Value Assignment	51
5.34	Documentation	51
5.35	Enum to String Functions	52
5.36	Using the Count Sentinel	52
5.37	Switch Statement Pattern	53
5.38	Forward Declaration	53
5.39	Naming Convention Summary	54
5.40	General Principles	54

5.41	ESP-IDF Error Codes	54
5.42	ESP-IDF Validation Macros	55
5.42.1	ESP_RETURN_ON_FALSE	55
5.42.2	ESP_GOTO_ON_ERROR	55
5.42.3	ESP_ERROR_CHECK	57
5.43	NULL Safety Patterns	57
5.44	Mutex Error Handling	58
5.45	Error Interface Dependency Injection	58
5.46	Cleanup Patterns	59
5.47	Error Logging	60
5.48	Summary	60
5.49	Return Value Conventions	61
5.50	Private and Exposed Private Functions	61
5.50.1	Private Functions (static)	61
5.50.2	Exposed Private Functions (priv_)	62
5.51	Parameter Validation	62
5.52	Function Declaration Formatting	63
5.53	Doxygen Documentation	63
5.54	Private Function Documentation	64
5.55	Function Implementation Patterns	64
5.55.1	Single Exit Point with Cleanup	64
5.55.2	Factory Function Pattern	65
5.56	Callback Functions	66
5.57	Summary	67
5.58	General Principles	67
5.59	Static File-Scoped Variables (s_ prefix)	67
5.60	Why static const Over #define	68
5.61	When to Use #define	68
5.62	True Global Variables (g_ prefix)	69
5.63	Dependency Injection Alternative	69
5.64	Static Variables for Module State	70
5.65	Thread Safety for Shared State	70
5.66	Summary	71
5.67	General Principles	71
5.68	Include Guard Naming Convention	72
5.69	C++ Compatibility	72
5.70	Header File Structure	72
5.71	Include Order	74
5.72	Forward Declarations	74
5.73	Separating Types from Functions	75
5.74	Doxygen File Documentation	75
5.75	Complete Header Template	76
5.76	Summary	77
5.77	General Rules	77
5.78	Space After Control Keywords	78
5.79	No Space After Function Names	78
5.80	Binary Operators	79
5.81	Unary Operators	79

5.82	Pointer Declarations	79
5.83	Comma and Semicolon Spacing	80
5.84	Parentheses Spacing	80
5.85	Parameter Alignment	80
5.86	Variable and Comment Alignment	81
5.87	Struct Initializer Alignment	81
5.88	Summary	82
5.89	General Rules	82
5.90	Include Order	82
5.91	Complete Example from Codebase	83
5.92	FreeRTOS Include Order	84
5.93	System Includes	84
5.94	ESP-IDF Includes	84
5.95	Project Includes	85
5.96	Header File Includes	85
5.97	Conditional Includes	85
5.98	Include Comments	86
5.99	Avoiding Circular Dependencies	86
5.100	Summary	86
5.101	General Rules	87
5.102	Functions and Variables	87
5.103	Module Prefixes	88
5.104	Constants and Macros	88
5.105	Type Names	89
5.106	Enum Members	89
5.107	Static Variables (File-Scoped)	90
5.108	Global Variables	90
5.109	Private Functions	90
5.109.1	File-Scoped Static Functions (<code>internal_</code> prefix)	90
5.109.2	Exposed Private Functions (<code>priv_</code> prefix)	91
5.110	Summary Table	92
5.111	Typedef Struct Pattern	92
5.111.1	Standard Pattern	92
5.111.2	Error Handler Example	93
5.112	Union Usage for Protocol Configurations	93
5.112.1	Protocol-Specific Structures	94
5.113	Callback Function Types	95
5.114	Interface Patterns (Dependency Injection)	95
5.114.1	Interface Usage Pattern	96
5.115	Opaque Handle Patterns	97
5.116	Operations Structure Pattern	97
5.117	Manager Structure Pattern	98
5.118	Best Practices	98
5.119	Library Directory Structure	99
5.119.1	Example: <code>star_bus</code> Library	99
5.120	CMakeLists.txt Structure	99
5.121	Header File Organization	100
5.121.1	Header File Template	100

5.121.2	Header Section Order	101
5.122	Include Guard Naming Convention	102
5.123	Source File Organization	102
5.123.1	Source File Template	102
5.123.2	Source Section Order	104
5.124	Types Header Separation	104
5.124.1	Controller Header Pattern	105
5.125	Project Root Structure	105
5.126	Best Practices	106
5.127	Overview	106
5.128	SPI Bus Terminology	106
5.128.1	COPI/CIPO (Not MOSI/MISO)	106
5.129	I2C and General Bus Terminology	107
5.129.1	Controller/Peripheral (Not Master/Slave)	107
5.130	Mapping ESP-IDF Legacy APIs	108
5.131	Documentation Guidelines	109
5.132	Code Review Checklist	109
5.133	Common Patterns	109
5.134	Summary	110
5.135	Dependency Injection (DIP)	110
5.135.1	Interface Definition	110
5.135.2	Interface Implementation	111
5.135.3	Dependency Injection Pattern	111
5.135.4	NULL Interface Handling	112
5.136	Bus Abstraction Pattern	112
5.136.1	Unified Configuration	112
5.136.2	Factory Functions	113
5.136.3	Manager Pattern	113
5.137	Operations Structure Pattern	114
5.138	Thread-Safe Callback Pattern	114
5.139	Thread Safety Patterns	115
5.139.1	Mutex Timeout Convention	115
5.139.2	Mutex with Cleanup (goto pattern)	116
5.140	Error Handler Ownership Pattern	116
5.141	Linked List Management	117
5.142	Best Practices Summary	118
5.143	Prefer Alternatives to Macros	118
5.144	When Macros Are Necessary	119
5.145	Macro Safety Rules	119
5.145.1	Wrap Statement-Like Macros in <code>do-while(0)</code>	119
5.145.2	Parenthesize All Macro Arguments	120
5.145.3	Beware of Side Effects and Multiple Evaluation	120
5.146	Multi-line Macro Formatting	120
5.147	Naming Conventions	121
5.148	Include Guards	121
5.149	Conditional Compilation	121
5.150	Overview	122
5.151	Benefits	122

5.152	Standard Unit Suffixes	123
5.153	Usage Examples	123
5.153.1	Distance and Velocity	123
5.153.2	Angular Measurements	123
5.153.3	Temperature	124
5.153.4	Time	124
5.153.5	Electrical Measurements	124
5.153.6	Percentage	125
5.154	Protocol Buffer Integration	125
5.155	Conversion Functions	125
5.156	Summary	126
5.157	No Dynamic Allocation	126
5.157.1	Rationale	126
5.157.2	Alternatives	127
5.157.3	ESP-IDF Heap Functions	127
5.158	Avoid Large Stack Allocations	128
5.158.1	Stack Size Constraints	128
5.158.2	Examples	128
5.158.3	Stack Usage Guidelines	129
5.159	Use <code>const</code> for Flash Storage	129
5.159.1	Rationale	129
5.159.2	Examples	129
5.159.3	Pointer Declarations	130
5.160	Validate All Buffers	130
5.160.1	Rationale	130
5.160.2	Validation Patterns	130
5.160.3	DMA Buffer Validation	131
5.160.4	Ring Buffer Validation	131
5.161	Summary	132
5.162	Interrupt Service Routines (ISRs)	132
5.162.1	Minimize ISR Work	133
5.162.2	ISR Constraints	133
5.163	No Blocking Delays	134
5.164	Real-Time and Safety	135
5.164.1	Timing Constraints	135
5.164.2	Watchdog Timers	135
5.164.3	Sensor Timeouts	136
5.164.4	Fail-Safe Design	136
5.164.5	CRC Validation	137
5.165	Hardware Abstraction	138
5.165.1	Use Abstract Bus Interfaces	138
5.165.2	Debounce Mechanical Inputs	138
5.165.3	Enable Brown-Out Detection	139
5.165.4	Flash Wear-Leveling	139
5.165.5	Always Use Pull-Up/Pull-Down	140
5.166	Performance and Real-Time	140
5.166.1	Benchmark, Don't Guess	140
5.166.2	Avoid Blocking Operations	140

5.167	Power Management	141
5.167.1	Use Sleep Modes	141
5.167.2	Use DMA to Reduce CPU Load	141
5.168	Testing and Debugging	141
5.168.1	Use Mock Drivers	141
5.168.2	Meaningful Logging	142
5.169	Reliability and Fault Tolerance	142
5.169.1	Reset Peripherals if Stuck	142
5.170	Summary	142
5.171	nanopb Overview	143
5.172	Why Static Allocation?	143
5.173	Field Size Configuration	143
5.174	Standard Field Size Constraints	143
5.175	Examples	143
5.175.1	String Fields	143
5.175.2	Repeated Fields	144
5.175.3	Bytes Fields	144
5.176	Sizing Guidelines	144
5.176.1	String Sizes	144
5.176.2	Repeated Field Counts	145
5.176.3	Bytes Field Sizes	145
5.177	Memory Impact	145
5.177.1	Buffer Size Calculation	145
5.177.2	RAM Usage Example	145
5.178	Best Practices	146
5.178.1	Right-Size Your Buffers	146
5.178.2	Use Fixed-Size Types	146
5.178.3	Validate Message Sizes	146
5.179	Integration with Code Generation	146
5.179.1	buf.gen.yaml Configuration	146
5.179.2	Options File Naming	147
5.180	Common Pitfalls	147
5.180.1	Missing Options File	147
5.180.2	Mismatched Field Names	147
5.180.3	Forgetting Array Count	147
5.181	Summary	148
5.182	NASA Power of 10: Rules for Safety-Critical Code	148
5.182.1	Compliance Status Legend	148
5.182.2	Rule 1: Simplify Control Flow	148
5.182.3	Rule 2: Fixed Loop Upper-Bounds	149
5.182.4	Rule 3: No Dynamic Memory After Initialization	149
5.182.5	Rule 4: Keep Functions Short	150
5.182.6	Rule 5: Use Assertions Generously	150
5.182.7	Rule 6: Declare Data at Smallest Scope	151
5.182.8	Rule 7: Check All Return Values and Parameters	151
5.182.9	Rule 8: Limit Preprocessor Use	152
5.182.10	Rule 9: Restrict Pointer Use	152
5.182.11	Rule 10: Compile with Maximum Warnings	153

5.182.12	Summary: STAR Compliance Matrix	153
5.182.13	References	153
6	Gateway Architecture	155
6.1	Overview	155
6.2	Protocol Stack Implementation	155
6.3	Directory Structure	155
6.4	Frame Package	156
6.4.1	Frame Constants	156
6.4.2	Frame Types	156
6.5	Transport Package	156
6.5.1	SPI Configuration	157
6.5.2	Transport Interface	157
6.6	ARQ Package	157
6.6.1	ARQ Parameters	157
6.6.2	ARQ State Machine	157
6.7	gRPC Services	157
6.8	Build and Test	158
6.9	Dependencies	158

1 Nanopb (Protobuf) Protocol with ARQ and CRC for SPI Communication

1.1 Executive Summary

Overview: Protocol Buffers (nanopb) encapsulated in frames with CRC-32 and Automatic Repeat Request (ARQ) for reliable SPI communication between RPi5 and ESP32-S3.

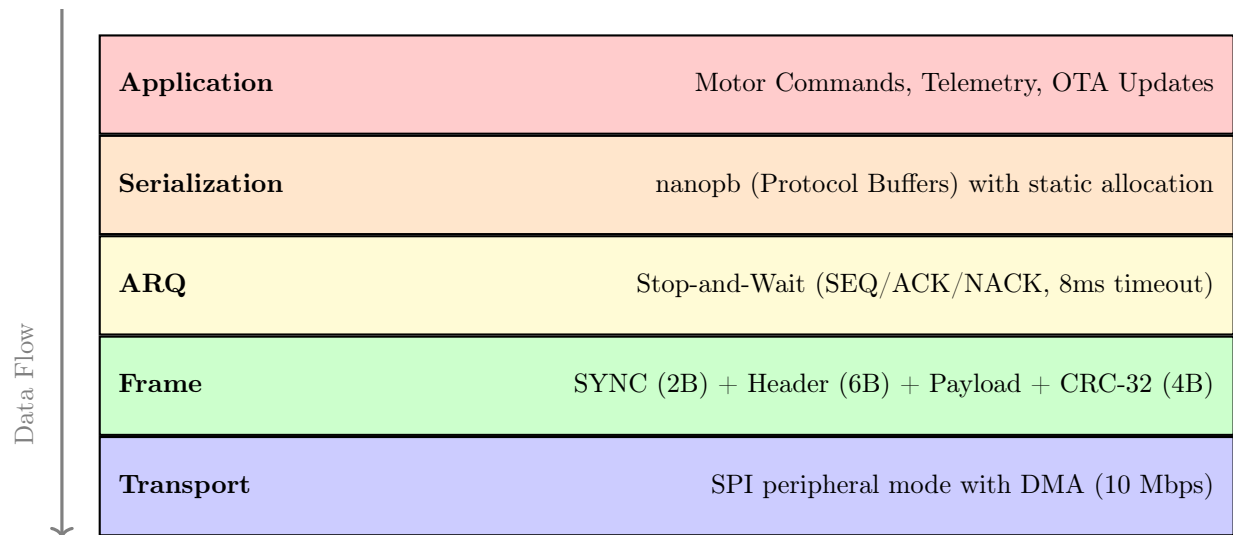


System Constraints:

- ESP32-S3-WROOM-1-N16: 16MB flash, 512KB SRAM, NO PSRAM
- 100Hz SPI communication (10ms period)
- 250Hz motor control loop (4ms period)
- Static allocation only (no malloc)

1.2 Key Design Decisions

1.2.1 Protocol Stack



1.2.2 Byte Order and Standards Compliance

The protocol follows established standards for byte ordering to ensure cross-platform compatibility:

Table 1: Byte Order Standards by Protocol Field

Field	Byte Order	Standard	Rationale
SYNC	Big-endian	RFC 1700	Network byte order for headers
SEQ	Big-endian	RFC 1700	Network byte order for headers
LEN	Big-endian	RFC 1700	Network byte order for headers
TYPE	N/A (1 byte)	–	Single byte, no endianness
FLAGS	N/A (1 byte)	–	Single byte, no endianness
CRC-32	Little-endian	IEEE 802.3	Standard CRC-32 polynomial

SYNC Magic Word (0x55AA): The frame synchronization marker uses the distinctive bit pattern 0x55AA (binary: 0101010101010101 in big-endian wire format). This alternating pattern is easily recognizable and unlikely to appear in random data, making it ideal for frame alignment. The value 0x55AA is transmitted as bytes [0x55, 0xAA] on the wire.

RFC 1700 (Network Byte Order): All multi-byte header fields (SYNC, SEQ, LEN) use big-endian format following RFC 1700 “Assigned Numbers” standard. This ensures compatibility when communicating between different architectures (ARM RPi5 and Xtensa ESP32, both little-endian internally).

IEEE 802.3 (CRC-32): The frame checksum uses IEEE 802.3 polynomial (0x04C11DB7) in little-endian format, matching the ESP32 ROM CRC implementation and standard Ethernet CRC libraries. This allows zero-copy CRC validation using hardware accelerators where available.

Key Insight: Mixed byte orders are intentional — headers follow network standards for cross-platform compatibility, while CRC follows IEEE 802.3 for hardware acceleration and library compatibility.

1.2.3 Memory Budget

Table 2: SRAM Usage (512 KB total)

Component	Size (KB)	Usage (%)
DMA double-buffer	8.0	1.56
RX/TX buffers	2.0	0.39
Protobuf structs	1.5	0.29
ARQ state	4.5	0.88
Total	16.0	3.12

Table 3: Flash Usage (16 MB total)

Component	Size (KB)	Usage (%)
Generated protobuf	80	0.49
Nanopb runtime	12	0.07
Frame + ARQ layers	8	0.05
Total	100	0.61

Memory Budget Calculations:

SRAM (512 KB total):

- **DMA double-buffer:** $2 \times 4092 \text{ bytes} = 8184 \text{ bytes} = 8.0 \text{ KB}$
Calculation: Two ping-pong DMA buffers at STAR_SPI_DMA_MAX_SIZE (4092 bytes each)
Usage: $\frac{8.0}{512} \times 100 = 1.56\%$
- **RX/TX buffers:** $1024 + 1024 = 2048 \text{ bytes} = 2.0 \text{ KB}$
Calculation: 1 KB RX buffer + 1 KB TX buffer for staging protobuf messages
Usage: $\frac{2.0}{512} \times 100 = 0.39\%$

- **Protobuf structs:** 1536 bytes = 1.5 KB
Calculation: Largest message structures (SetVelocityResponse $\approx$ 600 bytes, TelemetryData $\approx$ 400 bytes)
Usage: $\frac{1.5}{512} \times 100 = 0.29\%$
- **ARQ state:** 4 + 4 + 8 + 4 + 4092 = 4112 bytes $\approx$ 4.5 KB
Calculation: Sequence number (4B) + retry counter (4B) + timestamp (8B) + state (4B) + retry buffer (4092B)
Usage: $\frac{4.5}{512} \times 100 = 0.88\%$
- **Total SRAM:** 8.0 + 2.0 + 1.5 + 4.5 = 16.0 KB
Total usage: $\frac{16.0}{512} \times 100 = 3.12\%$
Flash (16 MB = 16384 KB total):
- **Generated protobuf:** 80 KB
Calculation: 6 proto files (common, motor_control, telemetry, battery_management, configuration, firmware_update) $\times$ 13 KB avg compiled size
Usage: $\frac{80}{16384} \times 100 = 0.49\%$
- **Nanopb runtime:** 12 KB
Calculation: Compiled size of pb_encode.c, pb_decode.c, pb_common.c
Usage: $\frac{12}{16384} \times 100 = 0.07\%$
- **Frame + ARQ layers:** 8 KB
Calculation: Framing logic + CRC-32 + ARQ state machine + retransmission logic
Usage: $\frac{8}{16384} \times 100 = 0.05\%$
- **Total Flash:** 80 + 12 + 8 = 100 KB
Total usage: $\frac{100}{16384} \times 100 = 0.61\%$

1.3 Implementation Requirements

1.3.1 Protocol Stack Architecture

The protocol is implemented as a layered stack on both the RPi5 (SPI controller) and ESP32 (SPI peripheral). Each layer has specific responsibilities:

Layer	Responsibility	Implementation
5. Application	Motor commands, telemetry, control logic	Different per platform
4. Protobuf	Message encode/decode (nanopb)	Both sides
3. ARQ	Sequence numbering, ACK/NACK, retry logic	Both sides (symmetric)
2. Framing	Header/footer, CRC-32 verification	Both sides (identical)
1. SPI+DMA	Physical transport with double-buffering	Both sides (different roles)

Table 4: Protocol stack layers and implementation requirements

Key Insight: Layers 2-4 are nearly identical on both platforms. Only the application layer (Layer 5) and SPI role (Layer 1: controller vs peripheral) differ.

1.3.2 ESP32 (SPI Peripheral) Implementation

Layer 1: SPI Peripheral with DMA

- Configure SPI3 in peripheral mode using ESP-IDF
- Set up DMA with double-buffering: 2×4092 bytes
- Register DMA interrupt callbacks for buffer swapping
- Wait for chip select from RPi5, respond to clock

Layer 2: Frame Layer (Custom Code)

RX Path:

- Parse frame header (magic bytes, length, sequence number)
- Verify CRC-32 over header + payload
- If CRC fails $\rightarrow$ discard packet, send NACK
- If CRC valid $\rightarrow$ extract payload, pass to ARQ layer

TX Path:

- Build frame: [Header | Payload | CRC32 | Footer]
- Calculate CRC-32 over header + payload
- Queue complete frame to DMA buffer for transmission

Layer 3: ARQ Layer (Custom Code)

RX Path:

- Check sequence number against `last_acked_rx`
- If duplicate (`seq  $\leq$  last_acked_rx`) $\rightarrow$ send ACK, discard payload
- If expected (`seq = last_acked_rx + 1`) $\rightarrow$ accept, send ACK
- If out-of-order $\rightarrow$ send NACK, request retransmission

TX Path:

- Assign `tx_sequence_number` to outgoing frame
- Store frame in retry buffer
- Start timeout timer (e.g., 10ms)
- On timeout $\rightarrow$ retransmit (up to 3 attempts)
- On ACK received $\rightarrow$ clear retry buffer, increment `tx_seq`

Layer 4: Protobuf (nanopb library)

- Decode incoming bytes $\rightarrow$ `SetVelocityRequest`, `EmergencyStopRequest`
- Encode outgoing structs $\rightarrow$ `SetVelocityResponse`, `TelemetryData`
- Uses `star.v1` package following Boston Dynamics style guide
- Static allocation (no malloc) with `.options` file constraints

Layer 5: Application

- Handle motor velocity commands
- Read encoder data via multiplexer
- Execute PID control loop (250 Hz per motor)
- Send telemetry responses (encoder feedback, current, temperature)

1.3.3 RPi5 (SPI Controller) Implementation

Layer 1: SPI Controller with DMA

- Use Linux `/dev/spidev` interface
- Configure SPI3: 10 Mbps, mode 0, 8-bit words
- Use `ioctl(SPI_IOC_MESSAGE)` for DMA transfers
- Implement userspace double-buffering for continuous operation

Layer 2: Frame Layer (*Custom Code - SAME AS ESP32*)

RX Path: Parse frame header, verify CRC-32, extract payload or discard on CRC failure

TX Path: Build frame [**Header** | **Payload** | **CRC32** | **Footer**], calculate CRC-32, send via SPI

Layer 3: ARQ Layer (*Custom Code - SAME AS ESP32*)

RX Path: Check sequence number, detect duplicates → ACK and discard, detect out-of-order → NACK

TX Path: Assign sequence number, store in retry buffer, timeout and retry logic, handle ACKs

Layer 4: Protobuf (Go with `protoc-gen-go`)

- Encode: `SetVelocityRequest`, `EmergencyStopRequest`
- Decode: `SetVelocityResponse`, `TelemetryData`, `EncoderData`
- Uses `star.v1` package following Boston Dynamics style guide

Layer 5: Application (`star-gateway`)

- Send motor velocity commands from Nav2 (autonomous) or UI (manual)
- All control flows through Nav2 stack for consistent behavior
- Receive encoder feedback for odometry calculations
- Bridge ROS2 topics to/from ESP32 via SPI

1.3.4 ARQ Symmetry and State Management

The ARQ layer uses **Stop-and-Wait** protocol (also called alternating bit protocol). This means each side sends one frame at a time, waits for acknowledgment (ACK) with a timeout, and retries up to 3 times before declaring failure. This approach provides:

- Simple implementation suitable for embedded systems
- Predictable worst-case latency for real-time control
- Low memory overhead (single retry buffer, not a window of frames)
- Sufficient throughput for 10 Mbps SPI with $<1\text{ms}$ frame time

The ARQ protocol is **symmetric** — both sides implement identical Stop-and-Wait logic but maintain separate state for their own transmissions and receptions.

Each Side Maintains:

- **TX sequence number** (`tx_seq`): For packets it sends

- **RX last acknowledged** (`rx_last_ack`): For packets it receives
- **Retry buffer**: Stores unacknowledged outgoing packets
- **Timeout timer**: Triggers retransmission if no ACK received

Responsibility	ESP32 (Peripheral)	RPi5 (Controller)
Sends	Telemetry, responses	Commands, requests
TX sequence numbering	Own <code>tx_seq</code>	Own <code>tx_seq</code>
ACK/NACK generation	Sends ACK for RPi5	Sends ACK for ESP32
Retry logic	Retries if no ACK	Retries if no ACK
Duplicate detection	Checks RPi5's <code>seq</code>	Checks ESP32's <code>seq</code>

Table 5: ARQ role distribution between controller and peripheral

Important: The SPI controller/peripheral roles (Layer 1) are **hardware-level only**. The ARQ layer (Layer 3) treats both sides as equals — each can send, ACK, and retry independently.

1.3.5 Example Transaction: RPi5 Sends Velocity Command

1. **RPi5 Application:** Create `SetVelocityRequest` (left=1.0 m/s, right=1.0 m/s)
2. **RPi5 Protobuf:** Encode to binary (nanopb)
3. **RPi5 ARQ:** Assign `seq=42`, save to retry buffer, start 10ms timeout
4. **RPi5 Frame:** Build packet with header, CRC-32, footer
5. **RPi5 SPI:** DMA transfer to ESP32 (controller initiates)
6. **ESP32 SPI:** DMA receives packet in double-buffer
7. **ESP32 Frame:** Verify CRC-32 → valid
8. **ESP32 ARQ:** Check `seq=42 = rx_last_ack+1` → accept
9. **ESP32 Protobuf:** Decode to `SetVelocityRequest` struct
10. **ESP32 Application:** Update motor setpoints, execute PID control
11. **ESP32 ARQ:** Build ACK packet with `ack_seq=42`
12. **ESP32 Frame:** Wrap ACK in frame with CRC-32
13. **ESP32 SPI:** DMA transmits ACK (peripheral responds)
14. **RPi5 SPI:** DMA receives ACK packet
15. **RPi5 Frame:** Verify CRC-32 → valid
16. **RPi5 ARQ:** Got ACK(42) → clear retry buffer, cancel timeout

Outcome: Command successfully delivered with guaranteed delivery via ARQ. If ACK was lost or CRC failed, RPi5 would retransmit after 10ms timeout.

1.4 ARQ Communication Flow

1.4.1 Normal Operation (No Retransmission)

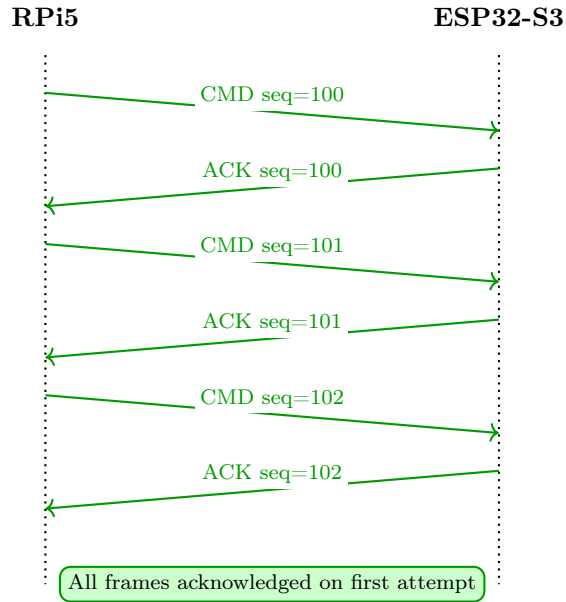


Figure 1: Normal ARQ Flow - Three Successful Transactions

1.4.2 Error Recovery Scenarios

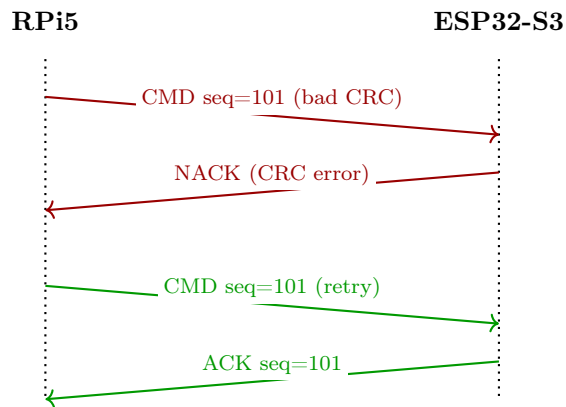


Figure 2: CRC Error - Retry with Same Sequence Number

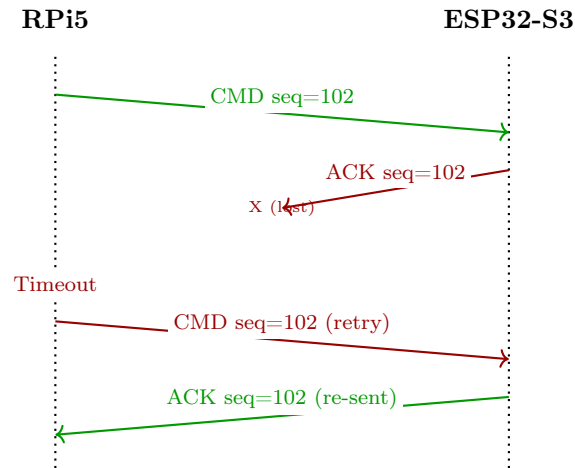


Figure 3: Duplicate Packet Scenario (Lost ACK)

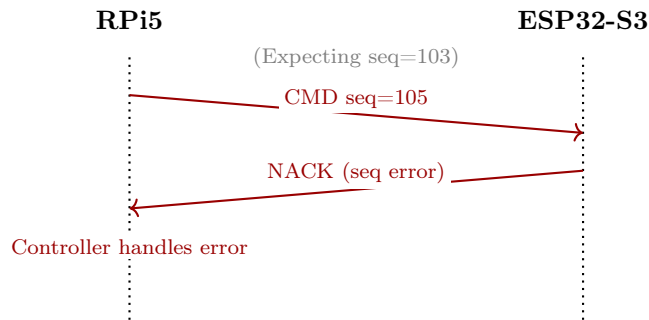


Figure 4: Out-of-Order Packet Scenario

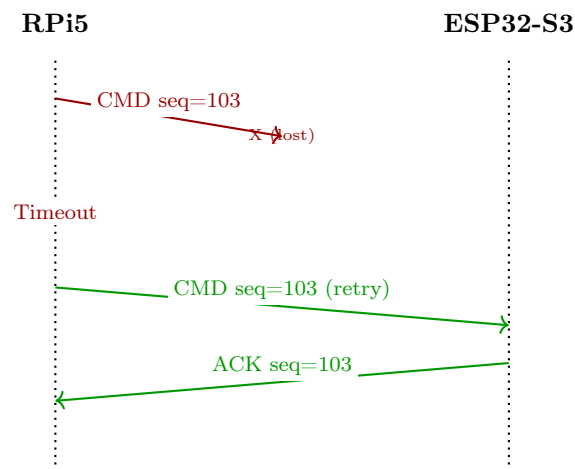


Figure 5: Lost Packet Scenario (Timeout)

1.5 Error Handling Strategies

CRC Mismatch

Action: NACK with CRC error reason
Recovery: Controller retry
Logging: Increment `crc_errors`

Sequence Error

Duplicate: Re-send ACK
Out-of-order: NACK + reject
Recovery: Controller retry

Timeout

RX timeout: Continue next cycle
500ms watchdog: Emergency stop
Retry fail: Log error

Invalid Protobuf

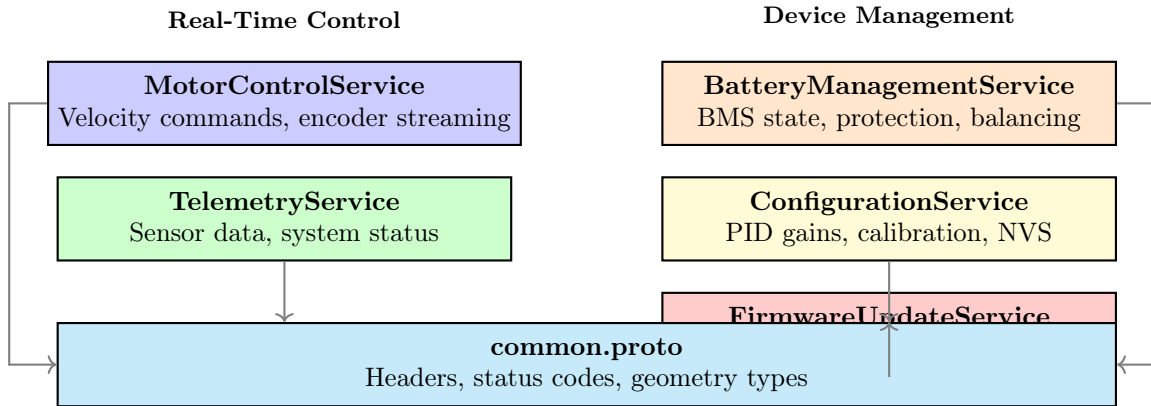
Action: NACK protocol error
Validate: IDs, ranges, offsets
Logging: Increment `proto_errors`

2 Protocol Buffer Schema Architecture

This section documents the Protocol Buffer schema architecture for RPi5↔ESP32 communication. All schemas follow the **Boston Dynamics Protobuf Style Guide** for consistency, maintainability, and industry best practices.

2.1 Architecture Overview

Overview: Six gRPC services define the complete API surface for robot control, organized by functional domain. Services communicate via the nanopb-serialized frames described in Section 1.



Key Design Principles

- **Versioned Package:** All services use `star.v1` namespace for API evolution
- **Request/Response Headers:** Every RPC includes tracing, timestamps, and status
- **Streaming Support:** Continuous data uses server streaming (up to 100Hz)
- **Static Allocation:** All messages sized for nanopb constraints (no dynamic memory)

Note: Naming conventions, unit suffixes, and coding standards are consolidated in Section ?? (Style Guide and Conventions).

2.2 Service Specifications

2.2.1 MotorControlService

Purpose: Low-latency differential drive control with real-time encoder feedback.

Target Latency: <10ms round-trip for velocity commands.

Table 6: MotorControlService RPCs

RPC	Type	Description
SetVelocity	Unary	Set left/right wheel velocities (m/s)
EmergencyStop	Unary	Immediate stop, priority over all commands
SetMotorPower	Unary	Direct PWM control (bypass PID, testing only)
StreamEncoders	Server stream	Encoder ticks and velocity at 1-100Hz
ControlStream	Bidirectional	Velocity in, encoder feedback out

Key Messages:

- **VelocityCommand** – Left/right velocity (m/s), sequence number, timestamp
- **EncoderData** – Motor ID, tick count (341.2 PPR), velocity, timestamp
- **MotorStatus** – Duty cycle, velocity, temperature, current, fault flags
- **PidConfig** – Kp, Ki, Kd gains with output saturation limits

2.2.2 TelemetryService

Purpose: Real-time sensor data aggregation and system health monitoring.

Table 7: TelemetryService RPCs

RPC	Type	Description
GetTelemetry	Unary	Snapshot of all sensor data
StreamTelemetry	Server stream	Continuous sensor updates at 1-100Hz
GetSystemStatus	Unary	Firmware version, uptime, connection states

Key Messages:

- **TelemetryData** – IMU, GPS, battery %, WiFi signal, CPU usage, temperature
- **ImuData** – Pitch/roll/yaw (rad), acceleration (m/s²), angular velocity (rad/s)
- **GpsData** – Lat/lon (degrees), altitude (m), accuracy, satellite count, fix type
- **SystemStatus** – Connection states, operating mode, free heap, uptime

Enumerations:

- **RobotMode** – IDLE, MANUAL, AUTONOMOUS, MAPPING, EMERGENCY_STOP
- **ConnectionStatus** – CONNECTED, DISCONNECTED, CONNECTING, ERROR
- **GpsFix** – NONE, 2D, 3D, DGPS, RTK_FLOAT, RTK_FIXED

2.2.3 BatteryManagementService

Purpose: BQ7850 BMS monitoring for lithium-ion battery pack safety and management.

Hardware: TI BQ7850 16-cell BMS with I2C interface.

Table 8: BatteryManagementService RPCs

RPC	Type	Description
GetBatteryState	Unary	Complete battery snapshot (cells, temps, SOC)
StreamBatteryState	Server stream	Continuous battery monitoring
Get/SetProtectionThresholds	Unary	OV/UV/OC/OT protection limits
Enable/DisableCellBalancing	Unary	Per-cell passive balancing control
ControlFets	Unary	Charge/discharge FET enable/disable
GetDeviceInfo	Unary	Manufacturer, serial, chemistry, versions

Key Messages:

- **BatteryState** – Composite of all battery data
- **CellData** – Individual cell voltages (mV), pack voltage, valid cell count
- **BatteryTemperatureData** – Thermistor readings (0.1°C), average
- **StateOfChargeData** – Relative/absolute SOC (%), capacity (mAh), cycle count
- **ProtectionThresholds** – OV/UV voltage limits, OC current limits, OT temperatures

2.2.4 ConfigurationService

Purpose: Runtime parameter management with NVS persistence across power cycles.

Table 9: ConfigurationService RPCs

RPC	Type	Description
GetConfiguration	Unary	Read current system configuration
SetConfiguration	Unary	Apply configuration (optional NVS persist)
ResetToDefaults	Unary	Restore factory defaults
ValidateConfiguration	Unary	Validate without applying
SaveConfiguration	Unary	Persist in-memory config to NVS
Get/SetMotorPidConfig	Unary	Per-motor PID tuning

Key Messages:

- **SystemConfiguration** – Complete config with version and CRC32
- **MotorPidConfig** – Per-motor PID gains (4 motors supported)
- **CurrentSensorCalibration** – Offset and scale for current sensing
- **SafetyThresholds** – Overcurrent, thermal shutdown, cell imbalance limits
- **TimingConfig** – Motor period, telemetry period, communication timeout

2.2.5 FirmwareUpdateService

Purpose: Over-the-air (OTA) firmware updates with validation and rollback support.

Chunk Size: 1KB optimal for SPI transfer efficiency.

Table 10: FirmwareUpdateService RPCs

RPC	Type	Description
BeginUpdate	Unary	Initialize OTA partition, set expected size/CRC
WriteChunk	Unary	Write single firmware chunk
StreamChunks	Client stream	Bulk chunk transfer
FinalizeUpdate	Unary	Validate firmware, prepare for reboot
AbortUpdate	Unary	Cancel and cleanup
Get/StreamUpdateProgress	Unary/Stream	Transfer progress and state
Reboot	Unary	Boot into new firmware
Rollback	Unary	Revert to previous firmware
MarkValid	Unary	Confirm firmware stability

Key Messages:

- **FirmwareChunk** – Data bytes, sequence number, offset
- **FirmwareUpdateProgress** – Total size, bytes received, state, CRC32
- **FirmwareInfo** – Version string, build date, git hash

Update States: IDLE → RECEIVING → VALIDATING → COMPLETE (or ERROR)

2.3 common.proto – Shared Types

All services import `common.proto` for standardized request/response handling.

Table 11: Standard Headers

Message	Fields
RequestHeader	request_id (UUID), client_timestamp, client_version, timeout
ResponseHeader	request_id (echo), server_timestamp, status, error_message, latency_us

Status Codes: UNKNOWN, OK, INVALID_REQUEST, INTERNAL_ERROR, NOT_FOUND, TIMEOUT, INVALID_STATE, ESTOP_ACTIVE

Geometric Types:

- **Vector3** – Generic 3D vector (x, y, z)
- **Quaternion** – Orientation (w, x, y, z)
- **SE2Pose** – 2D pose (x_m, y_m, angle_rad)
- **SE2Velocity** – 2D velocity (vx_mps, vy_mps, omega_rad_per_s)

2.4 Code Generation and Integration

Table 12: Code Generation Targets

Target	Generator	Output	Usage
Go nanopb	protoc-gen-go + grpc nanopb_generator	gen/go/ gen/nanopb/	star-gateway (RPi5) ESP32 firmware

nanopb Constraints: ESP32 requires static allocation. Field sizes are configured in `.options` files:

- String fields: `max_size:64` (`request_id`), `max_size:128` (`error_message`)
- Repeated fields: `max_count:16` (cell voltages), `max_count:4` (motors)
- Bytes fields: `max_size:1024` (firmware chunks)

Build Integration:

- **Linting:** `buf lint` enforces style guide compliance
- **Breaking Changes:** `buf breaking` detects API incompatibilities
- **CI Pipeline:** Automatic generation and testing on PR

3 Hardware Pin Connections

3.1 ESP32-S3-WROOM-1-N16 Pinout

Critical Hardware Note: This module is the ESP32-S3-WROOM-1-**N16** variant (does NOT end with R_), which means it has **NO PSRAM**. We are allowed to use IO35-IO37 for SPI2 communication with motor drivers.

3.1.1 Main GPIO Assignments

Pin Number	GPIO Label	Signal Name / Function
27	IO0	GND
39	IO1	MOTOR_DRIVER_MOTOR1_IPROPI (ADC)
38	IO2	MOTOR_DRIVER_MOTOR2_IPROPI (ADC)
15	IO3	JTAG signal source strapping pin (NC)
4	IO4	MOTOR_DRIVER_MOTOR3_IPROPI (ADC)
5	IO5	MOTOR_DRIVER_MOTOR4_IPROPI (ADC)
6	IO6	MOTOR_DRIVER_MOTOR1_EN_IN1 (MCPWM)
7	IO7	MOTOR_DRIVER_MOTOR1_PH_IN2 (MCPWM)
12	IO8	FUEL_GUAGE_SMBD_SDA
17	IO9	FUEL_GUAGE_SMBC_SCL
18	IO10	ESP_PI_SPI3_PM_CS
19	IO11	ESP_PI_SPI3_PM_COPI
20	IO12	ESP_PI_SPI3_PM_SCLK
21	IO13	ESP_PI_SPI3_PM_CIPO
22	IO14	MOTOR_DRIVER_MOTOR2_EN_IN1 (MCPWM)
8	IO15	MOTOR_DRIVER_MOTOR2_PH_IN2 (MCPWM)
9	IO16	MOTOR_DRIVER_MOTOR3_EN_IN1 (MCPWM)
10	IO17	MOTOR_DRIVER_MOTOR3_PH_IN2 (MCPWM)
11	IO18	MOTOR_DRIVER_MOTOR4_EN_IN1 (MCPWM)
13	USB_D-	USB_C_N
14	USB_D+	USB_C_P
23	IO21	MOTOR_DRIVER_MOTOR4_PH_IN2 (MCPWM)
28	IO35	MOTOR_DRIVER_SPI2_CM_COPI
29	IO36	MOTOR_DRIVER_SPI2_CM_SCLK
30	IO37	MOTOR_DRIVER_SPI2_CM_CIPO
31	IO38	Decoder Input 1A0 (for Motor nCS selection)
32	IO39	Decoder Input 1A1 (for Motor nCS selection)
33	IO40	MOTOR_ENCODER_SEL0
34	IO41	MOTOR_ENCODER_SEL1
35	IO42	MOTOR_ENCODER_OUT0
26	IO45	Strapping pin for VDD_SPI voltage (NC)
16	IO46	Boot-mode / ROM-print strapping pin
24	IO47	MOTOR_ENCODER_OUT1
25	IO48	TEMPERATURE_SENSOR (DS18B20+)
37	TXD0	ESP_PI_SPI3_PM_DRDY1 (Interrupt to Pi)
36	RXD0	ESP_PI_SPI3_PM_DRDY2 (Interrupt to Pi)

Table 13: ESP32-S3 GPIO pin assignments

3.1.2 Pin Function Categories

SPI3 (Peripheral Mode - Communication with RPi5):

- IO10: Chip Select (CS)
- IO11: Controller Out Peripheral In (COPI)
- IO12: Serial Clock (SCLK)
- IO13: Controller In Peripheral Out (CIPO)
- TXD0 (IO1): Data Ready 1 interrupt to Pi
- RXD0 (IO3): Data Ready 2 interrupt to Pi

SPI2 (Controller Mode - Communication with Motor Drivers):

- IO35: Controller Out Peripheral In (COPI)
- IO36: Serial Clock (SCLK)
- IO37: Controller In Peripheral Out (CIPO)
- Chip selects generated via 74HC238 decoder (see below)

MCPWM (Motor Control):

- Motor 1: IO6 (EN), IO7 (PH)
- Motor 2: IO14 (EN), IO15 (PH)
- Motor 3: IO16 (EN), IO17 (PH)
- Motor 4: IO18 (EN), IO21 (PH)

ADC (Current Sensing - IPROPI):

- IO1: Motor 1 current
- IO2: Motor 2 current
- IO4: Motor 3 current
- IO5: Motor 4 current

Encoder Selection and Reading:

- IO40, IO41: Multiplexer select lines
- IO42, IO47: Encoder output channels

3.2 74HC238 Decoder (Motor Driver Chip Select)

The 74HC238 3-to-8 line decoder generates individual chip select signals for the four motor drivers using only two GPIO pins (IO38, IO39).

Pin Number	Pin Name	Signal Label / Connection	Source/Destination
1	1E	GND (Always Enabled)	Ground
2	1A0	Connected to IO38	ESP32-S3
3	1A1	Connected to IO39	ESP32-S3
4	1Y0	MOTOR_DRIVER_SPI2_MOTOR1_nCS	SPI2 Chip Select 1
5	1Y1	MOTOR_DRIVER_SPI2_MOTOR2_nCS	SPI2 Chip Select 2
6	1Y2	MOTOR_DRIVER_SPI2_MOTOR3_nCS	SPI2 Chip Select 3
7	1Y3	MOTOR_DRIVER_SPI2_MOTOR4_nCS	SPI2 Chip Select 4
8	GND	GND	Ground
9-14	2Y3-2A0	NC (Not Connected)	Marked with "X"
15	2E	P3V3_BUCK (Disabled)	Power Rail
16	VCC	P3V3_BUCK	Power Rail

Table 14: 74HC238 decoder pinout for motor driver chip select generation

Design Rationale: Using a hardware decoder reduces GPIO usage from 4 pins to 2 pins while providing active-low chip select signals required by the DRV8243S motor drivers.

3.3 74HC4052 Multiplexer (Encoder Channel Selection)

The 74HC4052 dual 4-to-1 multiplexer allows reading all four encoder channels (A and B) using only four GPIO pins instead of eight.

Pin Number	Pin Name	Signal Label / Connection	Source/Destination
1	1G	GND (Always Enabled)	Ground
2	B	MOTOR_ENCODER_SEL1 (IO41)	ESP32-S3
3	1C3	MOTOR_DRIVER_ENCODER4_OUTPUT_A	Encoder 4
4	1C2	MOTOR_DRIVER_ENCODER3_OUTPUT_A	Encoder 3
5	1C1	MOTOR_DRIVER_ENCODER2_OUTPUT_A	Encoder 2
6	1C0	MOTOR_DRIVER_ENCODER1_OUTPUT_A	Encoder 1
7	1Y	MOTOR_ENCODER_OUT1 (IO47)	ESP32-S3
8	GND	GND	Ground
9	2Y	MOTOR_ENCODER_OUT0 (IO42)	ESP32-S3
10	2C0	MOTOR_DRIVER_ENCODER1_OUTPUT_B	Encoder 1
11	2C1	MOTOR_DRIVER_ENCODER2_OUTPUT_B	Encoder 2
12	2C2	MOTOR_DRIVER_ENCODER3_OUTPUT_B	Encoder 3
13	2C3	MOTOR_DRIVER_ENCODER4_OUTPUT_B	Encoder 4
14	A	MOTOR_ENCODER_SEL0 (IO40)	ESP32-S3
15	2G	GND (Always Enabled)	Ground
16	VCC	P3V3_BUCK	Power Rail

Table 15: 74HC4052 multiplexer pinout for encoder channel selection

Design Rationale: The dual multiplexer allows simultaneous reading of both A and B channels from the selected encoder, reducing GPIO usage from 8 pins to 4 pins while maintaining quadrature decoding capability.

3.4 Raspberry Pi 5 Pinout

Pin	RPi Label	Signal Name / Connection	Status / Note
1	3v3	Not Connected	Pi 3v3 Rail (Not used)
2	5v	Not Connected	Pi 5v Rail (Not used)
3	SDA	PI_SDA	I2C Data
4	5v	Not Connected	Pi 5v Rail (Not used)
5	SCL	PI_SCL	I2C Clock
7	IO4	HCSR04_TRIG1	Ultrasonic Trigger 1
8	TXD	LiDAR_RX_PI5_TX	UART Transmit to LiDAR
9	GND	GND	Common Ground
10	RXD	LiDAR_TX_PI5_RX	UART Receive from LiDAR
11	IO17	HCSR04_ECHO1	Ultrasonic Echo 1
12	IO18	HCSR04_TRIG4	Ultrasonic Trigger 4
13	IO27	HCSR04_TRIG2	Ultrasonic Trigger 2
15	IO22	HCSR04_ECHO2	Ultrasonic Echo 2
16	IO23	IMU_RST	IMU Reset
17	3v3	Not Connected	Pi 3v3 Rail (Not used)
18	IO24	IMU_INT	IMU Interrupt
19	COPI	ESP_PI_SPI3_PM_COPI	SPI Controller Out
21	CIPO	ESP_PI_SPI3_PM_CIPO	SPI Controller In
22	IO25	HCSR04_ECHO4	Ultrasonic Echo 4
23	SCLK	ESP_PI_SPI3_PM_SCLK	SPI Clock
24	CE0	ESP_PI_SPI3_PM_CS	SPI Chip Enable 0
26	CE1	HCSR04_TRIG5	Ultrasonic Trigger 5
27	ID_SD	MOTOR_DRIVER_MOTOR4_nFAULT	Motor 4 Fault
28	ID_SC	HCSR04_ECHO5	Ultrasonic Echo 5
29	IO5	MOTOR_DRIVER_MOTOR3_nFAULT	Motor 3 Fault
31	IO6	MOTOR_DRIVER_MOTOR2_nFAULT	Motor 2 Fault
32	IO12	HCSR04_TRIG6	Ultrasonic Trigger 6
33	IO13	MOTOR_DRIVER_MOTOR1_nFAULT	Motor 1 Fault
35	IO19	HCSR04_TRIG3	Ultrasonic Trigger 3
36	IO16	HCSR04_ECHO6	Ultrasonic Echo 6
37	IO26	HCSR04_ECHO3	Ultrasonic Echo 3
38	IO20	ESP_PI_SPI3_PM_DRDY1	SPI Data Ready 1
40	IO21	ESP_PI_SPI3_PM_DRDY2	SPI Data Ready 2

Table 16: Raspberry Pi 5 GPIO pin assignments

3.4.1 RPi5 Pin Function Categories

SPI3 (Controller Mode - Communication with ESP32):

- Pin 24 (CE0): Chip Select
- Pin 19 (COPI): Controller Out Peripheral In
- Pin 23 (SCLK): Serial Clock
- Pin 21 (CIPO): Controller In Peripheral Out
- Pin 38 (IO20): Data Ready 1 interrupt from ESP32
- Pin 40 (IO21): Data Ready 2 interrupt from ESP32

I2C (IMU and Stereo Cameras):

- Pin 3 (SDA): I2C Data
- Pin 5 (SCL): I2C Clock
- Pin 16 (IO23): IMU Reset
- Pin 18 (IO24): IMU Interrupt

UART (RPLiDAR C1):

- Pin 8 (TXD): LiDAR RX (Pi TX)
- Pin 10 (RXD): LiDAR TX (Pi RX)

Motor Driver Fault Lines:

- Pin 33 (IO13): Motor 1 nFAULT
- Pin 31 (IO6): Motor 2 nFAULT
- Pin 29 (IO5): Motor 3 nFAULT
- Pin 27 (ID_SD): Motor 4 nFAULT

HC-SR04 Ultrasonic Sensors (6 sensors):

- Sensor 1: Pin 7 (TRIG), Pin 11 (ECHO)
- Sensor 2: Pin 13 (TRIG), Pin 15 (ECHO)
- Sensor 3: Pin 35 (TRIG), Pin 37 (ECHO)
- Sensor 4: Pin 12 (TRIG), Pin 22 (ECHO)
- Sensor 5: Pin 26 (TRIG), Pin 28 (ECHO)
- Sensor 6: Pin 32 (TRIG), Pin 36 (ECHO)

3.5 Additional Peripheral Connections

3.5.1 IMU (Inertial Measurement Unit)

Hardware:

- BNO055: 9-axis absolute orientation sensor
- BMP280: Barometric pressure sensor

Protocol: I2C communication

Connection: Connects to RPi5 via Header J10:

- PI_SCL (Clock) routes to RPi5 Pin 5
- PI_SDA (Data) routes to RPi5 Pin 3
- IMU_RST routes to RPi5 Pin 16 (IO23)
- IMU_INT routes to RPi5 Pin 18 (IO24)

3.5.2 Stereo Cameras

Hardware: External stereo camera modules

Protocol: I2C communication

Connection: Connects via Header J12 to the same I2C bus (PI_SCL and PI_SDA), allowing the Pi to address them alongside the IMU using different I2C addresses.

3.5.3 Temperature Sensor

Hardware: DS18B20+ 1-Wire digital temperature sensor

Protocol: 1-Wire communication

Connection: Connected to ESP32-S3 IO48

4 Protocol Buffer Style Guide

This section defines naming standards and conventions for all Protocol Buffer definitions in the STAR project, following Boston Dynamics style guidelines.

4.1 Terminology Standard

The project uses inclusive terminology following OSHWA (Open Source Hardware Association) standards.

Table 17: Communication Bus Terminology

Use	Avoid	Context
Controller/Peripheral	Master/Slave	I2C, SPI, 1-Wire bus roles
COP1	MOSI	Controller Out, Peripheral In
C1PO	MISO	Controller In, Peripheral Out
Primary/Main	Master	Configuration structures

4.2 Protocol Buffer Naming Conventions (Boston Dynamics Style)

The STAR project follows Boston Dynamics' Protocol Buffer style guide for all .proto definitions.

Table 18: Protocol Buffer Naming Conventions

Element	Convention	Example
Package	Versioned namespace	<code>star.v1</code>
Messages	PascalCase	<code>VelocityCommand</code> , <code>BatteryState</code>
Fields	snake_case + unit suffix	<code>velocity_mps</code> , <code>temperature_celsius</code>
Enums	SCREAMING_SNAKE_CASE	<code>MOTOR_STATE_RUNNING</code>
Enum Zero Value	_UNKNOWN suffix	<code>STATUS_UNKNOWN = 0</code>
Enum Values	Type name prefix	<code>ROBOT_MODE_MANUAL</code>
RPC Pattern	{Verb}Request/Response	<code>SetVelocityRequest</code>

4.3 Unit Suffixes (MKS System)

All numeric fields use SI units with explicit suffixes. This eliminates unit ambiguity and enables automatic documentation generation.

Table 19: Standard Unit Suffixes

Suffix	Unit	Proto Example	C Example
<code>_m</code>	meters	<code>x_m</code>	<code>distance_m</code>
<code>_mps</code>	meters/second	<code>velocity_mps</code>	<code>speed_mps</code>
<code>_mps2</code>	m/s^2	<code>accel_x_mps2</code>	<code>acceleration_mps2</code>
<code>_rad</code>	radians	<code>angle_rad</code>	<code>heading_rad</code>
<code>_rad_per_s</code>	rad/second	<code>omega_rad_per_s</code>	<code>angular_vel_rad_per_s</code>
<code>_celsius</code>	degrees C	<code>temperature_celsius</code>	<code>motor_temp_celsius</code>
<code>_deci_celsius</code>	0.1°C	<code>temp_deci_celsius</code>	(BMS native format)
<code>_us</code>	microseconds	<code>timestamp_us</code>	<code>elapsed_us</code>
<code>_ms</code>	milliseconds	<code>timeout_ms</code>	<code>period_ms</code>
<code>_ma</code>	milliamps	<code>current_ma</code>	<code>motor_current_ma</code>
<code>_mv</code>	millivolts	<code>cell_mv</code>	<code>pack_voltage_mv</code>
<code>_mw</code>	milliwatts	<code>power_mw</code>	<code>consumption_mw</code>
<code>_percent</code>	0–100%	<code>soc_percent</code>	<code>duty_cycle_percent</code>

4.4 General Documentation Standards

- Use Doxygen-compatible comments for all public APIs
- Document the “why” not just the “what”
- Keep configuration centralized in header files
- Avoid magic numbers – use named constants with comments
- Version hardware and firmware together using git tags

5 ESP32 Firmware Style Guide

This section defines comprehensive style guidelines for ESP32-S3 firmware development in the STAR project. All ESP32 firmware code must follow these conventions to ensure consistency, maintainability, and safety in embedded systems development.

In this project, **Assembly code** and **Inline Assembly** should be used with extreme caution. While it can offer performance improvements or enable access to low-level hardware features, its use can make code harder to understand, maintain, and port. Always prefer writing in C unless absolutely necessary.

5.1 General Guidelines for ASM and Inline ASM

- **Avoid ASM When Possible:** Always try to write in C before resorting to assembly language. C compilers are highly optimized and can often produce efficient machine code that rivals hand-written assembly.
- **Document ASM Code Thoroughly:** Assembly code is often harder to understand than C, so it should be accompanied by thorough comments that explain **why** ASM was used, as well as what the code does.

```
1 /* Set register to zero using ASM */
2 mov r0, #0      /* Initialize r0 to 0 */
```

- **Use Inline ASM Sparingly:** Inline assembly allows you to embed assembly instructions directly in C code. It should only be used when absolutely necessary, such as when interacting with hardware registers or performing CPU-specific optimizations.
- **Constrain the Scope of Inline ASM:** When using inline assembly, constrain its scope to as few lines as possible and avoid intermixing it with complex C logic. This helps isolate potential issues and reduces the risk of bugs.

```
1 void set_flag(void)
2 {
3     asm volatile("mov r0, #1" : : : "r0"); /* Set flag in register r0 */
4 }
```

- **Always Use volatile for Inline ASM:** Inline assembly should be marked as `volatile` to prevent the compiler from optimizing it out.
- **Do Not Use Inline ASM for Optimizations:** Modern C compilers perform aggressive optimizations. Inline ASM should not be used to try and optimize code performance unless absolutely necessary, as it is often more error-prone than compiler optimizations.
- **Use Constraints Correctly:** When using inline ASM in C, use the appropriate constraints to inform the compiler how the assembly instructions interact with C variables.


```

1 int add(int a, int b)
2 {
3     int result;
4     asm volatile(
5         "add %[r], %[a], %[b]"
6         : [r] "=r" (result)      /* Output */
7         : [a] "r" (a), [b] "r" (b) /* Inputs */
8     );
9     return result;
10 }

```

5.2 When to Use ASM and Inline ASM

- **Hardware Access:** Use ASM when direct hardware access is required and C alone cannot provide the necessary control (e.g., setting or clearing specific registers).
- **Performance Critical Code:** ASM may be justified in performance-critical sections where C compiler optimizations fall short, but this should be documented and used as a last resort.
- **CPU-Specific Instructions:** Use ASM for executing instructions that are unique to the target CPU (e.g., specific ARM or x86 instructions that are not exposed in C).

In this project, **assertions** are used as a tool to catch programming errors and invalid states during development. They help ensure that assumptions made in the code are correct, and they provide valuable debugging information when things go wrong. However, assertions should never be relied on to handle runtime errors or expected conditions in production code.

5.3 Guidelines

- **Use Assertions for Debugging:** Assertions are intended to catch conditions that should never occur during normal program execution. Use them to assert assumptions about the state of the program, such as parameter validation or invariants within algorithms.

```

1 #include <assert.h>
2
3 void process_data(int* data)
4 {
5     assert(data != NULL); /* Assert that data is not NULL */
6     /* Continue processing */
7 }

```

5.4 ESP32-Specific Assertions

In ESP32 projects, you have two specific macros for error checking: `ESP_ERROR_CHECK` and `ESP_ERROR_CHECK_WITHOUT_ABORT`. These are commonly used to catch errors during development, especially when working with ESP-IDF functions.

- **ESP_ERROR_CHECK:** This macro checks the return value of an ESP-IDF function and aborts the program if the return code is not `ESP_OK`.

- **ESP_ERROR_CHECK_WITHOUT_ABORT:** This macro works similarly to `ESP_ERROR_CHECK`, but it will not abort the program when an error occurs.

Proper brace placement makes code cleaner and easier to read. This section establishes consistent brace conventions used throughout the STAR firmware codebase.

5.5 General Rules

- **Always Use Braces for Control Statements:** Every `if`, `for`, `while`, or similar control structure must have braces, even for single statements.
- **Same Line for Control Structures and Blocks:** For control structures like `if`, `for`, `while`, and blocks like structs and enums, always place the opening brace `{` on the same line as the statement.
- **Functions Have Braces on a New Line:** The only exception to the same-line rule is for functions, where the opening brace goes on its own line.

5.6 Control Structures

Opening brace on the **same line** as the control statement.

```

1  /* CORRECT - if/else with braces on same line */
2  if (ret != ESP_OK) {
3      ESP_LOGE(s_TAG, "Operation failed: %s", esp_err_to_name(ret));
4      return ret;
5  }
6
7  if (manager == NULL) {
8      return ESP_ERR_INVALID_ARG;
9  } else if (manager->mutex == NULL) {
10     return ESP_ERR_INVALID_STATE;
11 } else {
12     /* Proceed with operation */
13 }
14
15 /* CORRECT - Single statement still requires braces */
16 if (pin < 0 || pin >= GPIO_NUM_MAX) {
17     return ESP_OK; /* Not an error, pin is not used */
18 }
19
20 /* WRONG - Missing braces */
21 if (ret != ESP_OK)
22     return ret;
23
24 /* WRONG - Brace on new line */
25 if (ret != ESP_OK)
26 {
27     return ret;
28 }
```

5.6.1 For Loops

```

1  /* CORRECT - for loop with brace on same line */
2  for (star_bus_config_t* curr = manager->buses; curr = curr->next) {
3      if (curr->name && strcmp(curr->name, name) == 0) {
4          found_bus = curr;
5          break;
6      }
7  }
8
9  for (uint32_t i = 0; i < SPI_HOST_MAX; ++i) {
10     manager->spi_host_initialized[i] = false;
11     manager->spi_device_count[i] = 0;
12 }
13
14 /* WRONG */
15 for (uint32_t i = 0; i < count; ++i)
16 process_item(i); /* Missing braces */

```

5.6.2 While Loops

```

1  /* CORRECT - while loop with brace on same line */
2  while (curr) {
3      star_bus_config_t* next = curr->next;
4      esp_err_t destroy_result = star_bus_config_destroy(curr);
5      if (destroy_result != ESP_OK && first_error == ESP_OK) {
6          first_error = destroy_result;
7      }
8      curr = next;
9  }
10
11 while (error_handler_can_retry(&handler)) {
12     ret = attempt_operation();
13     if (ret == ESP_OK) {
14         break;
15     }
16 }
17
18 /* WRONG */
19 while (curr)
20 {
21     process_node(curr); /* Brace on new line */
22     curr = curr->next;
23 }

```

5.7 Functions

Opening brace on a **new line** for function definitions.

```

1  /* CORRECT - Function brace on new line */
2  esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
3  const char*          tag,

```

```

4 star_error_interface_t* error_iface,
5 star_pin_interface_t*   pin_iface)
6 {
7     ESP_RETURN_ON_FALSE(manager, ESP_ERR_INVALID_ARG, s_TAG,
8         "Manager pointer is NULL");
9
10    manager->buses          = NULL;
11    manager->error_iface    = error_iface;
12    manager->pin_iface      = pin_iface;
13
14    /* ... */
15
16    return ESP_OK;
17 }
18
19 static esp_err_t internal_register_bus_pin(star_pin_interface_t*
20     pin_iface,
21     gpio_num_t      pin,
22     const char*      bus_name,
23     const char*      usage,
24     bool             can_be_shared)
25 {
26     if (pin < 0 || pin >= GPIO_NUM_MAX) {
27         return ESP_OK;
28     }
29     /* ... */
30 }
31
32 /* WRONG - Function brace on same line */
33 esp_err_t star_bus_manager_init(star_bus_manager_t* manager, ...) {
34     /* ... */
35 }

```

5.8 Structs and Enums

Opening brace on the **same line** as the typedef/struct/enum keyword.

```

1 /* CORRECT - Struct brace on same line */
2 typedef struct star_bus_manager {
3     star_bus_config_t*    buses;
4     SemaphoreHandle_t     mutex;
5     const char*           tag;
6     star_error_interface_t* error_iface;
7     star_pin_interface_t* pin_iface;
8     bool                  spi_host_initialized[SPI_HOST_MAX];
9     int                   spi_device_count[SPI_HOST_MAX];
10 } star_bus_manager_t;
11
12 /* CORRECT - Enum brace on same line */
13 typedef enum {
14     k_star_bus_type_none,
15     k_star_bus_type_i2c,
16     k_star_bus_type_spi,
17     k_star_bus_type_gpio,

```

```

18     k_star_bus_type_adc,
19     k_star_bus_type_count,
20 } star_bus_type_t;
21
22 /* CORRECT - Union brace on same line */
23 union {
24     star_i2c_bus_config_t i2c;
25     star_spi_bus_config_t spi;
26     star_gpio_bus_config_t gpio;
27     star_adc_bus_config_t adc;
28 } proto;
29
30 /* WRONG - Brace on new line */
31 typedef struct star_bus_manager
32 {
33     /* ... */
34 } star_bus_manager_t;

```

5.9 Initializer Lists

Opening brace on the **same line** as the assignment.

```

1 /* CORRECT - Initializer brace on same line */
2 star_bus_event_t event = {
3     .bus_type = k_star_bus_type_i2c,
4     .bus_name = config->name,
5     .i2c = {
6         .port = port,
7         .address = address,
8         .is_write = true,
9         .len = len
10    }
11 };
12
13 spi_device_interface_config_t spi_dev_cfg = {
14     .clock_speed_hz = 10 * 1000 * 1000,
15     .mode = 0,
16     .spics_io_num = GPIO_NUM_5,
17     .queue_size = 7,
18     .pre_cb = NULL,
19     .post_cb = NULL,
20 };
21
22 /* CORRECT - Short initializers can be on one line */
23 i2c_config_t i2c_conf = {.mode = I2C_MODE_MASTER};
24
25 /* WRONG - Brace on new line */
26 star_bus_event_t event =
27 {
28     .bus_type = k_star_bus_type_i2c,
29     /* ... */
30 };

```

5.10 Closing Brace Placement

Closing braces are always on their own line, except for **else**, **else if**, and **while** in **do-while**.

```

1  /* CORRECT - else on same line as closing brace */
2  if (ret != ESP_OK) {
3      handle_error(ret);
4  } else {
5      continue_operation();
6  }
7
8  /* CORRECT - do-while */
9  do {
10     ret = try_operation();
11 } while (ret == ESP_ERR_TIMEOUT && retries-- > 0);
12
13 /* CORRECT - else if chain */
14 if (config->type == k_star_bus_type_i2c) {
15     /* Handle I2C */
16 } else if (config->type == k_star_bus_type_spi) {
17     /* Handle SPI */
18 } else {
19     /* Handle other */
20 }

```

5.11 Summary Table

Construct	Opening Brace Position
Functions	New line
if/else/else if	Same line
for	Same line
while	Same line
do-while	Same line
switch	Same line
struct	Same line
enum	Same line
union	Same line
Initializer lists	Same line

Switch statements follow specific formatting rules to ensure consistency and readability throughout the STAR firmware codebase.

5.12 General Rules

- **Same Line for Switch and Opening Brace:** The opening brace { for a **switch** statement should be on the same line as the **switch** keyword.
- **Braces for Case Blocks:** Each **case** should have its own block of code enclosed in braces {}.
- **Break Statements:** Every case must end with **break**, **return**, or a fall-through comment.
- **Default Case:** Always include a **default** case, even if it only logs a warning.

5.13 Basic Switch Structure

```

1  /* CORRECT - from star_bus_manager.c */
2  switch (config->type) {
3      case k_star_bus_type_i2c: {
4          ret = internal_register_bus_pin(pin_iface,
5              config->proto.i2c.config.sda_io_num,
6              config->name,
7              "I2C SDA",
8              true);
9          if (ret != ESP_OK && first_error == ESP_OK) {
10             first_error = ret;
11         }
12
13         ret = internal_register_bus_pin(pin_iface,
14             config->proto.i2c.config.scl_io_num,
15             config->name,
16             "I2C SCL",
17             true);
18         if (ret != ESP_OK && first_error == ESP_OK) {
19             first_error = ret;
20         }
21         break;
22     }
23
24     case k_star_bus_type_spi: {
25         ret = internal_register_bus_pin(pin_iface,
26             config->proto.spi.copi_io_num,
27             config->name,
28             "SPI COPI",
29             true);
30         if (ret != ESP_OK && first_error == ESP_OK) {
31             first_error = ret;
32         }
33         /* ... more pin registrations ... */
34         break;
35     }
36
37     default: {
38         ESP_LOGD(s_TAG, "No pin registration for bus type: %d", config->type);
39         break;
40     }
41 }

```

5.14 Case Block Braces

Each case must have its own braces. This prevents variable scope issues and improves readability.

```

1  /* CORRECT - Each case has braces */
2  switch (bus_type) {
3      case k_star_bus_type_i2c: {
4          i2c_port_t port = config->proto.i2c.port; /* Local variable */
5          uint8_t address = config->proto.i2c.address;

```

```

6     ESP_LOGI(s_TAG, "I2C bus on port %d, address 0x%02X", port, address);
7     break;
8 }
9 case k_star_bus_type_spi: {
10     spi_host_device_t host = config->proto.spi.host;  /* Different scope
11     */
12     ESP_LOGI(s_TAG, "SPI bus on host %d", host);
13     break;
14 }
15 case k_star_bus_type_gpio: {
16     uint8_t pin_count = config->proto.gpio.pin_count;
17     ESP_LOGI(s_TAG, "GPIO bus with %d pins", pin_count);
18     break;
19 }
20 default: {
21     ESP_LOGW(s_TAG, "Unknown bus type: %d", bus_type);
22     break;
23 }
24 }
25 /* WRONG - Missing braces */
26 switch (bus_type) {
27     case k_star_bus_type_i2c:
28         handle_i2c();
29         break;
30     case k_star_bus_type_spi:
31         handle_spi();
32         break;
33 }

```

5.15 Break Statement Requirements

Every case must have a clear exit: `break`, `return`, or explicit fall-through.

```

1  /* CORRECT - Using break */
2  switch (config->type) {
3      case k_star_bus_type_i2c: {
4          process_i2c(config);
5          break;
6      }
7      case k_star_bus_type_spi: {
8          process_spi(config);
9          break;
10     }
11     default: {
12         break;
13     }
14 }
15
16 /* CORRECT - Using return */
17 const char* star_bus_type_to_string(star_bus_type_t type)
18 {
19     switch (type) {
20         case k_star_bus_type_none: {

```



```

21     return "NONE";
22 }
23 case k_star_bus_type_i2c: {
24     return "I2C";
25 }
26 case k_star_bus_type_spi: {
27     return "SPI";
28 }
29 case k_star_bus_type_gpio: {
30     return "GPIO";
31 }
32 case k_star_bus_type_adc: {
33     return "ADC";
34 }
35 case k_star_bus_type_count: {
36     return "COUNT";
37 }
38 default: {
39     return "UNKNOWN";
40 }
41 }
42 }
43
44 /* CORRECT - Explicit fall-through (rare, must be commented) */
45 switch (priority) {
46     case PRIORITY_CRITICAL: {
47         log_critical(message);
48         /* FALLTHROUGH */
49     }
50     case PRIORITY_HIGH: {
51         notify_admin(message);
52         /* FALLTHROUGH */
53     }
54     case PRIORITY_NORMAL: {
55         log_message(message);
56         break;
57     }
58     default: {
59         break;
60     }
61 }

```

5.16 Default Case Handling

Always include a default case. Use it for error handling or logging unexpected values.

```

1 /* CORRECT - Default case logs warning */
2 switch (curr->type) {
3     case k_star_bus_type_i2c: {
4         reg_result = register_pin_if_valid(curr->proto.i2c.config.sda_io_num,
5         bus_name, "I2C SDA", reg_func,
6         user_ctx, true);
7         /* ... */
8         break;

```

```

9   }
10  case k_star_bus_type_spi: {
11      reg_result = register_pin_if_valid(curr->proto.spi.copi_io_num,
12      bus_name, "SPI COPI", reg_func,
13      user_ctx, true);
14      /* ... */
15      break;
16  }
17  default: {
18      ESP_LOGW(manager->tag, "Skipping pin registration for "
19      "unsupported bus type: %d", curr->type);
20      break;
21  }
22 }
23
24 /* CORRECT - Default case returns error for invalid input */
25 esp_err_t process_bus_type(star_bus_type_t type)
26 {
27     switch (type) {
28         case k_star_bus_type_i2c: {
29             return process_i2c();
30         }
31         case k_star_bus_type_spi: {
32             return process_spi();
33         }
34         default: {
35             ESP_LOGE(s_TAG, "Invalid bus type: %d", type);
36             return ESP_ERR_INVALID_ARG;
37         }
38     }
39 }

```

5.17 Switch on Enum Types

When switching on an enum, consider handling all values explicitly.

```

1  /* CORRECT - All enum values handled */
2  switch (config->type) {
3      case k_star_bus_type_none: {
4          ESP_LOGW(s_TAG, "Bus type is NONE - skipping");
5          break;
6      }
7      case k_star_bus_type_i2c: {
8          /* Initialize I2C */
9          break;
10     }
11     case k_star_bus_type_spi: {
12         /* Initialize SPI */
13         break;
14     }
15     case k_star_bus_type_gpio: {
16         /* Initialize GPIO */
17         break;
18     }

```

```

19  case k_star_bus_type_adc: {
20      /* Initialize ADC */
21      break;
22  }
23  case k_star_bus_type_count: {
24      /* Sentinel value - should never be used */
25      ESP_LOGE(s_TAG, "Invalid bus type: count sentinel");
26      return ESP_ERR_INVALID_ARG;
27  }
28  /* No default needed when all enum values are handled */
29  /* Compiler warns if new enum values are added */
30  }

```

5.18 Indentation

Case labels are at the same indentation level as the switch. Case body is indented one level.

```

1  /* CORRECT indentation */
2  switch (state) {
3      case STATE_IDLE: {
4          /* Case body indented */
5          if (condition) {
6              /* Nested code further indented */
7          }
8          break;
9      }
10     case STATE_RUNNING: {
11         process_running();
12         break;
13     }
14     default: {
15         break;
16     }
17 }
18
19 /* WRONG - Case labels indented */
20 switch (state) {
21     case STATE_IDLE: {
22         break;
23     }
24 }

```

5.19 Summary

- Opening brace on same line as `switch`
- Each `case` has its own braces `{...}`
- Every case ends with `break`, `return`, or fall-through comment
- Always include `default` case
- Case labels at switch indentation level

- Case body indented one level inside braces

The maximum line length is **80 characters**. Lines longer than this should be split.

- **Break at Natural Points:** Break lines at natural points, such as after operators.
- **Indentation for Continuation Lines:** When breaking lines, ensure the continued line is indented properly.

All functions must include Doxygen comments. This section establishes comprehensive commenting standards for the STAR firmware codebase.

5.20 General Commenting Guidelines

- **Use Block Comments** (`/* */`): Always prefer block comments over line comments (`//`).
- **Explain Why, Not What:** Comments should explain the reasoning behind code, not what the code does (which should be clear from the code itself).
- **Keep Comments Updated:** Outdated comments are worse than no comments.
- **English Only:** All comments must be in English.

```

1  /* CORRECT - Block comment style */
2  /* Initialize the mutex for thread-safe access */
3  manager->mutex = xSemaphoreCreateMutex();
4
5  /* WRONG - Line comment style */
6  // Initialize the mutex for thread-safe access
7  manager->mutex = xSemaphoreCreateMutex();

```

5.21 File Header Comments

Every source and header file should begin with a file path comment, followed by include guards (for headers) and a `@file` Doxygen block for headers with significant content.

```

1  /* lib/star_bus/include/star_bus_manager.h */
2
3  #ifndef STAR_COMPONENT_BUS_MANAGER_H
4  #define STAR_COMPONENT_BUS_MANAGER_H
5
6  #ifdef __cplusplus
7  extern "C" {
8      #endif
9
10     #include "esp_err.h"
11     #include "star_bus_manager_types.h"
12
13     /**
14     * @file star_bus_manager.h
15     * @brief Unified bus manager for I2C, SPI, GPIO, and other bus protocols
16     *
17     * The bus manager provides a central registry for managing multiple bus

```

```

18  * configurations. It supports dependency injection of error handlers and
19  * pin validators for loose coupling.
20  *
21  * Key Features:
22  * - Thread-safe bus management with mutex protection
23  * - Support for multiple bus types (I2C, SPI, GPIO, DHT22, etc.)
24  * - Automatic pin registration and conflict detection
25  * - Safe bus access via callback pattern (prevents use-after-free)
26  *
27  * Example Usage:
28  * @code
29  * // === Basic Bus Manager Setup ===
30  *
31  * #include "star_bus_manager.h"
32  * #include "star_bus_config.h"
33  * #include "star_error_handler.h"
34  *
35  * // 1. Create error handler and pin validator interfaces
36  * error_handler_t error_handler;
37  * error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
38  *
39  * star_error_interface_t error_iface;
40  * star_pin_interface_t pin_iface;
41  * error_handler_get_interface(&error_iface, &error_handler);
42  * pin_validator_get_interface(&pin_iface);
43  *
44  * // 2. Initialize bus manager with dependency injection
45  * star_bus_manager_t bus_manager;
46  * star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
47  * @endcode
48  */

```

5.22 Source File Header

Source files begin with a file path comment.

```

1  /* lib/star_bus/src/star_bus_manager.c */
2
3  #include "star_bus_manager.h"
4
5  #include "freertos/FreeRTOS.h"
6  #include "freertos/semphr.h"
7  /* ... */

```

5.23 Doxygen Documentation

5.23.1 Function Documentation

All public functions must have Doxygen documentation in the header file.

```

1  /**
2  * @brief Initialize a bus manager instance.
3  *

```

```

4  * Allocates resources (like a mutex) for the manager. Must be called
5  * before any other manager functions.
6  *
7  * @param[out] manager      Pointer to bus manager structure to initialize.
8  *                          Must not be NULL.
9  * @param[in]  tag          Optional tag string for logging associated with
10 *                          this manager instance. If NULL or empty, a
11 *                          default tag is used.
12 * @param[in]  error_iface  Optional error handler interface (can be NULL).
13 * @param[in]  pin_iface   Optional pin validator interface (can be NULL).
14 * @return esp_err_t ESP_OK on success, ESP_ERR_INVALID_ARG if manager
15 *                  is NULL, ESP_ERR_NO_MEM if mutex creation fails.
16 */
17 esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
18 const char*          tag,
19 star_error_interface_t* error_iface,
20 star_pin_interface_t* pin_iface);

```

5.23.2 Parameter Direction Markers

Use [in], [out], or [in,out] to indicate parameter direction.

```

1  /**
2  * @brief Write data to an I2C device register.
3  *
4  * @param[in]  config      Pointer to I2C bus configuration.
5  * @param[in]  data        Pointer to data buffer to write.
6  * @param[in]  len         Number of bytes to write.
7  * @param[in]  reg_addr    Target register address.
8  * @param[out] bytes_written Pointer to store actual bytes written.
9  *                          Can be NULL if not needed.
10 * @return esp_err_t ESP_OK on success.
11 */
12 static esp_err_t internal_star_bus_i2c_write(const star_bus_config_t*
13 config,
14 const uint8_t*      data,
15 size_t              len,
16 uint8_t             reg_addr,
17 size_t*             bytes_written);
18
19 /**
20 * @brief Record an error and manage retry/reset logic
21 *
22 * @param[in,out] handler  Pointer to error handler structure
23 * @param[in]     error     Error code that occurred
24 * @param[in]     description Human-readable description of the error
25 * @param[in]     file      Source file where error occurred
26 * @param[in]     line      Line number where error occurred
27 * @param[in]     func      Function name where error occurred
28 * @return ESP_OK if error was handled, or the original error code.
29 */
30 esp_err_t error_handler_record_error(error_handler_t* handler,
31 esp_err_t      error,
32 const char*    description,

```

```

32 const char*      file,
33 int32_t          line,
34 const char*      func);

```

5.23.3 Code Examples with @code/@endcode

Include usage examples in documentation blocks.

```

1  /**
2  * @file star_error_interface.h
3  * @brief Error handler interface for dependency inversion
4  *
5  * Example Usage:
6  * @code
7  * // === Creating and Using Error Interface ===
8  *
9  * #include "star_error_interface.h"
10 * #include "star_error_handler.h"
11 *
12 * // 1. Create concrete error handler
13 * error_handler_t error_handler;
14 * error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
15 *
16 * // 2. Get interface from implementation
17 * star_error_interface_t error_iface;
18 * error_handler_get_interface(&error_iface, &error_handler);
19 *
20 * // 3. Inject into components that need error handling
21 * star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
22 *
23 * // 4. Use the interface
24 * if (error_iface.can_retry(error_iface.ctx)) {
25 *     // Retry operation...
26 * }
27 * @endcode
28 */

```

5.23.4 Struct Member Documentation

Document struct members inline with Doxygen.

```

1  /**
2  * @brief Bus configuration structure (common part).
3  */
4  typedef struct star_bus_config {
5      const char*      name;          /**< Unique name for this bus/device
6          instance */
7      star_bus_type_t  type;          /**< Type of bus (I2C, SPI, etc.) */
8      bool             initialized;   /**< Whether this bus has been initialized
9          */
10     void*             handle;        /**< Driver/device handle */
11     void*             user_ctx;      /**< User context pointer for callbacks */

```

```

11 union {
12     star_i2c_bus_config_t i2c; /**< I2C-specific configuration */
13     star_spi_bus_config_t spi; /**< SPI-specific configuration */
14     star_gpio_bus_config_t gpio; /**< GPIO-specific configuration */
15     star_adc_bus_config_t adc; /**< ADC-specific configuration */
16 } proto;
17
18 struct star_bus_config* next; /**< Next bus config in linked list */
19 } star_bus_config_t;

```

5.24 Special Comment Keywords

Use these keywords to mark special comments for tracking.

```

1  /* TODO: Implement retry logic for I2C bus reset */
2
3  /* FIXME: Memory leak when bus removal fails mid-operation */
4
5  /* BUG: Race condition possible if mutex timeout is too short */
6
7  /* XXX: This code assumes little-endian byte order */
8
9  /* NOTE: ESP-IDF uses mosi_io_num internally - we map to COPI here */
10
11 /* WARNING: Must hold mutex before calling this function */

```

5.24.1 Keyword Definitions

- **TODO:** Work that needs to be done but is not urgent.
- **FIXME:** Known bugs or issues that need to be fixed.
- **BUG:** Confirmed bugs in the code.
- **XXX:** Dangerous or hacky code that works but should be reviewed.
- **NOTE:** Important information about the code.
- **WARNING:** Critical information that must be understood before modifying.

5.25 Section Comments

Use section comments to organize code within files.

```

1  /* === Includes === */
2  #include "star_bus_manager.h"
3  #include "freertos/FreeRTOS.h"
4
5  /* === Constants === */
6  static const char* s_TAG = "STAR_BUS_MANAGER";
7
8  /* === Private Function Prototypes === */
9  static esp_err_t internal_register_bus_pin(...);

```



```

10
11 /* === Public Functions === */
12 esp_err_t star_bus_manager_init(...) { /* ... */ }
13
14 /* === Private Functions === */
15 static esp_err_t internal_register_bus_pin(...) { /* ... */ }
16
17 /* --- Pin Registration Helpers --- */
18 /* Subsection within a section */

```

5.26 Aligned Comments

Align comments after declarations for improved readability.

```

1 typedef struct error_handler {
2     uint32_t      error_count;           /**< Total errors recorded */
3     uint32_t      max_retries;           /**< Max retry attempts */
4     uint32_t      current_retry;         /**< Current retry counter */
5     uint32_t      base_retry_delay;      /**< Base delay (ms) */
6     uint32_t      max_retry_delay;       /**< Max delay (ms) */
7     uint32_t      current_retry_delay;   /**< Current delay with backoff
8     */
9     esp_err_t     last_error;            /**< Last recorded error code */
10    bool           in_error_state;         /**< Whether in error state */
11    void*          reset_context;          /**< Context for reset function
12    */
13    SemaphoreHandle_t mutex;              /**< Mutex for thread-safety */
14
15    esp_err_t (*reset_fn)(void* context); /**< Optional reset function */
16 } error_handler_t;

```

5.27 Implementation Comments

Comments within function bodies explain non-obvious logic.

```

1 esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
2 star_bus_config_t* config)
3 {
4     /* Validate inputs using ESP-IDF macro */
5     ESP_RETURN_ON_FALSE(manager && config, ESP_ERR_INVALID_ARG,
6 s_TAG, "Manager or config pointer is NULL");
7
8     esp_err_t ret = ESP_OK;
9
10    /* Acquire mutex with timeout to prevent deadlock */
11    if (xSemaphoreTake(manager->mutex,
12 pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
13 != pdTRUE) {
14        ESP_LOGE(manager->tag, "Timeout acquiring mutex");
15        return ESP_ERR_TIMEOUT;
16    }
17
18    /* Check for duplicate name - bus names must be unique */

```

```

19  for (star_bus_config_t* curr = manager->buses; curr; curr = curr->next)
20  {
21      if (curr->name && strcmp(curr->name, config->name) == 0) {
22          ret = ESP_ERR_INVALID_STATE;
23          goto add_give_mutex; /* Single exit point pattern */
24      }
25  }
26
27  /* Initialize hardware before adding to list
28  * This ensures manager's SPI host tracking is updated correctly */
29  if (!config->initialized) {
30      ret = star_bus_config_init(config, manager);
31      if (ret != ESP_OK) {
32          goto add_give_mutex;
33      }
34  }
35
36  /* Add to head of list (O(1) insertion) */
37  config->next = manager->buses;
38  manager->buses = config;
39
40  add_give_mutex:
41  xSemaphoreGive(manager->mutex);
42  return ret;
43 }

```

5.28 Private Function Documentation

Static functions should have brief documentation in the source file.

```

1  /**
2   * @brief Helper to register a single pin via the interface
3   */
4  static esp_err_t internal_register_bus_pin(star_pin_interface_t*
5      pin_iface,
6      gpio_num_t      pin,
7      const char*      bus_name,
8      const char*      usage,
9      bool              can_be_shared)
10 {
11     if (pin < 0 || pin >= GPIO_NUM_MAX) {
12         return ESP_OK; /* Not an error, pin is not used */
13     }
14
15     if (!pin_iface || !pin_iface->register_pin) {
16         return ESP_OK; /* No interface, skip registration */
17     }
18
19     /* Create description: "BusName: Usage" (e.g., "I2C0: SDA") */
20     char desc[64];
21     snprintf(desc, sizeof(desc), "%s: %s", bus_name, usage);
22
23     return pin_iface->register_pin(pin_iface->ctx, pin, desc,
24         can_be_shared);

```

23 }

5.29 Summary

- Use block comments (`/* */`), never line comments (`//`)
- Document all public functions with Doxygen in headers
- Use `[in]`, `[out]`, `[in,out]` for parameter direction
- Include `@code/@endcode` examples in file and complex function docs
- Use `TODO`, `FIXME`, `BUG`, `XXX`, `NOTE`, `WARNING` keywords for tracking
- Align comments after struct members for readability
- Use section comments to organize source files

This section outlines the best practices for ensuring thread-safe and efficient concurrent programming.

5.30 RTOS Types, Mutexes, Semaphores, and Atomic Operations

- **RTOS Types:** Use RTOS types when developing for embedded systems like ESP32.
- **Mutexes:** Use a mutex when you need **exclusive access** to a shared resource.
- **Semaphores:** Use semaphores when you need to manage **shared resources** that multiple threads can access simultaneously.
- **Atomic Operations:** Use atomic operations when working with **simple data types** such as counters or flags.

This section establishes the standards for defining and using enumerations in the STAR firmware codebase.

5.31 General Rules

- **Enum Names Should End in `_t`:** All enum types should end with `_t`.
- **Use Snake Case for Enum Values:** All enum values should follow `snake_case` naming conventions.
- **Enum Values Should Start with `k_`:** To distinguish enum members from other constants, values should start with `k_`.
- **Use `typedef`:** Always use `typedef` to define the enum type.
- **Include Count Sentinel:** Include a `_count` sentinel value for iteration.

5.32 Basic Enum Structure

```

1  /* CORRECT - from star_bus_common_types.h */
2  typedef enum {
3      k_star_bus_type_none,    /**< No bus type specified */
4      k_star_bus_type_i2c,    /**< I2C bus */
5      k_star_bus_type_spi,    /**< SPI bus */
6      k_star_bus_type_gpio,   /**< GPIO bus */
7      k_star_bus_type_adc,    /**< ADC bus */
8      k_star_bus_type_count,  /**< Number of bus types (sentinel) */
9  } star_bus_type_t;
10
11 /* WRONG - Various style violations */
12 typedef enum {
13     STAR_BUS_TYPE_I2C,        /* Missing k_ prefix */
14     StarBusTypeSpi,          /* PascalCase not allowed */
15     star_bus_type_gpio,      /* Missing k_ prefix */
16 } bad_enum_style;           /* Missing _t suffix */

```

5.33 Explicit Value Assignment

Assign explicit values when needed for stability or compatibility.

```

1  /* CORRECT - Explicit values for protocol compatibility */
2  typedef enum {
3      k_error_none = 0,    /**< No error */
4      k_error_minor = 1,   /**< Minor error, operation can continue */
5      k_error_major = 2,   /**< Major error, operation should stop */
6      k_error_fatal = 3,   /**< Fatal error, system needs reset */
7      k_error_count = 4,   /**< Number of error levels */
8  } error_level_t;
9
10 /* CORRECT - Bit flags with explicit values */
11 typedef enum {
12     k_bus_flag_none = 0x00, /**< No flags */
13     k_bus_flag_initialized = 0x01, /**< Bus is initialized */
14     k_bus_flag_busy = 0x02, /**< Bus is busy */
15     k_bus_flag_error = 0x04, /**< Bus has error */
16     k_bus_flag_dma = 0x08, /**< DMA enabled */
17 } bus_flags_t;

```

5.34 Documentation

Document each enum value with Doxygen comments.

```

1  /**
2  * @brief Types of supported bus interfaces.
3  */
4  typedef enum {
5      k_star_bus_type_none,    /**< No bus type specified */
6      k_star_bus_type_i2c,    /**< I2C bus */
7      k_star_bus_type_spi,    /**< SPI bus */
8      k_star_bus_type_gpio,   /**< GPIO bus */

```

```

9   k_star_bus_type_adc,    /**< ADC bus */
10  k_star_bus_type_count, /**< Number of bus types */
11 } star_bus_type_t;

```

5.35 Enum to String Functions

Provide a string conversion function for each enum type.

```

1  /* In header file */
2  const char* star_bus_type_to_string(star_bus_type_t type);
3
4  /* In source file */
5  const char* star_bus_type_to_string(star_bus_type_t type)
6  {
7      switch (type) {
8          case k_star_bus_type_none: {
9              return "NONE";
10             }
11             case k_star_bus_type_i2c: {
12                 return "I2C";
13             }
14             case k_star_bus_type_spi: {
15                 return "SPI";
16             }
17             case k_star_bus_type_gpio: {
18                 return "GPIO";
19             }
20             case k_star_bus_type_adc: {
21                 return "ADC";
22             }
23             case k_star_bus_type_count: {
24                 return "COUNT";
25             }
26             default: {
27                 return "UNKNOWN";
28             }
29         }
30     }
31
32     /* Usage */
33     ESP_LOGI(s_TAG, "Added bus config: %s (Type: %s)",
34     config->name, star_bus_type_to_string(config->type));

```

5.36 Using the Count Sentinel

The `_count` value enables iteration and array sizing.

```

1  /* Array indexed by enum */
2  static const char* s_bus_type_names[k_star_bus_type_count] = {
3      [k_star_bus_type_none] = "NONE",
4      [k_star_bus_type_i2c]  = "I2C",
5      [k_star_bus_type_spi]  = "SPI",
6      [k_star_bus_type_gpio] = "GPIO",

```

```

7   [k_star_bus_type_adc] = "ADC",
8 };
9
10  /* Iteration using count */
11  for (star_bus_type_t type = k_star_bus_type_none;
12  type < k_star_bus_type_count; ++type) {
13      ESP_LOGI(s_TAG, "Bus type %d: %s", type, s_bus_type_names[type]);
14  }
15
16  /* Validation using count */
17  if (type >= k_star_bus_type_count) {
18      ESP_LOGE(s_TAG, "Invalid bus type: %d", type);
19      return ESP_ERR_INVALID_ARG;
20  }

```

5.37 Switch Statement Pattern

Always handle all enum values or include a default case.

```

1  /* All values handled explicitly */
2  switch (config->type) {
3      case k_star_bus_type_none: {
4          ESP_LOGW(s_TAG, "Bus type is NONE - skipping");
5          break;
6      }
7      case k_star_bus_type_i2c: {
8          ret = init_i2c_bus(config);
9          break;
10     }
11     case k_star_bus_type_spi: {
12         ret = init_spi_bus(config);
13         break;
14     }
15     case k_star_bus_type_gpio: {
16         ret = init_gpio_bus(config);
17         break;
18     }
19     case k_star_bus_type_adc: {
20         ret = init_adc_bus(config);
21         break;
22     }
23     case k_star_bus_type_count: {
24         /* Sentinel value should never be used */
25         ESP_LOGE(s_TAG, "Invalid bus type: count sentinel");
26         return ESP_ERR_INVALID_ARG;
27     }
28     /* No default - compiler warns if new enum values added */
29 }

```

5.38 Forward Declaration

Enums can be forward-declared for header dependency reduction.

```

1  /* Forward declaration (C11 or later with explicit underlying type) */
2  typedef enum star_bus_type star_bus_type_t;
3
4  /* Or use typedef in header, full definition in source */
5  /* In header: */
6  typedef enum star_bus_type star_bus_type_t;
7
8  /* In source or types header: */
9  enum star_bus_type {
10     k_star_bus_type_none,
11     k_star_bus_type_i2c,
12     /* ... */
13 };

```

5.39 Naming Convention Summary

Element	Convention	Example
Type name	snake_case_t	star_bus_type_t
Member prefix	k_	k_star_bus_type_i2c
Member naming	snake_case	k_star_bus_type_i2c
Count sentinel	_count	k_star_bus_type_count
String function	_to_string	star_bus_type_to_string

This section establishes error handling patterns for the STAR firmware codebase, with emphasis on ESP-IDF specific patterns.

5.40 General Principles

- **Return Error Codes:** Functions should return error codes via `esp_err_t`.
- **Check All Return Values:** Never ignore return values from functions that can fail.
- **Use goto for Clean-up:** Use `goto` to jump to a clean-up section when multiple resources need release.
- **Fail Fast:** Return early on invalid arguments or precondition failures.

5.41 ESP-IDF Error Codes

Use standard ESP-IDF error codes throughout the codebase.

```

1  /* Standard ESP-IDF error codes */
2  ESP_OK /* Success */
3  ESP_FAIL /* Generic failure */
4  ESP_ERR_NO_MEM /* Out of memory */
5  ESP_ERR_INVALID_ARG /* Invalid argument */
6  ESP_ERR_INVALID_STATE /* Invalid state for operation */
7  ESP_ERR_INVALID_SIZE /* Invalid size */
8  ESP_ERR_NOT_FOUND /* Requested resource not found */
9  ESP_ERR_NOT_SUPPORTED /* Operation not supported */
10 ESP_ERR_TIMEOUT /* Operation timed out */
11

```

```

12  /* Usage example */
13  esp_err_t star_bus_manager_init(star_bus_manager_t* manager, ...)
14  {
15      if (manager == NULL) {
16          return ESP_ERR_INVALID_ARG;
17      }
18
19      manager->mutex = xSemaphoreCreateMutex();
20      if (manager->mutex == NULL) {
21          return ESP_ERR_NO_MEM;
22      }
23
24      return ESP_OK;
25  }

```

5.42 ESP-IDF Validation Macros

Use ESP-IDF macros for parameter validation and error checking.

5.42.1 ESP_RETURN_ON_FALSE

Returns an error if condition is false, with logging.

```

1  #include "esp_check.h"
2
3  esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
4  const char* tag,
5  star_error_interface_t* error_iface,
6  star_pin_interface_t* pin_iface)
7  {
8      /* Validate required parameter with automatic logging */
9      ESP_RETURN_ON_FALSE(manager, ESP_ERR_INVALID_ARG, s_TAG,
10 "Manager pointer is NULL");
11
12      /* Multiple validations */
13      ESP_RETURN_ON_FALSE(manager->mutex,
14 ESP_ERR_INVALID_STATE,
15 manager->tag,
16 "Manager mutex not initialized");
17
18      ESP_RETURN_ON_FALSE(config->type > k_star_bus_type_none &&
19 config->type < k_star_bus_type_count,
20 ESP_ERR_INVALID_ARG,
21 manager->tag,
22 "Invalid bus type in config");
23
24      return ESP_OK;
25  }

```

5.42.2 ESP_GOTO_ON_ERROR

Jumps to a label on error, with logging. Use for cleanup patterns.


```

1 static esp_err_t internal_star_bus_i2c_write(const star_bus_config_t*
      config,
2 const uint8_t*      data,
3 size_t              len,
4 uint8_t              reg_addr,
5 size_t*              bytes_written)
6 {
7     esp_err_t      ret = ESP_OK;
8     i2c_cmd_handle_t cmd = NULL;
9
10    cmd = i2c_cmd_link_create();
11    ESP_RETURN_ON_FALSE(cmd, ESP_ERR_NO_MEM, s_TAG,
12        "Failed to create I2C command link");
13
14    ret = i2c_master_start(cmd);
15    ESP_GOTO_ON_ERROR(ret, write_fail, s_TAG,
16        "Write '%s': START failed: %s",
17        config->name, esp_err_to_name(ret));
18
19    ret = i2c_master_write_byte(cmd, (address << 1) | I2C_MASTER_WRITE,
20        true);
21    ESP_GOTO_ON_ERROR(ret, write_fail, s_TAG,
22        "Write '%s': Address (W) failed: %s",
23        config->name, esp_err_to_name(ret));
24
25    ret = i2c_master_write(cmd, data, len, true);
26    ESP_GOTO_ON_ERROR(ret, write_fail, s_TAG,
27        "Write '%s': Data write failed: %s",
28        config->name, esp_err_to_name(ret));
29
30    ret = i2c_master_stop(cmd);
31    ESP_GOTO_ON_ERROR(ret, write_fail, s_TAG,
32        "Write '%s': STOP failed: %s",
33        config->name, esp_err_to_name(ret));
34
35    /* Execute command */
36    ret = i2c_master_cmd_begin(port, cmd, pdMS_TO_TICKS(1000));
37    ESP_GOTO_ON_ERROR(ret, write_fail, s_TAG,
38        "Write '%s': CMD failed", config->name);
39
40    /* Success path */
41    i2c_cmd_link_delete(cmd);
42    if (bytes_written) {
43        *bytes_written = len;
44    }
45    return ESP_OK;
46
47    write_fail:
48    if (cmd) {
49        i2c_cmd_link_delete(cmd);
50    }
51    return ret;
52 }

```

5.42.3 ESP_ERROR_CHECK

Use during development to catch fatal errors.

```

1  /* Development only - aborts on error */
2  ESP_ERROR_CHECK(i2c_driver_install(port, I2C_MODE_MASTER, 0, 0, 0));
3
4  /* Use ESP_ERROR_CHECK_WITHOUT_ABORT for non-fatal checks */
5  ESP_ERROR_CHECK_WITHOUT_ABORT(optional_init());

```

5.43 NULL Safety Patterns

Always validate pointer parameters.

```

1  /* At function start - validate all pointers */
2  esp_err_t star_bus_i2c_write(const star_bus_manager_t* manager,
3  const char*                name,
4  const uint8_t*             data,
5  size_t                     len,
6  uint8_t                    reg_addr,
7  size_t*                    bytes_written)
8  {
9      ESP_RETURN_ON_FALSE(manager && name && data,
10      ESP_ERR_INVALID_ARG,
11      s_TAG,
12      "Manager, name, or data is NULL");
13      ESP_RETURN_ON_FALSE(len > 0, ESP_ERR_INVALID_ARG, s_TAG,
14      "Write length must be > 0");
15
16      /* Clear output parameter initially */
17      if (bytes_written) {
18          *bytes_written = 0;
19      }
20
21      /* ... rest of implementation ... */
22  }
23
24  /* Check optional interface before use */
25  if (manager->error_iface && manager->error_iface->record_error) {
26      manager->error_iface->record_error(
27      manager->error_iface->ctx,
28      ret,
29      "Bus hardware initialization failed",
30      __FILE__,
31      __LINE__,
32      __func__);
33  }
34
35  /* Use convenience macro for safe interface calls */
36  #define STAR_IFACE_RECORD_ERROR(iface, err, msg) \
37  ((iface) && (iface)->record_error \
38  ? (iface)->record_error((iface)->ctx, (err), (msg), \
39  __FILE__, __LINE__, __func__) \
40  : (err))

```

```

41
42 /* Usage */
43 STAR_IFACE_RECORD_ERROR(manager->error_iface, ret, "Operation failed");

```

5.44 Mutex Error Handling

Handle mutex acquisition failures properly.

```

1 esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
2 star_bus_config_t* config)
3 {
4     /* Validate arguments first */
5     ESP_RETURN_ON_FALSE(manager && config, ESP_ERR_INVALID_ARG,
6 s_TAG, "Invalid arguments");
7     ESP_RETURN_ON_FALSE(manager->mutex, ESP_ERR_INVALID_STATE,
8 manager->tag, "Mutex not initialized");
9
10    esp_err_t ret = ESP_OK;
11
12    /* Acquire mutex with timeout */
13    if (xSemaphoreTake(manager->mutex,
14 pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
15 != pdTRUE) {
16        ESP_LOGE(manager->tag, "Timeout acquiring mutex for adding bus '%s'",
17 config->name);
18        return ESP_ERR_TIMEOUT;
19    }
20
21    /* Critical section */
22    /* ... operations ... */
23
24    /* Single exit point ensures mutex release */
25    add_give_mutex:
26    xSemaphoreGive(manager->mutex);
27    return ret;
28 }

```

5.45 Error Interface Dependency Injection

Components receive error handlers through dependency injection.

```

1 /* Interface definition */
2 typedef struct star_error_interface {
3     esp_err_t (*record_error)(void* ctx, esp_err_t error,
4 const char* message, const char* file,
5 int line, const char* func);
6     bool (*can_retry)(void* ctx);
7     esp_err_t (*reset_state)(void* ctx);
8     void* ctx;
9 } star_error_interface_t;
10
11 /* Injecting error handler into component */
12 esp_err_t star_bus_manager_init(star_bus_manager_t* manager,

```

```

13  const char*          tag,
14  star_error_interface_t* error_iface,
15  star_pin_interface_t*  pin_iface)
16  {
17      manager->error_iface = error_iface;  /* Can be NULL */
18      /* ... */
19  }
20
21  /* Using the error interface */
22  if (ret != ESP_OK) {
23      if (manager->error_iface && manager->error_iface->record_error) {
24          manager->error_iface->record_error(
25              manager->error_iface->ctx,
26              ret,
27              "Bus initialization failed",
28              __FILE__,
29              __LINE__,
30              __func__);
31      }
32  }

```

5.46 Cleanup Patterns

Use goto for consistent cleanup when multiple resources are acquired.

```

1  esp_err_t complex_init(complex_t* obj)
2  {
3      esp_err_t ret = ESP_OK;
4
5      /* Allocate resources */
6      obj->buffer = malloc(BUFFER_SIZE);
7      if (!obj->buffer) {
8          ret = ESP_ERR_NO_MEM;
9          goto cleanup;
10     }
11
12     obj->mutex = xSemaphoreCreateMutex();
13     if (!obj->mutex) {
14         ret = ESP_ERR_NO_MEM;
15         goto cleanup;
16     }
17
18     ret = hardware_init(obj);
19     if (ret != ESP_OK) {
20         goto cleanup;
21     }
22
23     return ESP_OK;
24
25     cleanup:
26     /* Release resources in reverse order */
27     if (obj->mutex) {
28         vSemaphoreDelete(obj->mutex);
29         obj->mutex = NULL;

```

```

30 }
31 if (obj->buffer) {
32     free(obj->buffer);
33     obj->buffer = NULL;
34 }
35 return ret;
36 }

```

5.47 Error Logging

Use appropriate log levels for different error severities.

```

1  /* ESP_LOGE - Errors that prevent operation */
2  ESP_LOGE(s_TAG, "Failed to initialize bus '%s': %s",
3  config->name, esp_err_to_name(ret));
4
5  /* ESP_LOGW - Warnings that don't prevent operation */
6  ESP_LOGW(manager->tag, "Failed to register pins for bus '%s': %s "
7  "(bus still added)", config->name, esp_err_to_name(ret));
8
9  /* ESP_LOGI - Informational messages */
10 ESP_LOGI(manager->tag, "Added bus config: %s (Type: %s)",
11 config->name, star_bus_type_to_string(config->type));
12
13 /* ESP_LOGD - Debug messages (compiled out in release) */
14 ESP_LOGD(s_TAG, "Write Success: %zu bytes to '%s' (Addr 0x%02X)",
15 len, config->name, address);

```

5.48 Summary

- Return `esp_err_t` from all functions that can fail
- Use `ESP_RETURN_ON_FALSE` for parameter validation
- Use `ESP_GOTO_ON_ERROR` for cleanup patterns
- Always validate pointers before dereferencing
- Handle mutex timeouts as `ESP_ERR_TIMEOUT`
- Use dependency injection for error handlers
- Release resources in reverse order of acquisition
- Log errors with appropriate severity levels

This section establishes standards for function design, implementation, and documentation in the STAR firmware codebase.

5.49 Return Value Conventions

- If the name of a function is an action (or an imperative command), it should return an **error-code integer** (`esp_err_t`).
- If the name is a predicate (a function that returns true/false), it should return a **succeeded boolean**.

```

1  /* Action functions return esp_err_t */
2  esp_err_t star_bus_manager_init(...);
3  esp_err_t star_bus_manager_add_bus(...);
4  esp_err_t star_bus_config_destroy(...);
5
6  /* Predicate functions return bool */
7  bool error_handler_can_retry(error_handler_t* handler);
8  bool is_bus_initialized(const star_bus_config_t* config);
9
10 /* Getter functions return the value or pointer */
11 star_bus_config_t* star_bus_manager_find_bus(...);
12 const char* star_bus_type_to_string(star_bus_type_t type);

```

5.50 Private and Exposed Private Functions

5.50.1 Private Functions (static)

Functions internal to a single source file should be declared `static` with the `internal_` prefix.

```

1  /* In .c file - private static functions */
2  static esp_err_t internal_register_bus_pin(star_pin_interface_t*
3      pin_iface,
4      gpio_num_t      pin,
5      const char*      bus_name,
6      const char*      usage,
7      bool             can_be_shared)
8  {
9      if (pin < 0 || pin >= GPIO_NUM_MAX) {
10         return ESP_OK; /* Not an error, pin is not used */
11     }
12     /* ... implementation ... */
13 }
14
15 static esp_err_t internal_i2c_execute_with_retry(const star_bus_config_t*
16     config,
17     i2c_port_t      port,
18     i2c_cmd_handle_t cmd,
19     const char*      operation_name)
20 {
21     /* ... implementation ... */
22 }
23
24 /* Declare prototypes at top of file */
25 static esp_err_t internal_register_bus_pin(...);
26 static esp_err_t internal_unregister_bus_pin(...);
27 static esp_err_t internal_register_config_pins(...);

```

5.50.2 Exposed Private Functions (priv_)

If a function cannot be `static` and must be exposed across files within a module (but not publicly), prefix its name with `priv_`.

```

1  /* In internal header (e.g., star_bus_internal.h) */
2  esp_err_t priv_star_bus_validate_config(const star_bus_config_t* config);
3  esp_err_t priv_star_bus_acquire_mutex(star_bus_manager_t* manager);
4  void      priv_star_bus_release_mutex(star_bus_manager_t* manager);

```

5.51 Parameter Validation

Validate all parameters at the start of public functions.

```

1  esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
2  star_bus_config_t* config)
3  {
4      /* Validate required parameters */
5      ESP_RETURN_ON_FALSE(manager && config,
6      ESP_ERR_INVALID_ARG,
7      s_TAG,
8      "Manager or config pointer is NULL");
9
10     /* Validate config contents */
11     ESP_RETURN_ON_FALSE(config->name && strlen(config->name) > 0,
12     ESP_ERR_INVALID_ARG,
13     manager->tag,
14     "Config name is NULL or empty");
15
16     /* Validate enum range */
17     ESP_RETURN_ON_FALSE(config->type > k_star_bus_type_none &&
18     config->type < k_star_bus_type_count,
19     ESP_ERR_INVALID_ARG,
20     manager->tag,
21     "Invalid bus type in config");
22
23     /* Validate state */
24     ESP_RETURN_ON_FALSE(manager->mutex,
25     ESP_ERR_INVALID_STATE,
26     manager->tag,
27     "Manager mutex not initialized");
28
29     /* ... rest of implementation ... */
30 }
31
32 esp_err_t star_bus_i2c_write(const star_bus_manager_t* manager,
33 const char*          name,
34 const uint8_t*       data,
35 size_t              len,
36 uint8_t              reg_addr,
37 size_t*              bytes_written)
38 {
39     /* Validate all required pointers */
40     ESP_RETURN_ON_FALSE(manager && name && data,

```

```

41 ESP_ERR_INVALID_ARG ,
42 s_TAG ,
43 "Manager, name, or data is NULL");
44
45 /* Validate numeric constraints */
46 ESP_RETURN_ON_FALSE(len > 0, ESP_ERR_INVALID_ARG, s_TAG,
47 "Write length must be > 0");
48
49 /* Clear output parameters initially */
50 if (bytes_written) {
51     *bytes_written = 0;
52 }
53
54 /* ... rest of implementation ... */
55 }

```

5.52 Function Declaration Formatting

Align parameters vertically when they span multiple lines.

```

1  /* CORRECT - Aligned parameters */
2  esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
3  const char*      tag,
4  star_error_interface_t* error_iface,
5  star_pin_interface_t*  pin_iface);
6
7  star_bus_config_t* star_bus_config_create_spi_device(
8  const char*      name,
9  spi_host_device_t      host,
10 gpio_num_t      copi_pin,
11 gpio_num_t      cipo_pin,
12 gpio_num_t      sclk_pin,
13 int32_t          dma_chan,
14 const spi_device_interface_config_t* dev_cfg);
15
16 /* Function pointers in structs */
17 typedef struct star_error_interface {
18     esp_err_t (*record_error)(void*      ctx,
19     esp_err_t  error,
20     const char* message,
21     const char* file,
22     int         line,
23     const char* func);
24     bool (*can_retry)(void* ctx);
25     esp_err_t (*reset_state)(void* ctx);
26     void* ctx;
27 } star_error_interface_t;

```

5.53 Doxygen Documentation

All public functions must have complete Doxygen documentation in header files.

```

1  /**

```



```

2  * @brief Add a new bus/device configuration to the manager.
3  *
4  * Adds a previously created configuration (via star_bus_config_create_*)
5  * to the manager's internal list. The manager takes ownership
6  * conceptually,
7  * but the configuration still needs to be destroyed eventually (typically
8  * via
9  * star_bus_manager_remove_bus or star_bus_manager_deinit).
10 *
11 * The configuration must have a unique name. This function is thread-safe.
12 *
13 * @param[in] manager Pointer to the initialized bus manager.
14 *                  Must not be NULL.
15 * @param[in] config  Pointer to the bus configuration structure to add.
16 *                  Must not be NULL and must have a valid name.
17 * @return esp_err_t ESP_OK on success,
18 *                  ESP_ERR_INVALID_ARG if manager or config is invalid,
19 *                  ESP_ERR_INVALID_STATE if a bus with the same name
20 *                  already exists or mutex error,
21 *                  ESP_ERR_TIMEOUT if mutex times out.
22 */
23 esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
24 star_bus_config_t* config);

```

5.54 Private Function Documentation

Static functions should have brief documentation in the source file.

```

1  /**
2  * @brief Helper: Execute I2C command with retry logic
3  *
4  * Wraps i2c_master_cmd_begin() with automatic retry for transient errors.
5  *
6  * @param[in] config      Pointer to I2C bus configuration
7  * @param[in] port        I2C port number
8  * @param[in] cmd         I2C command handle
9  * @param[in] operation_name Name of operation for logging
10 * @return esp_err_t Final result after retries
11 */
12 static esp_err_t internal_i2c_execute_with_retry(const star_bus_config_t*
13 config,
14 i2c_port_t port,
15 i2c_cmd_handle_t cmd,
16 const char* operation_name);

```

5.55 Function Implementation Patterns

5.55.1 Single Exit Point with Cleanup

```

1  esp_err_t star_bus_manager_remove_bus(star_bus_manager_t* manager,
2  const char* name)
3  {

```

```

4  ESP_RETURN_ON_FALSE(manager && name && strlen(name) > 0,
5  ESP_ERR_INVALID_ARG, s_TAG, "Invalid arguments");
6
7  star_bus_config_t* to_remove = NULL;
8  star_bus_config_t* prev      = NULL;
9  esp_err_t          ret       = ESP_ERR_NOT_FOUND;
10
11  /* Acquire mutex */
12  if (xSemaphoreTake(manager->mutex,
13  pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
14  != pdTRUE) {
15      return ESP_ERR_TIMEOUT;
16  }
17
18  /* Find and unlink */
19  for (star_bus_config_t* curr = manager->buses; curr;
20  prev = curr, curr = curr->next) {
21      if (curr->name && strcmp(curr->name, name) == 0) {
22          to_remove = curr;
23          if (prev == NULL) {
24              manager->buses = curr->next;
25          } else {
26              prev->next = curr->next;
27          }
28          to_remove->next = NULL;
29          ret = ESP_OK;
30          break;
31      }
32  }
33
34  if (ret == ESP_ERR_NOT_FOUND) {
35      goto remove_give_mutex;
36  }
37
38  /* Cleanup operations */
39  esp_err_t destroy_result = star_bus_config_destroy(to_remove);
40  if (destroy_result != ESP_OK) {
41      ret = destroy_result;
42  }
43
44  remove_give_mutex:
45  xSemaphoreGive(manager->mutex);
46  return ret;
47  }

```

5.55.2 Factory Function Pattern

```

1  star_bus_config_t* star_bus_config_create_i2c(const char* name,
2  i2c_port_t port,
3  uint8_t address,
4  gpio_num_t sda_pin,
5  gpio_num_t scl_pin,
6  uint32_t clk_speed)

```

```

7 {
8   if (name == NULL || strlen(name) == 0) {
9       ESP_LOGE(s_TAG, "Invalid name for I2C bus config");
10      return NULL;
11  }
12
13  star_bus_config_t* config = calloc(1, sizeof(star_bus_config_t));
14  if (config == NULL) {
15      ESP_LOGE(s_TAG, "Failed to allocate memory for I2C config");
16      return NULL;
17  }
18
19  /* Initialize common fields */
20  config->name = strdup(name);
21  config->type = k_star_bus_type_i2c;
22  config->initialized = false;
23  config->handle = NULL;
24  config->next = NULL;
25
26  /* Initialize I2C-specific fields */
27  config->proto.i2c.port = port;
28  config->proto.i2c.address = address;
29  config->proto.i2c.config.sda_io_num = sda_pin;
30  config->proto.i2c.config.scl_io_num = scl_pin;
31  config->proto.i2c.config.master.clk_speed = clk_speed;
32
33  /* Set default operations */
34  star_bus_i2c_init_default_ops(&config->proto.i2c.ops);
35
36  return config;
37 }

```

5.56 Callback Functions

```

1  /* Callback type definition */
2  typedef esp_err_t (*star_bus_callback_t)(star_bus_config_t* bus,
3  void* user_ctx);
4
5  /* Function using callback */
6  esp_err_t star_bus_manager_with_bus(star_bus_manager_t* manager,
7  const char* name,
8  star_bus_callback_t callback,
9  void* user_ctx)
10 {
11     ESP_RETURN_ON_FALSE(manager && name && callback,
12     ESP_ERR_INVALID_ARG, s_TAG, "Invalid arguments");
13
14     /* ... find bus and invoke callback while holding mutex ... */
15     esp_err_t callback_result = callback(found_bus, user_ctx);
16
17     xSemaphoreGive(manager->mutex);
18     return callback_result;
19 }

```

```

20
21 /* Using the callback pattern */
22 esp_err_t my_bus_callback(star_bus_config_t* bus, void* ctx) {
23     ESP_LOGI(TAG, "Bus type: %s", star_bus_type_to_string(bus->type));
24     return ESP_OK;
25 }
26
27 star_bus_manager_with_bus(&manager, "sensor_i2c", my_bus_callback, NULL);

```

5.57 Summary

- Action functions return `esp_err_t`; predicates return `bool`
- Use `static` with `internal_` prefix for file-local functions
- Use `priv_` prefix for exposed-but-not-public functions
- Validate all parameters at function start
- Align parameters vertically in multi-line declarations
- Document all public functions with complete Doxygen
- Use single exit point with cleanup for complex functions
- Use factory functions for object creation

Global variables should be avoided when possible. This section establishes patterns for when they are necessary.

5.58 General Principles

- **Avoid Globals:** Global variables are strongly discouraged. Use dependency injection instead.
- **File-Scoped When Necessary:** If a global is needed, limit its scope to a single file using `static`.
- **Use Prefix Conventions:** `s_` for static file-scoped, `g_` for true globals.
- **Prefer static const Over #define:** For typed constants, use `static const`.

5.59 Static File-Scoped Variables (`s_` prefix)

Use `static` with `s_` prefix for variables that should be accessible only within a single source file.

```

1 /* CORRECT - Static file-scoped variables */
2
3 /* Logging tag - most common pattern */
4 static const char* s_TAG = "STAR_BUS_MANAGER";
5
6 /* Configuration constants - prefer static const over #define */
7 static const uint32_t s_i2c_timeout_ms = 1000;
8 static const uint32_t s_default_retry_count = 3;
9

```

```

10  /* State variables scoped to this file */
11  static bool s_module_initialized = false;
12
13  /* From star_bus_i2c.c */
14  static const char* s_TAG = "STAR_BUS_I2C";
15  static const uint32_t s_i2c_timeout_ms = 1000;
16
17  /* From star_bus_config.c */
18  static const char* s_TAG = "STAR_BUS_CONFIG";

```

5.60 Why static const Over #define

static const provides type safety and scope control that #define lacks.

```

1  /* PREFERRED - Type-safe, scoped, debugger-visible */
2  static const uint32_t s_timeout_ms = 1000;
3  static const size_t s_buffer_size = 256;
4
5  /* Use the constant with type checking */
6  uint32_t timeout = s_timeout_ms; /* Compiler checks type */
7
8  /* AVOID - No type safety, global namespace pollution */
9  #define TIMEOUT_MS 1000
10 #define BUFFER_SIZE 256
11
12 /* Issues with #define: */
13 /* 1. No type checking */
14 /* 2. Can collide with other defines */
15 /* 3. Not visible in debugger */
16 /* 4. Can have unexpected macro expansion */

```

5.61 When to Use #define

Use #define for:

- Compile-time constants for array sizes
- Conditional compilation
- Macros that cannot be functions

```

1  /* Array sizes must be compile-time constants */
2  #define STAR_BUS_NAME_MAX_LEN (32)
3
4  char bus_name[STAR_BUS_NAME_MAX_LEN]; /* OK - compile-time constant */
5
6  /* Conditional compilation */
7  #ifdef CONFIG_STAR_ENABLE_DEBUG_LOGGING
8  ESP_LOGD(s_TAG, "Debug info: %d", value);
9  #endif
10
11 /* Macros with special features */
12 #define RECORD_ERROR(handler, code, desc) \

```

```

13 do { \
14 error_handler_record_error((handler), (code), (desc), \
15 __FILE__, __LINE__, __func__); \
16 } while (0)

```

5.62 True Global Variables (g_ prefix)

If a truly global variable is unavoidable, use the g_ prefix. This should be rare.

```

1  /* DISCOURAGED but sometimes necessary */
2  /* g_ prefix for true globals (visible across files) */
3  g_system_state_t g_system_state;
4
5  /* Prefer dependency injection instead */
6  /* BETTER approach: */
7  typedef struct app_context {
8      star_bus_manager_t  bus_manager;
9      error_handler_t     error_handler;
10     /* ... */
11 } app_context_t;
12
13 /* Pass context to functions that need it */
14 void app_process_sensors(app_context_t* ctx);

```

5.63 Dependency Injection Alternative

Replace globals with dependency injection.

```

1  /* WRONG - Global error handler */
2  error_handler_t g_error_handler; /* Global - bad */
3
4  void some_function(void) {
5      if (error) {
6          RECORD_ERROR(&g_error_handler, err, "Failed"); /* Uses global */
7      }
8  }
9
10 /* CORRECT - Dependency injection */
11 typedef struct star_bus_manager {
12     star_error_interface_t* error_iface; /* Injected dependency */
13     /* ... */
14 } star_bus_manager_t;
15
16 esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
17 const char* tag,
18 star_error_interface_t* error_iface, /* Injected */
19 star_pin_interface_t* pin_iface) /* Injected */
20 {
21     manager->error_iface = error_iface;
22     /* ... */
23 }
24
25 /* Usage */

```

```

26 error_handler_t error_handler;
27 error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
28
29 star_error_interface_t error_iface;
30 error_handler_get_interface(&error_iface, &error_handler);
31
32 star_bus_manager_t bus_manager;
33 star_bus_manager_init(&bus_manager, "main", &error_iface, NULL);

```

5.64 Static Variables for Module State

When a module needs to maintain state, prefer explicit initialization and deinitialization.

```

1  /* Module-private state */
2  static struct {
3      bool      initialized;
4      uint32_t transaction_count;
5      /* ... */
6  } s_module_state = {
7      .initialized = false,
8      .transaction_count = 0,
9  };
10
11 esp_err_t module_init(void)
12 {
13     if (s_module_state.initialized) {
14         return ESP_ERR_INVALID_STATE;
15     }
16
17     /* Initialize module */
18     s_module_state.initialized = true;
19     return ESP_OK;
20 }
21
22 esp_err_t module_deinit(void)
23 {
24     if (!s_module_state.initialized) {
25         return ESP_ERR_INVALID_STATE;
26     }
27
28     /* Clean up */
29     s_module_state.initialized = false;
30     s_module_state.transaction_count = 0;
31     return ESP_OK;
32 }

```

5.65 Thread Safety for Shared State

If static variables are accessed from multiple threads, protect with mutex.

```

1  /* Thread-safe module state */
2  static struct {
3      bool      initialized;

```

```

4 SemaphoreHandle_t mutex;
5 uint32_t          counter;
6 } s_shared_state;
7
8 esp_err_t safe_increment_counter(void)
9 {
10     if (xSemaphoreTake(s_shared_state.mutex, pdMS_TO_TICKS(1000)) !=
11         pdTRUE) {
12         return ESP_ERR_TIMEOUT;
13     }
14     s_shared_state.counter++;
15
16     xSemaphoreGive(s_shared_state.mutex);
17     return ESP_OK;
18 }

```

5.66 Summary

Type	Prefix	Usage
Static file-scoped	s_	Prefer this for module-internal state
True global	g_	Avoid; use dependency injection
static const	s_	Type-safe constants
#define	None	Array sizes, macros, conditional compilation

- Use **static const** for type-safe constants
- Use **s_TAG** for module logging tags
- Prefer dependency injection over global variables
- Protect shared state with mutexes
- Limit variable scope to the smallest necessary

This section establishes standards for header file organization and content in the STAR firmware codebase.

5.67 General Principles

- **Header Files Should Only Contain Declarations:** No implementations except for inline functions.
- **Minimal Includes in Headers:** Include only what is necessary for the declarations.
- **Self-Contained Headers:** Each header should compile independently.
- **One Purpose Per Header:** Split large headers into focused sub-headers.

5.68 Include Guard Naming Convention

All header files must use include guards following the pattern: `STAR_COMPONENT_FILENAME_H`

```

1  /* CORRECT - from star_bus_manager.h */
2  #ifndef STAR_COMPONENT_BUS_MANAGER_H
3  #define STAR_COMPONENT_BUS_MANAGER_H
4
5  /* ... declarations ... */
6
7  #endif /* STAR_COMPONENT_BUS_MANAGER_H */
8
9  /* CORRECT - from star_bus_config.h */
10 #ifndef STAR_COMPONENT_BUS_CONFIG_H
11 #define STAR_COMPONENT_BUS_CONFIG_H
12 /* ... */
13 #endif /* STAR_COMPONENT_BUS_CONFIG_H */
14
15 /* CORRECT - from star_error_interface.h */
16 #ifndef STAR_ERROR_INTERFACE_H
17 #define STAR_ERROR_INTERFACE_H
18 /* ... */
19 #endif /* STAR_ERROR_INTERFACE_H */
20
21 /* CORRECT - from star_pin_interface.h */
22 #ifndef STAR_PIN_INTERFACE_H
23 #define STAR_PIN_INTERFACE_H
24 /* ... */
25 #endif /* STAR_PIN_INTERFACE_H */

```

5.69 C++ Compatibility

All headers must include C++ extern “C” blocks for compatibility with C++ compilers.

```

1  #ifndef STAR_COMPONENT_BUS_MANAGER_H
2  #define STAR_COMPONENT_BUS_MANAGER_H
3
4  #ifdef __cplusplus
5  extern "C" {
6      #endif
7
8      /* All declarations go here */
9
10     #ifdef __cplusplus
11     }
12 #endif
13
14 #endif /* STAR_COMPONENT_BUS_MANAGER_H */

```

5.70 Header File Structure

Headers should follow this organization:

```

1  /* lib/star_bus/include/star_bus_manager.h */
2
3  #ifndef STAR_COMPONENT_BUS_MANAGER_H
4  #define STAR_COMPONENT_BUS_MANAGER_H
5
6  #ifdef __cplusplus
7  extern "C" {
8      #endif
9
10     /* === 1. Includes === */
11     #include "esp_err.h"
12     #include "star_bus_manager_types.h"
13
14     /* === 2. File Documentation === */
15     /**
16      * @file star_bus_manager.h
17      * @brief Unified bus manager for I2C, SPI, GPIO, and other bus protocols
18      *
19      * The bus manager provides a central registry for managing multiple
20      * bus configurations. It supports dependency injection of error handlers
21      * and pin validators for loose coupling.
22      *
23      * Example Usage:
24      * @code
25      * // Initialize bus manager with dependency injection
26      * star_bus_manager_t bus_manager;
27      * star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
28      * @endcode
29      */
30
31     /* === 3. Forward Declarations (if needed) === */
32     /* Forward declarations reduce header dependencies */
33
34     /* === 4. Macros and Constants === */
35     #define STAR_BUS_NAME_MAX_LEN (32)
36
37     /* === 5. Type Definitions === */
38     /* (Usually in separate _types.h header) */
39
40     /* === 6. Function Declarations === */
41
42     /**
43      * @brief Initialize a bus manager instance.
44      *
45      * @param[out] manager Pointer to bus manager to initialize.
46      * @param[in] tag Optional tag string for logging.
47      * @param[in] error_iface Optional error handler interface.
48      * @param[in] pin_iface Optional pin validator interface.
49      * @return esp_err_t ESP_OK on success.
50      */
51     esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
52     const char* tag,
53     star_error_interface_t* error_iface,

```

```

54     star_pin_interface_t*    pin_iface);
55
56     /* ... more function declarations ... */
57
58     #ifdef __cplusplus
59 }
60 #endif
61
62 #endif /* STAR_COMPONENT_BUS_MANAGER_H */

```

5.71 Include Order

Includes should be ordered in groups, separated by blank lines:

```

1  /* 1. Primary header (in .c file, include corresponding .h first) */
2  #include "star_bus_manager.h"
3
4  /* 2. FreeRTOS includes */
5  #include "freertos/FreeRTOS.h"
6  #include "freertos/semphr.h"
7  #include "freertos/task.h"
8
9  /* 3. Standard library includes */
10 #include <inttypes.h>
11 #include <stdbool.h>
12 #include <stddef.h>
13 #include <stdint.h>
14 #include <stdio.h>
15 #include <stdlib.h>
16 #include <string.h>
17
18 /* 4. ESP-IDF includes */
19 #include "driver/gpio.h"
20 #include "driver/i2c.h"
21 #include "esp_check.h"
22 #include "esp_err.h"
23 #include "esp_log.h"
24 #include "sdkconfig.h"
25
26 /* 5. Project includes */
27 #include "star_bus_config.h"
28 #include "star_bus_i2c.h"
29 #include "star_bus_types.h"
30 #include "star_error_interface.h"

```

5.72 Forward Declarations

Use forward declarations to minimize header dependencies.

```

1  /* In star_bus_config.h - forward declare instead of including */
2  /* Forward declaration */
3  struct star_bus_manager; /* Don't need full definition */
4

```

```

5  /* Function using forward-declared type */
6  esp_err_t star_bus_config_init(star_bus_config_t*      config,
7  struct star_bus_manager* manager);
8
9  /* In star_bus_common_types.h */
10 /* Forward declarations for pointer-only usage */
11 typedef struct star_bus_config star_bus_config_t;
12 typedef struct star_bus_manager star_bus_manager_t;

```

5.73 Separating Types from Functions

For complex modules, split type definitions into dedicated headers.

```

1  /* File organization for star_bus module */
2  star_bus/include/
3  star_bus_manager.h      /* Main API (functions) */
4  star_bus_manager_types.h /* Types for manager */
5  star_bus_common_types.h /* Shared types */
6  star_bus_protocol_types.h /* Protocol-specific types */
7  star_bus_function_types.h /* Function pointer types */
8  star_bus_event_types.h   /* Event structures */
9  star_bus_types.h         /* Controller - includes all */
10
11 /* Controller header pattern */
12 /* star_bus_types.h */
13 #ifndef STAR_COMPONENT_BUS_TYPES_H
14 #define STAR_COMPONENT_BUS_TYPES_H
15
16 #ifdef __cplusplus
17 extern "C" {
18     #endif
19
20     /* Include all type headers */
21     #include "star_bus_common_types.h"
22     #include "star_bus_event_types.h"
23     #include "star_bus_function_types.h"
24     #include "star_bus_manager_types.h"
25     #include "star_bus_protocol_types.h"
26
27     #ifdef __cplusplus
28 }
29 #endif
30
31 #endif /* STAR_COMPONENT_BUS_TYPES_H */

```

5.74 Doxygen File Documentation

Include @file documentation for headers with significant content.

```

1  /**
2  * @file star_error_handler.h
3  * @brief Error handling with retry logic and exponential backoff
4  *

```

```

5  * This module provides a concrete implementation of the error interface
6  * with automatic retry management, exponential backoff delays, and
7  * optional reset function callbacks.
8  *
9  * Key Features:
10 * - Configurable max retries and backoff delays
11 * - Exponential backoff between retries
12 * - Optional custom reset function for recovery
13 * - Thread-safe with mutex protection
14 *
15 * Example Usage:
16 * @code
17 * error_handler_t error_handler;
18 * error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
19 *
20 * if (operation_failed()) {
21 *     RECORD_ERROR(&error_handler, ESP_ERR_TIMEOUT, "Timeout");
22 * }
23 * @endcode
24 */

```

5.75 Complete Header Template

```

1  /* lib/star_module/include/star_module.h */
2
3  #ifndef STAR_COMPONENT_MODULE_H
4  #define STAR_COMPONENT_MODULE_H
5
6  #ifdef __cplusplus
7  extern "C" {
8      #endif
9
10     /* === Includes === */
11     #include <stdbool.h>
12     #include <stdint.h>
13
14     #include "esp_err.h"
15
16     /* === File Documentation === */
17     /**
18      * @file star_module.h
19      * @brief Brief description of this module
20      *
21      * Detailed description of what this module provides and how to use it.
22      */
23
24     /* === Forward Declarations === */
25     typedef struct star_module star_module_t;
26
27     /* === Macros === */
28     #define STAR_MODULE_DEFAULT_TIMEOUT_MS (1000)
29
30     /* === Type Definitions === */

```

```

31 typedef struct star_module {
32     bool        initialized;
33     uint32_t     timeout_ms;
34     void*        user_ctx;
35 } star_module_t;
36
37 /* === Function Declarations === */
38
39 /**
40  * @brief Initialize the module.
41  *
42  * @param[out] module Pointer to module structure to initialize.
43  * @param[in]  timeout Timeout in milliseconds.
44  * @return esp_err_t ESP_OK on success.
45  */
46 esp_err_t star_module_init(star_module_t* module, uint32_t timeout);
47
48 /**
49  * @brief Deinitialize the module.
50  *
51  * @param[in,out] module Pointer to module structure to deinitialize.
52  * @return esp_err_t ESP_OK on success.
53  */
54 esp_err_t star_module_deinit(star_module_t* module);
55
56 #ifdef __cplusplus
57 }
58 #endif
59
60 #endif /* STAR_COMPONENT_MODULE_H */

```

5.76 Summary

- Use include guards: `STAR_COMPONENT_FILENAME_H`
- Include C++ extern “C” blocks
- Order includes: FreeRTOS, standard library, ESP-IDF, project
- Use forward declarations to minimize dependencies
- Split types into separate `_types.h` headers for complex modules
- Document files with `@file` Doxygen blocks
- Keep headers focused on declarations, not implementations

This section establishes standards for horizontal spacing in the STAR firmware codebase.

5.77 General Rules

- **Space After Keywords:** Add space after `if`, `for`, `while`, `switch`.
- **Binary Operators:** Add a single space around binary operators.

- **No Space After Function Names:** No space between function name and opening parenthesis.
- **Alignment:** Align related declarations and comments.

5.78 Space After Control Keywords

Add a space between keywords and opening parenthesis.

```

1  /* CORRECT - Space after keywords */
2  if (ret != ESP_OK) {
3      return ret;
4  }
5
6  for (star_bus_config_t* curr = manager->buses; curr; curr = curr->next) {
7      process(curr);
8  }
9
10 while (retry_count > 0) {
11     attempt_operation();
12     retry_count--;
13 }
14
15 switch (config->type) {
16     case k_star_bus_type_i2c: {
17         /* ... */
18         break;
19     }
20 }
21
22 /* WRONG - No space after keywords */
23 if(ret != ESP_OK) {           /* Missing space */
24 return ret;
25 }
26
27 for(int i = 0; i < 10; i++) { /* Missing space */
28 /* ... */
29 }

```

5.79 No Space After Function Names

Do not add space between function name and opening parenthesis.

```

1  /* CORRECT - No space after function name */
2  esp_err_t ret = star_bus_manager_init(&manager, "main", NULL, NULL);
3  ESP_LOGI(s_TAG, "Initialized bus manager");
4  free(buffer);
5
6  /* WRONG - Space after function name */
7  esp_err_t ret = star_bus_manager_init (&manager, "main", NULL, NULL);
8  ESP_LOGI (s_TAG, "Initialized bus manager");
9  free (buffer);

```

5.80 Binary Operators

Add a single space around binary operators.

```

1  /* CORRECT - Spaces around binary operators */
2  int result = a + b;
3  bool is_valid = (count > 0) && (size <= MAX_SIZE);
4  uint32_t mask = flags | FLAG_ENABLED;
5  int shifted = value << 2;
6  size_t remaining = total - processed;
7
8  /* Assignment operators */
9  count += 1;
10 flags |= FLAG_INITIALIZED;
11 value >>= 4;
12
13 /* Comparison operators */
14 if (ret != ESP_OK) { /* ... */ }
15 if (count >= threshold) { /* ... */ }
16 if (ptr == NULL) { /* ... */ }
17
18 /* WRONG - Missing spaces */
19 int result = a+b;
20 bool is_valid = (count>0)&&(size<=MAX_SIZE);
21 if (ret!=ESP_OK) { /* ... */ }

```

5.81 Unary Operators

No space between unary operator and operand.

```

1  /* CORRECT - No space with unary operators */
2  count++;
3  --remaining;
4  bool is_null = !ptr;
5  int negative = -value;
6  uint32_t complement = ~mask;
7  size_t size = sizeof(star_bus_config_t);
8  int* ptr = &value;
9  int deref = *ptr;
10
11 /* WRONG - Space with unary operators */
12 count ++;
13 -- remaining;
14 bool is_null = ! ptr;

```

5.82 Pointer Declarations

Place asterisk with the type, not the variable name.

```

1  /* CORRECT - Asterisk with type */
2  star_bus_config_t* config;
3  const char* name;
4  void* user_ctx;
5

```



```

6  /* Function parameters */
7  esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
8  const char*          tag,
9  star_error_interface_t* error_iface,
10 star_pin_interface_t* pin_iface);
11
12 /* Multiple pointers on same line (avoid when possible) */
13 int *a, *b; /* Both are pointers */
14
15 /* WRONG - Various incorrect styles */
16 star_bus_config_t *config; /* Asterisk with variable */
17 star_bus_config_t * config; /* Spaces around asterisk */

```

5.83 Comma and Semicolon Spacing

Space after comma and semicolon, not before.

```

1  /* CORRECT */
2  star_bus_manager_init(&manager, "main", &error_iface, &pin_iface);
3
4  for (int i = 0; i < count; i++) {
5      /* ... */
6  }
7
8  int array[] = {1, 2, 3, 4, 5};
9
10 /* WRONG */
11 star_bus_manager_init(&manager, "main", &error_iface, &pin_iface);
12 for (int i = 0 ; i < count ; i++) { /* ... */ }
13 int array[] = {1 ,2 ,3 ,4 ,5};

```

5.84 Parentheses Spacing

No space after opening parenthesis or before closing parenthesis.

```

1  /* CORRECT */
2  if (ret != ESP_OK) {
3      return ret;
4  }
5
6  result = calculate(a, b, c);
7
8  /* WRONG - Spaces inside parentheses */
9  if ( ret != ESP_OK ) {
10     return ret;
11 }
12
13 result = calculate( a, b, c );

```

5.85 Parameter Alignment

Align parameters vertically in multi-line function declarations and calls.

```

1  /* CORRECT - Aligned parameters */
2  esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
3  const char*      tag,
4  star_error_interface_t* error_iface,
5  star_pin_interface_t* pin_iface);
6
7  star_bus_config_t* star_bus_config_create_spi_device(
8  const char*      name,
9  spi_host_device_t host,
10 gpio_num_t      copi_pin,
11 gpio_num_t      cipo_pin,
12 gpio_num_t      sclk_pin,
13 int32_t          dma_chan,
14 const spi_device_interface_config_t* dev_cfg);
15
16 /* Long function calls */
17 ESP_RETURN_ON_FALSE(manager && config,
18 ESP_ERR_INVALID_ARG,
19 s_TAG,
20 "Manager or config pointer is NULL");

```

5.86 Variable and Comment Alignment

Align related variables and their comments.

```

1  /* CORRECT - Aligned declarations and comments */
2  typedef struct error_handler {
3      uint32_t      error_count;          /**< Total errors recorded */
4      uint32_t      max_retries;         /**< Max retry attempts */
5      uint32_t      current_retry;       /**< Current retry counter */
6      uint32_t      base_retry_delay;    /**< Base delay (ms) */
7      uint32_t      max_retry_delay;     /**< Max delay (ms) */
8      uint32_t      current_retry_delay; /**< Current delay with backoff
9      */
10     esp_err_t      last_error;          /**< Last recorded error code */
11     bool           in_error_state;       /**< Whether in error state */
12     void*          reset_context;        /**< Context for reset function
13     */
14     SemaphoreHandle_t mutex;             /**< Mutex for thread-safety */
15 } error_handler_t;
16
17 /* Aligned local variables */
18 i2c_port_t      port      = config->proto.i2c.port;
19 uint8_t         address   = config->proto.i2c.address;
20 esp_err_t       ret       = ESP_OK;
21 i2c_cmd_handle_t cmd      = NULL;

```

5.87 Struct Initializer Alignment

Align designated initializers.

```

1  /* CORRECT - Aligned initializers */

```

```

2 star_bus_event_t event = {
3     .bus_type = k_star_bus_type_i2c,
4     .bus_name = config->name,
5     .i2c      = {
6         .port      = port,
7         .address   = address,
8         .is_write  = true,
9         .len       = len
10    }
11 };
12
13 spi_device_interface_config_t spi_cfg = {
14     .clock_speed_hz = 10 * 1000 * 1000,
15     .mode           = 0,
16     .spics_io_num   = GPIO_NUM_5,
17     .queue_size     = 7,
18     .pre_cb         = NULL,
19     .post_cb        = NULL,
20 };

```

5.88 Summary

Context	Rule
After if, for, while, switch	Space
After function name	No space
Around binary operators	Space
With unary operators	No space
After comma/semicolon	Space
Inside parentheses	No space
Pointer asterisk	With type
Multi-line parameters	Align vertically

This section establishes standards for include statement organization and ordering in the STAR firmware codebase.

5.89 General Rules

- **Use Angle Brackets for System/Library Headers:** <stdio.h>.
- **Use Quotation Marks for Project Headers:** "my_project.h".
- **Minimal Includes in Header Files:** Only include what is necessary.
- **Group and Order Includes:** Follow consistent ordering.

5.90 Include Order

Includes should be ordered in these groups, separated by blank lines:

1. Primary header (in .c files)
2. FreeRTOS includes

3. Standard library includes
4. ESP-IDF includes
5. Project includes

```

1  /* lib/star_bus/src/star_bus_i2c.c */
2
3  /* 1. Primary header - always first in .c file */
4  #include "star_bus_i2c.h"
5
6  /* 2. FreeRTOS includes */
7  #include "freertos/FreeRTOS.h"
8
9  /* 3. Standard library includes */
10 #include <inttypes.h>
11 #include <string.h>
12
13 /* 4. ESP-IDF includes */
14 #include "esp_check.h"
15 #include "esp_log.h"
16
17 /* 5. Project includes */
18 #include "star_bus_manager.h"
19 #include "star_bus_types.h"

```

5.91 Complete Example from Codebase

```

1  /* lib/star_bus/src/star_bus_manager.c */
2
3  /* Primary header */
4  #include "star_bus_manager.h"
5
6  /* FreeRTOS */
7  #include "freertos/FreeRTOS.h"
8  #include "freertos/semphr.h"
9
10 /* Standard library */
11 #include <stdio.h>
12 #include <stdlib.h>
13 #include <string.h>
14
15 /* ESP-IDF */
16 #include "esp_check.h"
17 #include "esp_log.h"
18 #include "sdkconfig.h"
19
20 /* Project includes */
21 #include "star_bus_adc.h"
22 #include "star_bus_config.h"
23 #include "star_bus_gpio.h"
24 #include "star_bus_i2c.h"
25 #include "star_bus_spi.h"

```

5.92 FreeRTOS Include Order

FreeRTOS requires specific include ordering. Always include `FreeRTOS.h` first.

```
1  /* CORRECT - FreeRTOS.h must come first */
2  #include "freertos/FreeRTOS.h"
3  #include "freertos/semphr.h"
4  #include "freertos/task.h"
5  #include "freertos/queue.h"
6  #include "freertos/event_groups.h"
7
8  /* WRONG - Other FreeRTOS headers before FreeRTOS.h */
9  #include "freertos/semphr.h" /* Error: FreeRTOS.h not included first */
10 #include "freertos/FreeRTOS.h"
```

5.93 System Includes

Use angle brackets for system and standard library headers.

```
1  /* Standard C library */
2  #include <inttypes.h>
3  #include <stdbool.h>
4  #include <stddef.h>
5  #include <stdint.h>
6  #include <stdio.h>
7  #include <stdlib.h>
8  #include <string.h>
```

5.94 ESP-IDF Includes

ESP-IDF headers use quotation marks but are treated as a separate group.

```
1  /* ESP-IDF driver includes */
2  #include "driver/gpio.h"
3  #include "driver/i2c.h"
4  #include "driver/spi_master.h"
5  #include "driver/uart.h"
6
7  /* ESP-IDF system includes */
8  #include "esp_check.h"
9  #include "esp_err.h"
10 #include "esp_log.h"
11 #include "esp_system.h"
12
13 /* ESP-IDF ADC includes */
14 #include "esp_adc/adc_cali.h"
15 #include "esp_adc/adc_cali_scheme.h"
16 #include "esp_adc/adc_oneshot.h"
17
18 /* Configuration */
19 #include "sdkconfig.h"
```

5.95 Project Includes

Use quotation marks for project headers. Alphabetize within each group.

```

1  /* Project includes - alphabetized */
2  #include "star_bus_adc.h"
3  #include "star_bus_config.h"
4  #include "star_bus_gpio.h"
5  #include "star_bus_i2c.h"
6  #include "star_bus_manager.h"
7  #include "star_bus_spi.h"
8  #include "star_bus_types.h"
9  #include "star_error_interface.h"
10 #include "star_pin_interface.h"

```

5.96 Header File Includes

Header files should minimize includes and use forward declarations when possible.

```

1  /* star_bus_manager.h - Header file includes */
2  #ifndef STAR_COMPONENT_BUS_MANAGER_H
3  #define STAR_COMPONENT_BUS_MANAGER_H
4
5  #ifdef __cplusplus
6  extern "C" {
7      #endif
8
9      /* Only include what's needed for declarations */
10     #include "esp_err.h"
11     #include "star_bus_manager_types.h"
12
13     /* Use forward declarations for pointer-only usage */
14     struct star_bus_config; /* Don't include star_bus_config.h */
15
16     /* Function using forward-declared type */
17     esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
18     struct star_bus_config* config);
19
20     #ifdef __cplusplus
21 }
22 #endif
23
24 #endif /* STAR_COMPONENT_BUS_MANAGER_H */

```

5.97 Conditional Includes

Use `#ifdef` for platform-specific or optional includes.

```

1  /* Platform-specific includes */
2  #ifdef CONFIG_IDF_TARGET_ESP32S3
3  #include "esp32s3/rom/gpio.h"
4  #else
5  #include "esp32/rom/gpio.h"
6  #endif

```

```

7
8  /* Optional feature includes */
9  #ifdef CONFIG_STAR_ENABLE_DEBUG_LOGGING
10 #include "esp_debug_helpers.h"
11 #endif

```

5.98 Include Comments

Add comments for non-obvious includes explaining why they are needed.

```

1  #include "star_bus_manager.h"
2
3  #include "freertos/FreeRTOS.h"
4
5  #include <inttypes.h>
6  #include <string.h>
7
8  #include "esp_check.h"
9  #include "esp_log.h"
10 #include "star_bus_manager.h" /* Needed for star_bus_manager_find_bus */
11 #include "star_bus_types.h" /* Needed for star_bus_config_t and union */

```

5.99 Avoiding Circular Dependencies

Structure your includes to avoid circular dependencies.

```

1  /* Problem: Circular dependency */
2  /* star_bus_manager.h includes star_bus_config.h */
3  /* star_bus_config.h includes star_bus_manager.h */
4
5  /* Solution: Use forward declarations */
6  /* star_bus_config.h */
7  struct star_bus_manager; /* Forward declare, don't include */
8
9  esp_err_t star_bus_config_init(star_bus_config_t* config,
10 struct star_bus_manager* manager);
11
12 /* star_bus_config.c - Include full header here */
13 #include "star_bus_config.h"
14 #include "star_bus_manager.h" /* Full definition needed in .c */

```

5.100 Summary

Order	Category	Style
1	Primary header (.c files)	"..."
2	FreeRTOS	"freertos/..."
3	Standard library	<...>
4	ESP-IDF	"..."
5	Project	"..."

- Always include FreeRTOS.h before other FreeRTOS headers

- Separate groups with blank lines
- Alphabetize within each group
- Minimize includes in headers; use forward declarations
- Add comments for non-obvious include dependencies

This project follows the rule of **2 spaces** for indentation, without using tabs.

Unix-style line endings ('LF', represented as '\n') are mandatory.

Logging is handled using the ESP-IDF logging macros.

- **ESP_LOGE** - Error
- **ESP_LOGW** - Warning
- **ESP_LOGI** - Info
- **ESP_LOGD** - Debug
- **ESP_LOGV** - Verbose

This section establishes comprehensive naming standards for all identifiers in the STAR firmware codebase. Consistent naming improves readability, maintainability, and reduces cognitive load when navigating the codebase.

5.101 General Rules

- **Language:** All identifiers must be in English.
- **Clarity:** Names should be descriptive and self-documenting.
- **Abbreviations:** Avoid abbreviations unless they are widely understood (e.g., GPIO, I2C, SPI).

5.102 Functions and Variables

Functions and variables use **snake_case**—all lowercase with words separated by underscores.

```

1  /* CORRECT - snake_case for functions and variables */
2  esp_err_t star_bus_manager_init(star_bus_manager_t* manager, const char*
   tag);
3  uint32_t error_count;
4  size_t bytes_written;
5  bool is_initialized;
6
7  /* WRONG - other naming styles */
8  esp_err_t StarBusManagerInit(); /* PascalCase */
9  esp_err_t starBusManagerInit(); /* camelCase */
10 uint32_t ErrorCount; /* PascalCase */

```


5.103 Module Prefixes

All public functions must use the module prefix pattern: `star_[module]_[function]`.

```

1  /* From star_bus_manager.h */
2  esp_err_t star_bus_manager_init(...);
3  esp_err_t star_bus_manager_add_bus(...);
4  esp_err_t star_bus_manager_remove_bus(...);
5  esp_err_t star_bus_manager_deinit(...);
6
7  /* From star_bus_config.h */
8  star_bus_config_t* star_bus_config_create_i2c(...);
9  star_bus_config_t* star_bus_config_create_spi_device(...);
10 esp_err_t          star_bus_config_destroy(...);
11
12 /* From star_bus_i2c.h */
13 esp_err_t star_bus_i2c_write(...);
14 esp_err_t star_bus_i2c_read(...);
15 void      star_bus_i2c_init_default_ops(...);
16
17 /* From star_error_handler.h */
18 esp_err_t error_handler_init(...);
19 esp_err_t error_handler_record_error(...);
20 bool      error_handler_can_retry(...);

```

5.104 Constants and Macros

Constants and macros use `SCREAMING_SNAKE_CASE`—all uppercase with words separated by underscores.

```

1  /* CORRECT - SCREAMING_SNAKE_CASE for macros and constants */
2  #define STAR_BUS_NAME_MAX_LEN          (32)
3  #define CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS (1000)
4  #define I2C_MASTER_WRITE               0
5  #define I2C_MASTER_READ                1
6  #define GPIO_NUM_MAX                   48
7
8  /* Convenience macro with proper naming */
9  #define RECORD_ERROR(handler, code, desc) \
10 do { \
11     error_handler_record_error((handler), (code), (desc), \
12     __FILE__, __LINE__, __func__); \
13 } while (0)
14
15 #define STAR_IFACE_RECORD_ERROR(iface, err, msg) \
16 ((iface) && (iface)->record_error \
17 ? (iface)->record_error((iface)->ctx, (err), (msg), \
18     __FILE__, __LINE__, __func__) \
19 : (err))
20
21 /* WRONG */
22 #define star_bus_name_max_len  32  /* lowercase */
23 #define StarBusNameMaxLen     32  /* PascalCase */

```

5.105 Type Names

All custom types must use `snake_case` with the `_t` suffix.

```

1  /* CORRECT - snake_case_t for type names */
2  typedef struct star_bus_config  star_bus_config_t;
3  typedef struct star_bus_manager star_bus_manager_t;
4  typedef struct error_handler    error_handler_t;
5  typedef enum star_bus_type      star_bus_type_t;
6
7  /* Callback function types also use _t suffix */
8  typedef esp_err_t (*star_bus_callback_t)(star_bus_config_t* bus, void*
    user_ctx);
9  typedef esp_err_t (*pin_registration_function_t)(gpio_num_t pin_num,
10 const char* bus_name,
11 const char* pin_usage,
12 void*      user_ctx);
13
14 /* WRONG */
15 typedef struct StarBusConfig StarBusConfig; /* PascalCase */
16 typedef struct star_bus_config star_bus_config; /* missing _t suffix */

```

5.106 Enum Members

Enum members use the `k_` prefix followed by `snake_case`.

```

1  /* CORRECT - k_prefix with snake_case */
2  typedef enum {
3      k_star_bus_type_none,    /**< No bus type specified */
4      k_star_bus_type_i2c,    /**< I2C bus */
5      k_star_bus_type_spi,    /**< SPI bus */
6      k_star_bus_type_gpio,   /**< GPIO bus */
7      k_star_bus_type_adc,    /**< ADC bus */
8      k_star_bus_type_count,  /**< Number of bus types (sentinel) */
9  } star_bus_type_t;
10
11 /* Usage in switch statements */
12 switch (config->type) {
13     case k_star_bus_type_i2c: {
14         /* Handle I2C */
15         break;
16     }
17     case k_star_bus_type_spi: {
18         /* Handle SPI */
19         break;
20     }
21     default: {
22         ESP_LOGW(s_TAG, "Unknown bus type: %d", config->type);
23         break;
24     }
25 }
26
27 /* WRONG */
28 typedef enum {

```

```

29  STAR_BUS_TYPE_I2C,      /* SCREAMING_SNAKE_CASE without k_ */
30  StarBusTypeI2c,        /* PascalCase */
31  starBusTypeI2c,        /* camelCase */
32 } bad_enum_t;

```

5.107 Static Variables (File-Scoped)

Static file-scoped variables use the `s_` prefix followed by `snake_case`.

```

1  /* CORRECT - s_ prefix for static file-scoped variables */
2  static const char* s_TAG = "STAR_BUS_MANAGER";
3  static const uint32_t s_i2c_timeout_ms = 1000;
4
5  /* From star_bus_i2c.c */
6  static const char* s_TAG = "STAR_BUS_I2C";
7  static const uint32_t s_i2c_timeout_ms = 1000;
8
9  /* From star_bus_config.c */
10 static const char* s_TAG = "STAR_BUS_CONFIG";
11
12 /* WRONG - missing s_ prefix */
13 static const char* TAG = "MODULE";
14 static uint32_t timeout_ms = 1000;

```

5.108 Global Variables

Global variables use the `g_` prefix followed by `snake_case`. However, global variables are **strongly discouraged**. Prefer dependency injection or static file-scoped variables.

```

1  /* CORRECT (but discouraged) - g_ prefix for globals */
2  g_system_state_t g_system_state;
3  uint32_t g_error_count;
4
5  /* PREFERRED - use dependency injection instead */
6  typedef struct {
7      star_error_interface_t* error_iface; /* Injected */
8      star_pin_interface_t* pin_iface; /* Injected */
9  } star_bus_manager_t;
10
11 /* Initialize with injected dependencies */
12 esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
13 const char* tag,
14 star_error_interface_t* error_iface,
15 star_pin_interface_t* pin_iface);

```

5.109 Private Functions

5.109.1 File-Scoped Static Functions (`internal_` prefix)

Static functions that are internal to a single source file use the `internal_` prefix.

```

1  /* From star_bus_manager.c - internal helper functions */
2  static esp_err_t internal_register_bus_pin(star_pin_interface_t*
      pin_iface,
3  gpio_num_t          pin,
4  const char*         bus_name,
5  const char*         usage,
6  bool                can_be_shared)
7  {
8      if (pin < 0 || pin >= GPIO_NUM_MAX) {
9          return ESP_OK; /* Not an error, pin is not used */
10     }
11     /* ... implementation ... */
12 }
13
14 static esp_err_t internal_unregister_bus_pin(star_pin_interface_t*
      pin_iface,
15 gpio_num_t          pin,
16 const char*         bus_name,
17 const char*         usage)
18 {
19     /* ... implementation ... */
20 }
21
22 static esp_err_t internal_register_config_pins(star_pin_interface_t*
      pin_iface,
23 const star_bus_config_t* config)
24 {
25     /* ... implementation ... */
26 }
27
28 /* From star_bus_i2c.c */
29 static esp_err_t internal_star_bus_i2c_write(const star_bus_config_t*
      config,
30 const uint8_t*       data,
31 size_t              len,
32 uint8_t              reg_addr,
33 size_t*              bytes_written);
34
35 static esp_err_t internal_i2c_execute_with_retry(const star_bus_config_t*
      config,
36 i2c_port_t           port,
37 i2c_cmd_handle_t     cmd,
38 const char*          operation_name);

```

5.109.2 Exposed Private Functions (priv_ prefix)

Functions that must be exposed across files within a module (but are not part of the public API) use the `priv_` prefix.

```

1  /* In private header (e.g., star_bus_internal.h) */
2  /* These functions are used by multiple .c files in the same module
3  but should not be called by external code */
4

```

```

5  esp_err_t priv_star_bus_validate_config(const star_bus_config_t* config);
6  esp_err_t priv_star_bus_acquire_mutex(star_bus_manager_t* manager);
7  void      priv_star_bus_release_mutex(star_bus_manager_t* manager);
8
9  /* Used in implementation files within the module */
10 esp_err_t star_bus_config_init(star_bus_config_t* config,
11 struct star_bus_manager* manager)
12 {
13     /* Validate using private helper */
14     esp_err_t ret = priv_star_bus_validate_config(config);
15     if (ret != ESP_OK) {
16         return ret;
17     }
18     /* ... rest of implementation ... */
19 }

```

5.110 Summary Table

Identifier Type	Convention	Example
Functions	snake_case	star_bus_manager_init
Variables	snake_case	bytes_written
Constants/Macros	SCREAMING_SNAKE_CASE	STAR_BUS_NAME_MAX_LEN
Type names	snake_case_t	star_bus_config_t
Enum members	k_snake_case	k_star_bus_type_i2c
Static variables	s_snake_case	s_TAG
Global variables	g_snake_case	g_system_state
Internal functions	internal_snake_case	internal_register_bus_pin
Private functions	priv_snake_case	priv_star_bus_validate_config
Module prefix	star_[module]_	star_bus_, star_error_

This section describes the patterns and conventions for defining custom types in the STAR firmware codebase. Consistent type definition patterns improve code readability and enable better tooling support.

5.111 Typedef Struct Pattern

All structures should be defined using the typedef pattern with the `_t` suffix.

5.111.1 Standard Pattern

```

1  /* Forward declaration (in header for dependencies) */
2  typedef struct star_bus_config star_bus_config_t;
3  typedef struct star_bus_manager star_bus_manager_t;
4
5  /* Full definition (in header or implementation) */
6  typedef struct star_bus_config {
7      const char* name;          /**< Unique name for this bus/device
8          instance */
9      star_bus_type_t type;      /**< Type of bus (I2C, SPI, etc.) */
10     bool initialized; /**< Whether this bus has been initialized
11         */

```

```

10 void*          handle;          /**< Driver/device handle */
11 void*          user_ctx;        /**< User context pointer for callbacks */
12
13 /* Protocol-specific configuration (see union section) */
14 union {
15     star_i2c_bus_config_t  i2c;
16     star_spi_bus_config_t  spi;
17     star_gpio_bus_config_t gpio;
18     star_adc_bus_config_t  adc;
19 } proto;
20
21 struct star_bus_config* next; /**< Linked list pointer */
22 } star_bus_config_t;

```

5.111.2 Error Handler Example

```

1  /* From star_error_handler.h */
2  typedef struct error_handler {
3      uint32_t      error_count;          /**< Total number of errors
4          recorded */
5      uint32_t      max_retries;          /**< Maximum retry attempts
6          allowed */
7      uint32_t      current_retry;        /**< Current retry attempt
8          counter */
9      uint32_t      base_retry_delay;     /**< Base delay between retries
10         (ms) */
11      uint32_t      max_retry_delay;      /**< Maximum retry delay (ms) */
12      uint32_t      current_retry_delay;  /**< Current delay with backoff */
13
14      esp_err_t      last_error;          /**< Last recorded error code */
15      bool           in_error_state;      /**< Whether in error state */
16      void*          reset_context;       /**< Context for reset function */
17
18      SemaphoreHandle_t mutex;            /**< Mutex for thread-safety */
19
20      esp_err_t (*reset_fn)(void* context); /**< Optional reset function */
21 } error_handler_t;

```

5.112 Union Usage for Protocol Configurations

Unions are used to store protocol-specific configurations efficiently. This pattern allows a single configuration structure to support multiple bus types.

```

1  /* From star_bus_manager_types.h */
2  typedef struct star_bus_config {
3      const char*      name;
4      star_bus_type_t  type;  /* Discriminator for the union */
5      bool             initialized;
6      void*            handle;
7      void*            user_ctx;
8
9      /* Union holds protocol-specific configuration */

```

```

10 union {
11     star_i2c_bus_config_t i2c;    /**< I2C-specific configuration */
12     star_spi_bus_config_t spi;    /**< SPI-specific configuration */
13     star_gpio_bus_config_t gpio;  /**< GPIO-specific configuration */
14     star_adc_bus_config_t adc;    /**< ADC-specific configuration */
15 } proto;
16
17 struct star_bus_config* next;
18 } star_bus_config_t;
19
20 /* Accessing union members based on type discriminator */
21 switch (config->type) {
22     case k_star_bus_type_i2c: {
23         i2c_port_t port = config->proto.i2c.port;
24         uint8_t address = config->proto.i2c.address;
25         /* ... */
26         break;
27     }
28     case k_star_bus_type_spi: {
29         spi_host_device_t host = config->proto.spi.host;
30         gpio_num_t cop_i_pin = config->proto.spi.copi_io_num;
31         /* ... */
32         break;
33     }
34     /* ... other cases ... */
35 }

```

5.112.1 Protocol-Specific Structures

```

1  /* I2C bus configuration (from star_bus_protocol_types.h) */
2  typedef struct star_i2c_bus_config {
3      i2c_port_t port;    /**< I2C port number */
4      i2c_config_t config; /**< ESP-IDF I2C configuration */
5      uint8_t address;    /**< 7-bit I2C device address */
6      star_i2c_callbacks_t callbacks; /**< User callbacks */
7      star_i2c_ops_t ops;    /**< Operation function pointers */
8  } star_i2c_bus_config_t;
9
10 /* SPI bus configuration */
11 typedef struct star_spi_bus_config {
12     spi_host_device_t host;    /**< SPI host number */
13     bool is_peripheral; /**< True if SPI peripheral mode */
14     spi_bus_config_t bus_cfg;    /**< ESP-IDF SPI bus config */
15     spi_device_interface_config_t dev_cfg; /**< Device config */
16     star_spi_callbacks_t callbacks;
17     star_spi_ops_t ops;
18
19     /* Inclusive terminology pin names (see Terminology section) */
20     gpio_num_t cop_i_io_num;    /**< Controller Out, Peripheral In */
21     gpio_num_t cipo_io_num;    /**< Controller In, Peripheral Out */
22     gpio_num_t sclk_io_num;    /**< Serial Clock */
23     gpio_num_t cs_io_num;    /**< Chip Select */
24     /* ... */

```

```
25 } star_spi_bus_config_t;
```

5.113 Callback Function Types

Callback function pointer types follow a consistent pattern with the `_t` suffix.

```
1  /* From star_bus_common_types.h - Pin registration callback */
2  typedef esp_err_t (*pin_registration_function_t)(gpio_num_t pin_num,
3  const char* bus_name,
4  const char* pin_usage,
5  void* user_ctx);
6
7  /* From star_bus_manager.h - Bus access callback */
8  typedef esp_err_t (*star_bus_callback_t)(star_bus_config_t* bus,
9  void* user_ctx);
10
11 /* From star_bus_function_types.h - I2C operation callbacks */
12 typedef esp_err_t (*star_i2c_write_fn_t)(const star_bus_config_t* config,
13 const uint8_t* data,
14 size_t len,
15 uint8_t reg_addr,
16 size_t* bytes_written);
17
18 typedef esp_err_t (*star_i2c_read_fn_t)(const star_bus_config_t* config,
19 uint8_t* data,
20 size_t len,
21 uint8_t reg_addr,
22 size_t* bytes_read);
23
24 /* Event callbacks */
25 typedef void (*star_i2c_transfer_complete_cb_t)(const star_bus_event_t*
26 event,
27 void* user_ctx);
28
29 typedef void (*star_i2c_error_cb_t)(const star_bus_event_t* event,
30 esp_err_t error,
31 void* user_ctx);
32
33 /* Grouping related callbacks in a struct */
34 typedef struct star_i2c_callbacks {
35     star_i2c_transfer_complete_cb_t on_transfer_complete;
36     star_i2c_error_cb_t on_error;
37 } star_i2c_callbacks_t;
```

5.114 Interface Patterns (Dependency Injection)

Interfaces use structs with function pointers and an opaque context pointer. This pattern enables dependency injection and testability.

```
1  /* From star_error_interface.h */
2  typedef struct star_error_interface {
3      /**
4       * @brief Record an error
```



```

5  * @param ctx Implementation context
6  * @param error Error code
7  * @param message Error message
8  * @param file Source file
9  * @param line Line number
10 * @param func Function name
11 * @return ESP_OK on success
12 */
13 esp_err_t (*record_error)(void* ctx,
14 esp_err_t error,
15 const char* message,
16 const char* file,
17 int line,
18 const char* func);
19
20 /**
21 * @brief Check if retry is possible
22 * @param ctx Implementation context
23 * @return true if retry is allowed
24 */
25 bool (*can_retry)(void* ctx);
26
27 /**
28 * @brief Reset error state
29 * @param ctx Implementation context
30 * @return ESP_OK on success
31 */
32 esp_err_t (*reset_state)(void* ctx);
33
34 /**
35 * @brief Implementation context (opaque pointer)
36 */
37 void* ctx;
38 } star_error_interface_t;
39
40 /* From star_pin_interface.h */
41 typedef struct star_pin_interface {
42     esp_err_t (*register_pin)(void* ctx, int pin,
43     const char* description, bool shared);
44     esp_err_t (*unregister_pin)(void* ctx, int pin,
45     const char* description);
46     esp_err_t (*validate)(void* ctx);
47     void* ctx;
48 } star_pin_interface_t;

```

5.114.1 Interface Usage Pattern

```

1  /* Creating an interface from concrete implementation */
2  error_handler_t error_handler;
3  error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
4
5  star_error_interface_t error_iface;
6  error_handler_get_interface(&error_iface, &error_handler);

```

```

7
8  /* Injecting interfaces into components */
9  star_bus_manager_t bus_manager;
10 star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
11
12 /* Using interface through function pointers */
13 if (error_iface.can_retry(error_iface.ctx)) {
14     /* Retry operation */
15 }
16
17 /* Or using the convenience macro */
18 STAR_IFACE_RECORD_ERROR(&error_iface, ESP_ERR_TIMEOUT, "Operation timed
    out");

```

5.115 Opaque Handle Patterns

For complex objects that should not be directly manipulated, use opaque handles.

```

1  /* Forward declaration creates an opaque type */
2  typedef struct star_bus_manager star_bus_manager_t;
3
4  /* Public API only uses pointers to the opaque type */
5  esp_err_t star_bus_manager_init(star_bus_manager_t* manager, ...);
6  esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager, ...);
7
8  /* Full definition is only in the .c file or private header */
9  /* Users cannot access internal fields directly */
10
11 /* Example usage - users allocate storage but don't access internals */
12 star_bus_manager_t bus_manager; /* Stack allocation */
13 star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
14
15 /* Or heap allocation */
16 star_bus_manager_t* manager_ptr = malloc(sizeof(star_bus_manager_t));
17 star_bus_manager_init(manager_ptr, "main", &error_iface, &pin_iface);

```

5.116 Operations Structure Pattern

Group related function pointers in an operations structure for pluggable implementations.

```

1  /* From star_bus_protocol_types.h */
2  typedef struct star_i2c_ops {
3      star_i2c_write_fn_t      write;           /**< Write with register */
4      star_i2c_read_fn_t      read;            /**< Read with register */
5      star_i2c_write_command_fn_t write_command; /**< Write command byte */
6      star_i2c_read_raw_fn_t   read_raw;       /**< Read raw bytes */
7  } star_i2c_ops_t;
8
9  typedef struct star_spi_ops {
10     star_spi_transmit_fn_t transmit; /**< Transmit only */
11     star_spi_receive_fn_t receive;   /**< Receive only */
12     star_spi_transfer_fn_t transfer; /**< Full-duplex transfer */
13 } star_spi_ops_t;

```

```

14
15 typedef struct star_gpio_ops {
16     esp_err_t (*write_digital)(const struct star_bus_config* config,
17                               uint8_t pin_index, uint8_t value);
18     esp_err_t (*read_digital)(const struct star_bus_config* config,
19                               uint8_t pin_index, uint8_t* value);
20 } star_gpio_ops_t;
21
22 /* Initializing default operations */
23 void star_bus_i2c_init_default_ops(star_i2c_ops_t* ops)
24 {
25     if (ops == NULL) {
26         ESP_LOGE(s_TAG, "Cannot initialize NULL ops pointer");
27         return;
28     }
29     ops->write          = internal_star_bus_i2c_write;
30     ops->read           = internal_star_bus_i2c_read;
31     ops->write_command  = internal_star_bus_i2c_write_command;
32     ops->read_raw       = internal_star_bus_i2c_read_raw;
33 }

```

5.117 Manager Structure Pattern

Manager structures coordinate multiple resources with thread-safe access.

```

1 /* From star_bus_manager_types.h */
2 typedef struct star_bus_manager {
3     star_bus_config_t*    buses;          /**< Linked list of bus configs */
4     SemaphoreHandle_t     mutex;          /**< Mutex for thread-safety */
5     const char*           tag;            /**< Logging tag */
6     star_error_interface_t* error_iface;  /**< Injected error handler */
7     star_pin_interface_t* pin_iface;      /**< Injected pin validator */
8
9     /* Resource tracking */
10    bool spi_host_initialized[SPI_HOST_MAX];
11    int  spi_device_count[SPI_HOST_MAX];
12 } star_bus_manager_t;

```

5.118 Best Practices

- **Always use `_t` suffix:** Makes types easily distinguishable from variables.
- **Use forward declarations:** Minimize header dependencies and compilation time.
- **Document all fields:** Use Doxygen comments for struct members.
- **Keep unions discriminated:** Always pair unions with a type field.
- **Prefer static const over `#define`:** For typed constants within structures.
- **Use opaque types for encapsulation:** Hide implementation details from users.
- **Group related function pointers:** Use ops structs for pluggable implementations.

This section describes the standard file and directory structure for the STAR firmware codebase. Consistent organization improves navigability and maintainability.

5.119 Library Directory Structure

Each library follows a standard PlatformIO/ESP-IDF component structure:

```
lib/star_component/
  CMakeLists.txt      # CMake build configuration
  library.json         # PlatformIO library metadata
  include/            # Public headers (API)
    star_component.h
    star_component_types.h
  src/                # Implementation files
    star_component.c
```

5.119.1 Example: star_bus Library

```
lib/star_bus/
  CMakeLists.txt
  library.json
  include/
    star_bus_adc.h
    star_bus_common_types.h
    star_bus_config.h
    star_bus_event_types.h
    star_bus_function_types.h
    star_bus_gpio.h
    star_bus_i2c.h
    star_bus_manager.h
    star_bus_manager_types.h
    star_bus_protocol_types.h
    star_bus_spi.h
    star_bus_types.h
  src/
    star_bus_adc.c
    star_bus_config.c
    star_bus_gpio.c
    star_bus_i2c.c
    star_bus_manager.c
    star_bus_spi.c
    star_bus_types.c
```

5.120 CMakeLists.txt Structure

```
1 # lib/star_bus/CMakeLists.txt
2
3 idf_component_register(
4 SRCS
```

```

5 "src/star_bus_adc.c"
6 "src/star_bus_config.c"
7 "src/star_bus_gpio.c"
8 "src/star_bus_i2c.c"
9 "src/star_bus_manager.c"
10 "src/star_bus_spi.c"
11 "src/star_bus_types.c"
12 INCLUDE_DIRS
13 "include"
14 REQUIRES
15 driver
16 esp_adc
17 freertos
18 star_core
19 )

```

5.121 Header File Organization

Header files must follow a consistent internal structure.

5.121.1 Header File Template

```

1  /* lib/star_bus/include/star_bus_manager.h */
2
3  #ifndef STAR_COMPONENT_BUS_MANAGER_H
4  #define STAR_COMPONENT_BUS_MANAGER_H
5
6  #ifdef __cplusplus
7  extern "C" {
8      #endif
9
10     /* === Includes === */
11     #include "esp_err.h"
12     #include "star_bus_manager_types.h"
13
14     /**
15      * @file star_bus_manager.h
16      * @brief Unified bus manager for I2C, SPI, GPIO, and other bus protocols
17      *
18      * The bus manager provides a central registry for managing multiple
19      * bus configurations. It supports dependency injection of error handlers
20      * and pin validators for loose coupling.
21      *
22      * Key Features:
23      * - Thread-safe bus management with mutex protection
24      * - Support for multiple bus types (I2C, SPI, GPIO, etc.)
25      * - Automatic pin registration and conflict detection
26      * - Safe bus access via callback pattern
27      *
28      * Example Usage:
29      * @code
30      * // Initialize bus manager with dependency injection
31      * star_bus_manager_t bus_manager;

```

```

32  * star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
33  *
34  * // Create and add an I2C bus
35  * star_bus_config_t* i2c_bus = star_bus_config_create_i2c(
36  *     "sensor_i2c", I2C_NUM_0, 0x68, GPIO_NUM_21, GPIO_NUM_22, 400000);
37  * star_bus_manager_add_bus(&bus_manager, i2c_bus);
38  * @endcode
39  */
40
41  /* === Forward Declarations === */
42  /* (if needed beyond what's in included headers) */
43
44  /* === Public Functions === */
45
46  /**
47  * @brief Initialize a bus manager instance.
48  *
49  * @param[out] manager      Pointer to bus manager to initialize. Must
50  *                          not be NULL.
51  * @param[in]  tag          Optional tag string for logging. NULL uses
52  *                          default.
53  * @param[in]  error_iface  Optional error handler interface (can be
54  *                          NULL).
55  * @param[in]  pin_iface    Optional pin validator interface (can be
56  *                          NULL).
57  * @return esp_err_t ESP_OK on success, ESP_ERR_INVALID_ARG if manager
58  *         is NULL,
59  *         ESP_ERR_NO_MEM if mutex creation fails.
60  */
61  esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
62  const char* tag,
63  star_error_interface_t* error_iface,
64  star_pin_interface_t* pin_iface);
65
66  /* ... more function declarations ... */
67
68  #ifdef __cplusplus
69  }
70 #endif
71
72 #endif /* STAR_COMPONENT_BUS_MANAGER_H */

```

5.121.2 Header Section Order

1. **File path comment:** Identifies the file location
2. **Include guard:** #ifndef STAR_COMPONENT_FILENAME_H
3. **C++ extern “C” opening:** For C++ compatibility
4. **Includes:** System, then ESP-IDF, then project headers
5. **File documentation:** @file Doxygen block with overview

6. **Forward declarations:** If needed
7. **Constants and macros:** Public defines
8. **Type definitions:** Enums, structs, typedefs
9. **Function declarations:** Grouped by category
10. **C++ extern “C” closing:** Matches opening
11. **Include guard closing:** #endif /* ... */

5.122 Include Guard Naming Convention

Include guards follow the pattern: STAR_COMPONENT_FILENAME_H

```

1  /* star_bus_manager.h */
2  #ifndef STAR_COMPONENT_BUS_MANAGER_H
3  #define STAR_COMPONENT_BUS_MANAGER_H
4  /* ... */
5  #endif /* STAR_COMPONENT_BUS_MANAGER_H */
6
7  /* star_bus_config.h */
8  #ifndef STAR_COMPONENT_BUS_CONFIG_H
9  #define STAR_COMPONENT_BUS_CONFIG_H
10 /* ... */
11 #endif /* STAR_COMPONENT_BUS_CONFIG_H */
12
13 /* star_error_interface.h */
14 #ifndef STAR_ERROR_INTERFACE_H
15 #define STAR_ERROR_INTERFACE_H
16 /* ... */
17 #endif /* STAR_ERROR_INTERFACE_H */
18
19 /* star_pin_interface.h */
20 #ifndef STAR_PIN_INTERFACE_H
21 #define STAR_PIN_INTERFACE_H
22 /* ... */
23 #endif /* STAR_PIN_INTERFACE_H */

```

5.123 Source File Organization

Source files follow a consistent internal structure.

5.123.1 Source File Template

```

1  /* lib/star_bus/src/star_bus_manager.c */
2
3  /* === Primary Header === */
4  #include "star_bus_manager.h"
5
6  /* === System Includes === */
7  #include "freertos/FreeRTOS.h"

```

```

8  #include "freertos/semphr.h"
9
10 #include <stdio.h>
11 #include <stdlib.h>
12 #include <string.h>
13
14 /* === ESP-IDF Includes === */
15 #include "esp_check.h"
16 #include "esp_log.h"
17 #include "sdkconfig.h"
18
19 /* === Project Includes === */
20 #include "star_bus_adc.h"
21 #include "star_bus_config.h"
22 #include "star_bus_gpio.h"
23
24 /* === Constants === */
25
26 static const char* s_TAG = "STAR_BUS_MANAGER";
27
28 /* === Private Function Prototypes === */
29
30 static esp_err_t internal_register_bus_pin(star_pin_interface_t*
    pin_iface,
31 gpio_num_t          pin,
32 const char*         bus_name,
33 const char*         usage,
34 bool                can_be_shared);
35
36 static esp_err_t internal_unregister_bus_pin(star_pin_interface_t*
    pin_iface,
37 gpio_num_t          pin,
38 const char*         bus_name,
39 const char*         usage);
40
41 /* === Public Functions === */
42
43 esp_err_t star_bus_manager_init(star_bus_manager_t*    manager,
44 const char*          tag,
45 star_error_interface_t* error_iface,
46 star_pin_interface_t* pin_iface)
47 {
48     ESP_RETURN_ON_FALSE(manager, ESP_ERR_INVALID_ARG, s_TAG,
49         "Manager pointer is NULL");
50
51     manager->buses          = NULL;
52     manager->error_iface    = error_iface;
53     manager->pin_iface      = pin_iface;
54
55     /* ... rest of implementation ... */
56
57     return ESP_OK;
58 }
59

```



```

60  /* ... more public functions ... */
61
62  /* === Private Functions === */
63
64  static esp_err_t internal_register_bus_pin(star_pin_interface_t*
        pin_iface,
65  gpio_num_t          pin,
66  const char*         bus_name,
67  const char*         usage,
68  bool                can_be_shared)
69  {
70      if (pin < 0 || pin >= GPIO_NUM_MAX) {
71          return ESP_OK; /* Not an error, pin is not used */
72      }
73      /* ... implementation ... */
74  }

```

5.123.2 Source Section Order

1. **File path comment:** Identifies the file location
2. **Primary header:** The corresponding .h file for this .c file
3. **System includes:** FreeRTOS, standard library (<...>)
4. **ESP-IDF includes:** ESP-IDF components
5. **Project includes:** Other project headers ("...")
6. **Constants:** static const variables, including s_TAG
7. **Private function prototypes:** Forward declarations of static functions
8. **Public functions:** Implementation of API functions
9. **Private functions:** Implementation of static helper functions

5.124 Types Header Separation

For complex modules, separate type definitions into dedicated headers:

```

star_bus/include/
    star_bus_manager.h          # Main API (functions)
    star_bus_manager_types.h    # Types for manager
    star_bus_common_types.h     # Shared types (enums, forward decls)
    star_bus_protocol_types.h   # Protocol-specific types
    star_bus_function_types.h   # Function pointer types
    star_bus_event_types.h      # Event structures
    star_bus_types.h            # Controller header (includes all)

```

5.124.1 Controller Header Pattern

```

1  /* star_bus_types.h - includes all type headers */
2  #ifndef STAR_COMPONENT_BUS_TYPES_H
3  #define STAR_COMPONENT_BUS_TYPES_H
4
5  #ifdef __cplusplus
6  extern "C" {
7      #endif
8
9      /* Controller include file for all public types */
10     #include "star_bus_common_types.h"
11     #include "star_bus_event_types.h"
12     #include "star_bus_function_types.h"
13     #include "star_bus_manager_types.h"
14     #include "star_bus_protocol_types.h"
15
16     #ifdef __cplusplus
17     }
18     #endif
19
20 #endif /* STAR_COMPONENT_BUS_TYPES_H */

```

5.125 Project Root Structure

```

star-esp32-firmware/
    CLAUDE.md                # Claude Code instructions
    platformio.ini           # PlatformIO configuration
    sdkconfig.defaults       # ESP-IDF default config
    lib/                     # Libraries/components
        star_core/           # Core interfaces
        star_bus/            # Bus abstraction
        star_error_handler/
        star_pin_validator/
        star_sensor_*/       # Sensor drivers
        star_motor/
        star_pid/
    src/                     # Main application
        main.c
    partitions/              # Partition tables
        partitions_4mb.csv
        partitions_16mb.csv
    docs/                    # Documentation
        style_guide.tex
        style_guide_sections/
    scripts/                 # Build/utility scripts
    test/                    # Test files

```

5.126 Best Practices

- **One header per component:** Each component has its primary public header.
- **Minimize header dependencies:** Use forward declarations to reduce includes.
- **Self-contained headers:** Each header should compile independently.
- **Consistent naming:** Files match the component they contain.
- **Group related files:** Keep related headers and sources together.
- **Separate types from functions:** For complex modules, split into `_types.h`.
- **Document file purpose:** Use `@file` Doxygen blocks.

This section establishes the OSHWA (Open Source Hardware Association) inclusive terminology standards used throughout the STAR firmware codebase. These standards replace legacy terminology with more inclusive alternatives.

5.127 Overview

The STAR project follows inclusive terminology guidelines to ensure our codebase is welcoming to all contributors and avoids language with harmful historical connotations. This is particularly relevant in embedded systems where certain legacy terms have been deeply entrenched.

5.128 SPI Bus Terminology

5.128.1 COPI/CIPO (Not MOSI/MISO)

STAR Term	Legacy Term	Description
COPI	MOSI	Controller Out, Peripheral In
CIPO	MISO	Controller In, Peripheral Out
SCLK	SCLK	Serial Clock (unchanged)
CS	SS	Chip Select (unchanged)

```

1  /* CORRECT - OSHWA terminology */
2  typedef struct star_spi_bus_config {
3      gpio_num_t copi_io_num; /**< Controller Out, Peripheral In */
4      gpio_num_t cipo_io_num; /**< Controller In, Peripheral Out */
5      gpio_num_t sclk_io_num; /**< Serial Clock */
6      gpio_num_t cs_io_num;   /**< Chip Select */
7      /* ... */
8  } star_spi_bus_config_t;
9
10 /* Factory function uses inclusive naming */
11 star_bus_config_t* star_bus_config_create_spi_device(
12     const char*      name,
13     spi_host_device_t host,
14     gpio_num_t       copi_pin,   /* NOT mosi_pin */
15     gpio_num_t       cipo_pin,   /* NOT miso_pin */
16     gpio_num_t       sclk_pin,
17     int32_t          dma_chan,

```

```

18 const spi_device_interface_config_t* dev_cfg);
19
20 /* In documentation and comments */
21 /* CORRECT */
22 reg_result = register_pin_if_valid(curr->proto.spi.copi_io_num,
23 bus_name,
24 "SPI COPI", /* Descriptive usage */
25 reg_func,
26 user_ctx,
27 true);
28
29 /* WRONG */
30 reg_result = register_pin_if_valid(curr->proto.spi.mosi_io_num,
31 bus_name,
32 "SPI MOSI", /* Legacy terminology */
33 reg_func,
34 user_ctx,
35 true);

```

5.129 I2C and General Bus Terminology

5.129.1 Controller/Peripheral (Not Master/Slave)

STAR Term	Legacy Term	Description
Controller	Master	Device that initiates communication
Peripheral	Slave	Device that responds to controller
Primary	Master	For configuration structures
Main	Master	For general reference

```

1  /* CORRECT - Controller/Peripheral terminology */
2  typedef struct star_spi_bus_config {
3      spi_host_device_t host;
4      bool is_peripheral; /*< True if operating as SPI peripheral */
5      /* ... */
6  } star_spi_bus_config_t;
7
8  /* Factory function for peripheral mode */
9  star_bus_config_t* star_bus_config_create_spi_peripheral(
10 const char*      name,
11 spi_host_device_t host,
12 gpio_num_t      copi_pin,
13 gpio_num_t      cipo_pin,
14 gpio_num_t      sclk_pin,
15 gpio_num_t      cs_pin,
16 int32_t          queue_size,
17 uint8_t          mode);
18
19 /* In comments and documentation */
20 /**
21  * @brief I2C Controller operations
22  *
23  * The controller device initiates all transactions on the bus.

```

```

24  * Peripheral devices respond when addressed by the controller.
25  */
26
27  /* WRONG */
28  bool is_slave;           /* Use is_peripheral */
29  star_bus_config_create_spi_slave(); /* Use _peripheral */

```

5.130 Mapping ESP-IDF Legacy APIs

ESP-IDF still uses legacy terminology internally. When interfacing with ESP-IDF, document the mapping clearly.

```

1  /* From star_bus_protocol_types.h */
2  typedef struct star_spi_bus_config {
3      /* ... */
4
5      /* SPI pin naming follows OSHWA standard:
6      * - COPI (Controller Out, Peripheral In) - replaces MOSI
7      * - CIPO (Controller In, Peripheral Out) - replaces MISO
8      * This modern terminology removes master/slave language.
9      * The ESP-IDF still uses mosi_io_num/miso_io_num internally,
10     * which we map here.
11     */
12     gpio_num_t copi_io_num; /**< GPIO for COPI (maps to ESP-IDF
13         mosi_io_num) */
14     gpio_num_t cipo_io_num; /**< GPIO for CIPO (maps to ESP-IDF
15         miso_io_num) */
16     /* ... */
17 } star_spi_bus_config_t;
18
19 /* When setting up ESP-IDF structures */
20 static esp_err_t internal_setup_spi_bus(star_bus_config_t* config)
21 {
22     spi_bus_config_t bus_cfg = {
23         /* Map our inclusive terminology to ESP-IDF legacy names */
24         .mosi_io_num = config->proto.spi.copi_io_num,
25         .miso_io_num = config->proto.spi.cipo_io_num,
26         .sclk_io_num = config->proto.spi.sclk_io_num,
27         .quadwp_io_num = -1,
28         .quadhd_io_num = -1,
29         .max_transfer_sz = config->proto.spi.max_transfer_sz,
30     };
31     /* ... */
32 }
33
34 /* Similarly for I2C */
35 /* ESP-IDF uses I2C_MODE_SLAVE, but we document as peripheral mode */
36 i2c_config_t i2c_conf = {
37     .mode = I2C_MODE_MASTER, /* Controller mode in STAR terminology */
38     /* or */
39     .mode = I2C_MODE_SLAVE, /* Peripheral mode in STAR terminology */
40     /* ... */
41 };

```

5.131 Documentation Guidelines

When writing comments, documentation, or user-facing text:

```

1  /* CORRECT - Use inclusive terminology in all documentation */
2
3  /**
4   * @brief Configure ESP32 as SPI peripheral device
5   *
6   * The peripheral will respond to transactions initiated by an
7   * external SPI controller. Data flows:
8   * - COPI: Data received from controller
9   * - CIP0: Data sent to controller
10  *
11  * @param[in] name Unique name for this SPI peripheral instance
12  * @param[in] host SPI host device
13  * @param[in] copi_pin GPIO for Controller Out, Peripheral In
14  * @param[in] cipo_pin GPIO for Controller In, Peripheral Out
15  * ...
16  */
17 star_bus_config_t* star_bus_config_create_spi_peripheral(...);
18
19 /* WRONG - Legacy terminology */
20 /**
21  * @brief Configure ESP32 as SPI slave device
22  *
23  * The slave will respond to transactions initiated by the
24  * SPI master. Data flows:
25  * - MOSI: Data from master to slave
26  * - MISO: Data from slave to master
27  */

```

5.132 Code Review Checklist

When reviewing code, check for:

- **Variable names:** No “master”, “slave”, “mosi”, “miso”
- **Function names:** Use “controller”, “peripheral”, “copi”, “cipo”
- **Comments:** Update any legacy terminology
- **Documentation:** Ensure Doxygen uses inclusive terms
- **String literals:** Check log messages and error strings
- **ESP-IDF mapping:** Document when interfacing with legacy APIs

5.133 Common Patterns

```

1  /* Logging with inclusive terminology */
2  ESP_LOGI(s_TAG, "SPI peripheral initialized on host %d", host);
3  ESP_LOGI(s_TAG, "I2C controller configured on port %d", port);
4

```

```

5  /* Error messages */
6  ESP_LOGE(s_TAG, "Failed to initialize SPI peripheral: %s",
7  esp_err_to_name(ret));
8  ESP_LOGE(s_TAG, "I2C controller transaction failed");
9
10 /* Pin registration descriptions */
11 internal_register_bus_pin(pin_iface, config->proto.spi.copi_io_num,
12 config->name, "SPI COPI", true);
13 internal_register_bus_pin(pin_iface, config->proto.spi.cipo_io_num,
14 config->name, "SPI CIPO", true);

```

5.134 Summary

Use This	Not This
Controller	Master
Peripheral	Slave
COPI	MOSI
CIPO	MISO
Primary	Master (for configs)
Main	Master (general)
Allowlist	Whitelist
Blocklist	Blacklist

Following these guidelines ensures our codebase is inclusive and welcoming while maintaining technical clarity. The terminology accurately describes the functional roles of devices without relying on harmful historical language.

This section describes the key architectural patterns used throughout the STAR firmware codebase. These patterns enable loose coupling, testability, and maintainability in embedded systems.

5.135 Dependency Injection (DIP)

The Dependency Inversion Principle (DIP) is the foundational architectural pattern of the STAR framework. High-level modules depend on abstract interfaces rather than concrete implementations.

5.135.1 Interface Definition

Interfaces are defined as structs containing function pointers and an opaque context pointer.

```

1  /* From star_error_interface.h */
2  typedef struct star_error_interface {
3      esp_err_t (*record_error)(void* ctx, esp_err_t error,
4      const char* message, const char* file,
5      int line, const char* func);
6      bool (*can_retry)(void* ctx);
7      esp_err_t (*reset_state)(void* ctx);
8      void* ctx; /* Opaque context pointer */
9  } star_error_interface_t;
10
11 /* From star_pin_interface.h */
12 typedef struct star_pin_interface {
13     esp_err_t (*register_pin)(void* ctx, int pin,

```

```

14  const char* description, bool shared);
15  esp_err_t (*unregister_pin)(void* ctx, int pin,
16  const char* description);
17  esp_err_t (*validate)(void* ctx);
18  void* ctx;
19  } star_pin_interface_t;

```

5.135.2 Interface Implementation

Concrete implementations provide the actual functionality.

```

1  /* Create error handler and get interface */
2  error_handler_t error_handler;
3  error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
4
5  star_error_interface_t error_iface;
6  error_handler_get_interface(&error_iface, &error_handler);
7
8  /* Implementation of get_interface */
9  void error_handler_get_interface(star_error_interface_t* iface,
10 error_handler_t* handler)
11 {
12     iface->record_error = adapter_record_error; /* Adapter function */
13     iface->can_retry     = adapter_can_retry;
14     iface->reset_state  = adapter_reset_state;
15     iface->ctx           = handler; /* Concrete handler as context */
16 }

```

5.135.3 Dependency Injection Pattern

Components receive their dependencies through initialization functions.

```

1  /* Component declares interface dependencies */
2  typedef struct star_bus_manager {
3      star_bus_config_t*      buses;
4      SemaphoreHandle_t      mutex;
5      const char*            tag;
6      star_error_interface_t* error_iface; /* Injected */
7      star_pin_interface_t*  pin_iface;   /* Injected */
8      /* ... */
9  } star_bus_manager_t;
10
11 /* Initialization receives dependencies */
12 esp_err_t star_bus_manager_init(star_bus_manager_t* manager,
13 const char* tag,
14 star_error_interface_t* error_iface,
15 star_pin_interface_t* pin_iface)
16 {
17     ESP_RETURN_ON_FALSE(manager, ESP_ERR_INVALID_ARG, s_TAG,
18 "Manager pointer is NULL");
19
20     manager->buses = NULL;
21     manager->error_iface = error_iface; /* Can be NULL */

```



```

22     manager->pin_iface    = pin_iface;    /* Can be NULL */
23     /* ... */
24 }
25
26 /* Usage with injected dependencies */
27 error_handler_t error_handler;
28 error_handler_init(&error_handler, 3, 100, 5000, NULL, NULL);
29
30 star_error_interface_t error_iface;
31 star_pin_interface_t pin_iface;
32 error_handler_get_interface(&error_iface, &error_handler);
33 pin_validator_get_interface(&pin_iface);
34
35 star_bus_manager_t bus_manager;
36 star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);

```

5.135.4 NULL Interface Handling

Components gracefully handle NULL interfaces.

```

1  /* Check interface before use */
2  if (manager->error_iface && manager->error_iface->record_error) {
3      manager->error_iface->record_error(manager->error_iface->ctx,
4      ret, "Operation failed",
5      __FILE__, __LINE__, __func__);
6  }
7
8  /* Or use convenience macro */
9  #define STAR_IFACE_RECORD_ERROR(iface, err, msg) \
10 ((iface) && (iface)->record_error \
11 ? (iface)->record_error((iface)->ctx, (err), (msg), \
12 __FILE__, __LINE__, __func__) \
13 : (err))

```

5.136 Bus Abstraction Pattern

The bus abstraction provides a unified interface for different communication protocols.

5.136.1 Unified Configuration

```

1  /* Single config type with protocol union */
2  typedef struct star_bus_config {
3      const char*      name;
4      star_bus_type_t type; /* Discriminator */
5      bool             initialized;
6      void*            handle;
7      void*            user_ctx;
8
9      union {
10         star_i2c_bus_config_t i2c;
11         star_spi_bus_config_t spi;
12         star_gpio_bus_config_t gpio;

```

```

13     star_adc_bus_config_t  adc;
14 } proto;
15
16 struct star_bus_config* next;
17 } star_bus_config_t;

```

5.136.2 Factory Functions

Factory functions create configurations for specific protocols.

```

1  /* I2C bus factory */
2  star_bus_config_t* star_bus_config_create_i2c(
3  const char* name,
4  i2c_port_t port,
5  uint8_t address,
6  gpio_num_t sda_pin,
7  gpio_num_t scl_pin,
8  uint32_t clk_speed);
9
10 /* SPI device factory */
11 star_bus_config_t* star_bus_config_create_spi_device(
12 const char* name,
13 spi_host_device_t host,
14 gpio_num_t copi_pin,
15 gpio_num_t cipo_pin,
16 gpio_num_t sclk_pin,
17 int32_t dma_chan,
18 const spi_device_interface_config_t* dev_cfg);
19
20 /* GPIO bus factory */
21 star_bus_config_t* star_bus_config_create_gpio(
22 const char* name,
23 gpio_num_t* pins,
24 uint8_t pin_count);

```

5.136.3 Manager Pattern

The bus manager coordinates multiple bus configurations.

```

1  /* Initialize manager */
2  star_bus_manager_t bus_manager;
3  star_bus_manager_init(&bus_manager, "main", &error_iface, &pin_iface);
4
5  /* Add buses */
6  star_bus_config_t* i2c_bus = star_bus_config_create_i2c(
7  "sensor_i2c", I2C_NUM_0, 0x68, GPIO_NUM_21, GPIO_NUM_22, 400000);
8  star_bus_manager_add_bus(&bus_manager, i2c_bus);
9
10 /* Find and use buses */
11 star_bus_config_t* bus = star_bus_manager_find_bus(&bus_manager,
12     "sensor_i2c");
13
14 /* Safe access with callback (prevents use-after-free) */

```

```

14 esp_err_t my_callback(star_bus_config_t* bus, void* ctx) {
15     /* Access bus safely while mutex is held */
16     return ESP_OK;
17 }
18 star_bus_manager_with_bus(&bus_manager, "sensor_i2c", my_callback, NULL);

```

5.137 Operations Structure Pattern

Group related function pointers for pluggable implementations.

```

1  /* Define operations structure */
2  typedef struct star_i2c_ops {
3      star_i2c_write_fn_t      write;
4      star_i2c_read_fn_t      read;
5      star_i2c_write_command_fn_t write_command;
6      star_i2c_read_raw_fn_t   read_raw;
7  } star_i2c_ops_t;
8
9  /* Initialize with default implementations */
10 void star_bus_i2c_init_default_ops(star_i2c_ops_t* ops)
11 {
12     if (ops == NULL) {
13         ESP_LOGE(s_TAG, "Cannot initialize NULL ops pointer");
14         return;
15     }
16     ops->write      = internal_star_bus_i2c_write;
17     ops->read       = internal_star_bus_i2c_read;
18     ops->write_command = internal_star_bus_i2c_write_command;
19     ops->read_raw    = internal_star_bus_i2c_read_raw;
20 }
21
22 /* Use through function pointers */
23 esp_err_t star_bus_i2c_write(const star_bus_manager_t* manager,
24 const char* name, ...)
25 {
26     star_bus_config_t* bus_config = star_bus_manager_find_bus(manager,
27 name);
28     /* ... validation ... */
29     return bus_config->proto.i2c.ops.write(bus_config, data, len,
30 reg_addr, bytes_written);
31 }

```

5.138 Thread-Safe Callback Pattern

Safely access shared resources through callbacks.

```

1  /* Callback type */
2  typedef esp_err_t (*star_bus_callback_t)(star_bus_config_t* bus,
3 void* user_ctx);
4
5  /* Thread-safe access function */
6  esp_err_t star_bus_manager_with_bus(star_bus_manager_t* manager,
7 const char* name,

```

```

8 star_bus_callback_t callback,
9 void* user_ctx)
10 {
11     ESP_RETURN_ON_FALSE(manager && name && callback,
12     ESP_ERR_INVALID_ARG, s_TAG, "Invalid arguments");
13
14     /* Acquire mutex */
15     if (xSemaphoreTake(manager->mutex,
16     pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
17     != pdTRUE) {
18         return ESP_ERR_TIMEOUT;
19     }
20
21     /* Find bus while holding mutex */
22     star_bus_config_t* found_bus = NULL;
23     for (star_bus_config_t* curr = manager->buses; curr; curr = curr->next)
24     {
25         if (curr->name && strcmp(curr->name, name) == 0) {
26             found_bus = curr;
27             break;
28         }
29     }
30
31     if (found_bus == NULL) {
32         xSemaphoreGive(manager->mutex);
33         return ESP_ERR_NOT_FOUND;
34     }
35
36     /* Invoke callback while holding mutex - prevents use-after-free */
37     esp_err_t callback_result = callback(found_bus, user_ctx);
38
39     xSemaphoreGive(manager->mutex);
40     return callback_result;
41 }

```

5.139 Thread Safety Patterns

5.139.1 Mutex Timeout Convention

Standard mutex timeout is 1000ms (configurable via Kconfig).

```

1  /* Standard mutex acquisition pattern */
2  if (xSemaphoreTake(manager->mutex,
3  pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
4  != pdTRUE) {
5      ESP_LOGE(manager->tag, "Timeout acquiring mutex");
6      return ESP_ERR_TIMEOUT;
7  }
8
9  /* Critical section */
10 /* ... protected operations ... */
11
12 xSemaphoreGive(manager->mutex);

```

5.139.2 Mutex with Cleanup (goto pattern)

```

1  esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
2  star_bus_config_t* config)
3  {
4      ESP_RETURN_ON_FALSE(manager && config, ESP_ERR_INVALID_ARG,
5      s_TAG, "Invalid arguments");
6
7      esp_err_t ret = ESP_OK;
8
9      if (xSemaphoreTake(manager->mutex,
10 pdMS_TO_TICKS(CONFIG_STAR_KCONFIG_BUS_MUTEX_TIMEOUT_MS))
11 != pdTRUE) {
12     return ESP_ERR_TIMEOUT;
13 }
14
15 /* Check for duplicate name */
16 for (star_bus_config_t* curr = manager->buses; curr; curr = curr->next)
17 {
18     if (curr->name && strcmp(curr->name, config->name) == 0) {
19         ret = ESP_ERR_INVALID_STATE;
20         goto add_give_mutex; /* Single exit point */
21     }
22 }
23
24 /* Initialize bus hardware */
25 ret = star_bus_config_init(config, manager);
26 if (ret != ESP_OK) {
27     goto add_give_mutex;
28 }
29
30 /* Add to list */
31 config->next = manager->buses;
32 manager->buses = config;
33
34 add_give_mutex:
35 xSemaphoreGive(manager->mutex);
36 return ret;
37 }

```

5.140 Error Handler Ownership Pattern

Components track whether they own their error handler for proper cleanup.

```

1  /* Sensor handle with ownership tracking */
2  typedef struct mpu6050_handle {
3      star_bus_manager_t*    bus_manager;
4      const char*           bus_name;
5      star_error_interface_t* error_iface;
6      bool                  owns_error_handler; /* Ownership flag */
7      SemaphoreHandle_t      mutex;
8      bool                  initialized;
9      /* ... */

```

```

10 } mpu6050_handle_t;
11
12 /* Initialization with optional interface */
13 esp_err_t star_sensor_mpu6050_init(mpu6050_handle_t* handle,
14 star_bus_manager_t* bus_manager,
15 const char* bus_name,
16 star_error_interface_t* error_iface,
17 const mpu6050_config_t* config)
18 {
19     if (error_iface != NULL) {
20         /* Use provided interface - we don't own it */
21         handle->error_iface = error_iface;
22         handle->owns_error_handler = false;
23     } else {
24         /* Create default - we own it */
25         handle->error_iface = star_error_interface_create_default();
26         handle->owns_error_handler = true;
27     }
28     /* ... */
29 }
30
31 /* Cleanup respects ownership */
32 esp_err_t star_sensor_mpu6050_deinit(mpu6050_handle_t* handle)
33 {
34     /* Clean up error interface if we own it */
35     star_error_interface_cleanup(handle->error_iface,
36 handle->owns_error_handler);
37     handle->error_iface = NULL;
38     /* ... */
39 }
40
41 /* Helper for cleanup */
42 void star_error_interface_cleanup(star_error_interface_t* error_iface,
43 bool owns_handler)
44 {
45     if (!owns_handler || !error_iface) {
46         return; /* Not our responsibility */
47     }
48     error_handler_t* handler = (error_handler_t*)error_iface->ctx;
49     error_handler_destroy_default(handler);
50     free(error_iface);
51 }

```

5.141 Linked List Management

Bus configurations are managed as a singly-linked list.

```

1 /* Add to head of list (O(1)) */
2 config->next = manager->buses;
3 manager->buses = config;
4
5 /* Remove from list (O(n)) */
6 star_bus_config_t* prev = NULL;

```

```

7 for (star_bus_config_t* curr = manager->buses; curr; prev = curr, curr =
  curr->next) {
8     if (strcmp(curr->name, name) == 0) {
9         if (prev == NULL) {
10             manager->buses = curr->next; /* Remove head */
11         } else {
12             prev->next = curr->next; /* Remove middle/tail */
13         }
14         curr->next = NULL;
15         /* ... cleanup curr ... */
16         break;
17     }
18 }
19
20 /* Iterate through list */
21 for (star_bus_config_t* curr = manager->buses; curr; curr = curr->next) {
22     /* Process each configuration */
23 }

```

5.142 Best Practices Summary

- **Use dependency injection:** Components receive interfaces, not concrete types.
- **Handle NULL interfaces:** Always check before dereferencing.
- **Track ownership:** Know who is responsible for cleanup.
- **Use callbacks for safe access:** Prevents use-after-free with shared resources.
- **Consistent mutex timeout:** 1000ms standard timeout.
- **Single exit point with goto:** Ensures mutex release on all paths.
- **Factory functions:** Create configurations with proper initialization.
- **Operations structures:** Enable pluggable implementations.

5.143 Prefer Alternatives to Macros

Rule: Avoid macros unless absolutely necessary. Use modern C alternatives:

- `static const` for constants
- `inline` functions for simple operations
- `enum` for related constants

Rationale: Macros lack type safety, scope control, and debugger visibility.

Good:

```

static const uint32_t k_max_buffer_size = 256;
static const float k_pi = 3.14159f;

static inline int max(int a, int b) {

```

```
    return (a > b) ? a : b;
}
```

Bad:

```
#define MAX_BUFFER_SIZE 256
#define PI 3.14159
#define MAX(a,b) ((a) > (b) ? (a) : (b))
```

5.144 When Macros Are Necessary

Use macros only for:

1. Array sizes (compile-time constants)
2. Conditional compilation (`#ifdef`, `#if`)
3. Token pasting and stringification
4. Platform-specific code paths

5.145 Macro Safety Rules

5.145.1 Wrap Statement-Like Macros in `do-while(0)`

Rule: Always wrap multi-statement macros in `do { ... } while(0)`.

Why: Prevents bugs when used in if-statements without braces.

Correct:

```
#define LOG_ERROR(msg)      \
    do {                    \
        log_timestamp();    \
        log_message(msg);  \
    } while (0)
```

Incorrect (causes bugs):

```
#define LOG_ERROR(msg)      \
    log_timestamp();        \
    log_message(msg)
```

```
/* Bug example: */
if (error)
    LOG_ERROR("fail"); /* Only log_timestamp() is conditional! */
else
    handle_success();
```


5.145.2 Parenthesize All Macro Arguments

Rule: Wrap all argument uses and the entire expression in parentheses.

Correct:

```
#define SQUARE(x) ((x) * (x))
#define MAX(a, b) (((a) > (b)) ? (a) : (b))
```

Incorrect (causes bugs):

```
#define SQUARE(x) x * x

int result = SQUARE(2 + 3); /* Expands to: 2 + 3 * 2 + 3 = 11, not 25! */
```

5.145.3 Beware of Side Effects and Multiple Evaluation

Rule: Document if macro arguments are evaluated multiple times. Never pass arguments with side effects.

Dangerous:

```
#define MAX(a, b) (((a) > (b)) ? (a) : (b))

int i = 5;
int result = MAX(i++, 10); /* i is incremented TWICE! */
```

Solution: Use inline functions or add warning comment:

```
/* WARNING: Arguments evaluated multiple times - no side effects! */
#define MAX(a, b) (((a) > (b)) ? (a) : (b))
```

5.146 Multi-line Macro Formatting

Rules:

- Use backslash continuation aligned consistently
- Indent continuation lines for readability
- Place opening brace on same line as do

Example:

```
#define RECORD_ERROR(handler, code, desc) \
do { \
    if ((handler)->error_iface != NULL) { \
        (handler)->error_iface->record_error( \
            (code), \
            (desc), \
            __FILE__, \
            __LINE__ \
        ); \
    } \
} while (0)
```

5.147 Naming Conventions

Rule: Use SCREAMING_SNAKE_CASE for all macros.

Examples:

```
/* Simple constants */
#define MAX_RETRY_COUNT (3)
#define MOTOR_PWM_FREQ_HZ (20000)

/* Function-like macros (always use parentheses) */
#define SQUARE(x) ((x) * (x))
#define MAX(a, b) (((a) > (b)) ? (a) : (b))

/* Multi-statement macros (always use do-while(0)) */
#define LOG_ERROR(msg) \
do { \
    log_timestamp(); \
    log_message(msg); \
} while (0)

#define RECORD_ERROR(handler, code, desc) \
do { \
    error_handler_record_error( \
        (handler), (code), (desc), \
        __FILE__, __LINE__, __func__ \
    ); \
} while (0)
```

5.148 Include Guards

Use traditional include guards for maximum portability:

```
#ifndef STAR_COMPONENT_FILENAME_H
#define STAR_COMPONENT_FILENAME_H

/* File contents */

#endif /* STAR_COMPONENT_FILENAME_H */
```

Do not use `#pragma once` (see Section 12 for details).

5.149 Conditional Compilation

Best Practices:

- Document why conditionals exist
- Keep conditional blocks small
- Use configuration headers instead of scattered `#ifdef`

Good:

```
/* Platform-specific GPIO initialization */
#if defined(CONFIG_STAR_BOARD_ESP32_S3)
    configure_esp32s3_gpio();
#elif defined(CONFIG_STAR_BOARD_ESP32_WROOM)
    configure_esp32_wroom_gpio();
#else
    #error "Unsupported board configuration"
#endif
```

This section establishes the unit suffix naming standard used throughout the STAR firmware codebase. All numeric fields use SI (International System of Units) units with explicit suffixes to eliminate unit ambiguity and enable automatic documentation generation.

5.150 Overview

The firmware uses the MKS (Meter-Kilogram-Second) system as the foundation for all physical measurements. Every numeric variable that represents a physical quantity must include a unit suffix that clearly identifies its meaning.

5.151 Benefits

- **Eliminates Unit Ambiguity:** No confusion about whether a value is in meters or millimeters, degrees or radians
- **Self-Documenting Code:** Variable names explicitly state their units
- **Protocol Buffer Compatibility:** Matches naming conventions used in `star-proto/` messages
- **Prevents Conversion Errors:** Explicit units reduce the risk of unit mismatch bugs

5.152 Standard Unit Suffixes

Table 20: Standard Unit Suffixes for STAR Firmware

Suffix	Unit	Proto Example	C Example
_m	meters	x_m	distance_m
_mps	meters/second	velocity_mps	speed_mps
_mps2	m/s ²	accel_x_mps2	acceleration_mps2
_rad	radians	angle_rad	heading_rad
_rad_per_s	rad/second	omega_rad_per_s	angular_vel_rad_per_s
_celsius	degrees C	temperature_celsius	motor_temp_celsius
_deci_celsius	0.1°C	temp_deci_celsius	(BMS native format)
_us	microseconds	timestamp_us	elapsed_us
_ms	milliseconds	timeout_ms	period_ms
_ma	milliamps	current_ma	motor_current_ma
_mv	millivolts	cell_mv	pack_voltage_mv
_mw	milliwatts	power_mw	consumption_mw
_percent	0-100%	soc_percent	duty_cycle_percent

5.153 Usage Examples

5.153.1 Distance and Velocity

```

1  /* CORRECT - Distance and velocity with units */
2  float distance_m = 1.5f;           /* 1.5 meters */
3  float velocity_mps = 2.0f;         /* 2.0 meters per second */
4  float acceleration_mps2 = 9.81f;   /* 9.81 m/s^2 (gravity) */
5
6  /* Motor velocity tracking */
7  float target_velocity_mps = 0.5f;
8  float current_velocity_mps;
9  star_encoder_get_velocity_rpm(&encoder, 10.0f, 500,
    &current_velocity_mps);
10
11 /* WRONG - No unit suffix */
12 float distance = 1.5f;             /* What unit? Meters? Millimeters? Feet? */
13 float velocity = 2.0f;             /* Meters per second? RPM? */

```

5.153.2 Angular Measurements

```

1  /* CORRECT - Angular measurements in radians */
2  float heading_rad = 1.57f;         /* 90 degrees in radians */
3  float angular_velocity_rad_per_s = 3.14f; /* pi radians per second */
4
5  /* Robot pose */
6  typedef struct {
7      float x_m;
8      float y_m;
9      float theta_rad; /* Heading angle */

```

```

10 } robot_pose_t;
11
12 /* WRONG - Ambiguous angle units */
13 float heading = 90.0f; /* Degrees or radians? */
14 float omega = 3.14f; /* What unit? */

```

5.153.3 Temperature

```

1 /* CORRECT - Temperature with units */
2 float motor_temp_celsius;
3 star_sensor_ds18b20_read_temp(&temp_sensor, &motor_temp_celsius);
4
5 /* BMS uses deci-celsius (0.1 degree resolution) */
6 int16_t cell_temp_deci_celsius = 235; /* 23.5 degrees C */
7
8 /* Thermal shutdown threshold */
9 #define MOTOR_THERMAL_SHUTDOWN_CELSIUS (85.0f)
10
11 if (motor_temp_celsius > MOTOR_THERMAL_SHUTDOWN_CELSIUS) {
12     star_drv8243_stop(&motor, true); /* Emergency brake */
13 }
14
15 /* WRONG */
16 float motor_temp = 75.0f; /* Celsius? Fahrenheit? Kelvin? */

```

5.153.4 Time

```

1 /* CORRECT - Time units */
2 uint32_t timeout_ms = 1000; /* 1 second timeout */
3 uint64_t timestamp_us; /* Microsecond timestamp */
4 star_encoder_get_timestamp_us(&encoder, &timestamp_us);
5
6 /* Timing for control loops */
7 const uint32_t control_period_ms = 1; /* 1ms = 1000Hz */
8 const uint32_t sensor_poll_us = 10000; /* 10ms in microseconds */
9
10 /* WRONG */
11 uint32_t timeout = 1000; /* Milliseconds? Seconds? Ticks? */
12 uint64_t timestamp; /* What resolution? */

```

5.153.5 Electrical Measurements

```

1 /* CORRECT - Electrical units */
2 uint32_t motor_current_ma;
3 star_drv8243_read_current(&motor, &motor_current_ma);
4
5 uint32_t cell_voltage_mv = 3700; /* 3.7V in millivolts */
6 uint32_t pack_voltage_mv = 11100; /* 11.1V (3S LiPo) */
7 uint32_t power_mw = 7400; /* 7.4W in milliwatts */
8

```

```

9  /* Current limiting */
10 #define MAX_MOTOR_CURRENT_MA (2000) /* 2A limit */
11
12 if (motor_current_ma > MAX_MOTOR_CURRENT_MA) {
13     ESP_LOGW(s_TAG, "Motor current limit exceeded: %lu mA",
14             motor_current_ma);
15     star_drv8243_set_speed(&motor, 0.0f);
16 }
17
18 /* WRONG */
19 uint32_t motor_current = 1500; /* Milliamps? Amps? */
20 uint32_t voltage = 3700; /* Millivolts? Volts? */

```

5.153.6 Percentage

```

1  /* CORRECT - Percentage (0-100 range) */
2  float duty_cycle_percent = 75.0f; /* 75% duty cycle */
3  uint8_t battery_soc_percent = 85; /* 85% state of charge */
4  float throttle_percent = 50.0f; /* 50% throttle */
5
6  star_drv8243_set_speed(&motor, duty_cycle_percent);
7
8  /* WRONG */
9  float duty_cycle = 0.75f; /* Normalized (0-1)? Percentage (0-100)? */
10 uint8_t battery_soc = 85; /* Percentage? Raw ADC value? */

```

5.154 Protocol Buffer Integration

The unit suffix convention matches the naming used in Protocol Buffer messages:

```

1  // From star-proto/proto/robot_command.proto
2  message VelocityCommand {
3      float velocity_mps = 1; // Meters per second
4      float angular_velocity_rad_per_s = 2; // Radians per second
5  }
6
7  message BatteryState {
8      repeated uint32 cell_mv = 1; // Cell voltages in millivolts
9      uint32 current_ma = 2; // Pack current in milliamps
10     uint8 soc_percent = 3; // State of charge (0-100%)
11     int16 temperature_celsius = 4; // Pack temperature
12 }

```

5.155 Conversion Functions

When converting between units, use descriptive function names:

```

1  /* CORRECT - Clear unit conversion */
2  float rpm_to_mps(float rpm, float wheel_diameter_m)
3  {
4      float wheel_circumference_m = 3.14159f * wheel_diameter_m;
5      float revolutions_per_second = rpm / 60.0f;

```

```
6     return revolutions_per_second * wheel_circumference_m;
7 }
8
9 float celsius_to_deci_celsius(float temp_celsius)
10 {
11     return temp_celsius * 10.0f;
12 }
13
14 float mv_to_volts(uint32_t voltage_mv)
15 {
16     return voltage_mv / 1000.0f;
17 }
18
19 /* WRONG - Ambiguous conversions */
20 float convert_rpm(float rpm); /* Convert to what? */
21 float scale_temp(float temp); /* What units? */
```

5.156 Summary

- **Always use unit suffixes** for physical quantities
- **Match Protocol Buffer conventions** for consistency
- **Use SI base units** when possible (meters, radians, seconds)
- **Document conversions** clearly in function names
- **Prefer millivolts/milliamps** over floating-point volts/amps for integer precision

This convention eliminates an entire class of bugs caused by unit ambiguity and makes the codebase more maintainable.

This section establishes memory management rules for ESP32 firmware. The ESP32-S3 has limited RAM (512KB SRAM, no PSRAM in our configuration), making proper memory management critical for system stability and performance.

5.157 No Dynamic Allocation

Rule: Never use `malloc()`, `calloc()`, `realloc()`, or `free()` in production firmware.

5.157.1 Rationale

- **Heap Fragmentation:** Repeated allocation/deallocation causes memory fragmentation over time
- **Memory Leaks:** Forgotten `free()` calls lead to gradual RAM exhaustion
- **Unpredictable Timing:** Allocation time is non-deterministic (breaks real-time constraints)
- **Out-of-Memory Failures:** No recovery mechanism when heap is exhausted
- **Debugging Difficulty:** Memory corruption bugs are extremely hard to track down

5.157.2 Alternatives

Use static allocation instead:

```

1  /* WRONG - Dynamic allocation */
2  void process_sensor_data(void)
3  {
4      float* samples = (float*)malloc(1000 * sizeof(float));
5      if (samples == NULL) {
6          ESP_LOGE(s_TAG, "Out of memory!");
7          return;
8      }
9
10     /* Process samples... */
11
12     free(samples); /* Easy to forget! */
13 }
14
15 /* CORRECT - Static allocation */
16 static float s_samples[1000]; /* Allocated once at compile time */
17
18 void process_sensor_data(void)
19 {
20     /* Use s_samples directly - no allocation, no leaks */
21     for (int i = 0; i < 1000; i++) {
22         s_samples[i] = read_sensor();
23     }
24 }

```

5.157.3 ESP-IDF Heap Functions

ESP-IDF provides heap functions (`heap_caps_malloc`, `esp_heap_caps_malloc`) that are **only acceptable** for:

- **One-time initialization:** Allocating resources during startup that persist for the lifetime of the program
- **FreeRTOS task stacks:** Created once during task initialization
- **Third-party libraries:** When required by external dependencies (document clearly)

```

1  /* ACCEPTABLE - One-time allocation during init */
2  esp_err_t sensor_driver_init(sensor_handle_t* handle)
3  {
4      /* Allocate persistent buffer once */
5      handle->rx_buffer = heap_caps_malloc(DMA_BUFFER_SIZE,
6      MALLOC_CAP_DMA);
7      if (handle->rx_buffer == NULL) {
8          return ESP_ERR_NO_MEM;
9      }
10     handle->owns_buffer = true; /* Track ownership */
11     return ESP_OK;
12 }

```



```

13
14 /* Must have corresponding cleanup */
15 esp_err_t sensor_driver_deinit(sensor_handle_t* handle)
16 {
17     if (handle->owns_buffer && handle->rx_buffer) {
18         free(handle->rx_buffer);
19         handle->rx_buffer = NULL;
20     }
21     return ESP_OK;
22 }

```

5.158 Avoid Large Stack Allocations

Rule: Never allocate large arrays on the stack. Use static or global storage instead.

5.158.1 Stack Size Constraints

- ESP32 default task stack: **3KB–8KB** (configurable in FreeRTOS)
- ISR stack: Even smaller (~1KB)
- Stack overflow causes silent corruption or hard faults

5.158.2 Examples

```

1 /* WRONG - Large stack allocation */
2 void process_image(void)
3 {
4     uint8_t frame_buffer[640 * 480]; /* 307KB on stack! */
5     /* Stack overflow guaranteed! */
6 }
7
8 /* WRONG - Nested large allocations */
9 void motor_control_loop(void)
10 {
11     float pid_state[100]; /* 400 bytes */
12     uint32_t encoder_samples[500]; /* 2000 bytes */
13     /* Total: 2400 bytes - may overflow 3KB stack */
14 }
15
16 /* CORRECT - Use static storage */
17 static uint8_t s_frame_buffer[640 * 480]; /* In .bss, not stack */
18
19 void process_image(void)
20 {
21     /* Use s_frame_buffer safely */
22     memset(s_frame_buffer, 0, sizeof(s_frame_buffer));
23 }
24
25 /* CORRECT - Small stack allocations are fine */
26 void calculate_checksum(const uint8_t* data, size_t len)
27 {
28     uint32_t crc = 0; /* 4 bytes - perfectly fine on stack */

```

```

29  /* ... */
30  }

```

5.158.3 Stack Usage Guidelines

Table 21: Stack Allocation Guidelines

Size	Allowed	Storage Location
< 256 bytes	Yes	Stack (automatic)
256 bytes – 1 KB	Caution	Prefer static if persistent
> 1 KB	No	Static or global only
> 10 KB	Never	Static with explicit comment

5.159 Use const for Flash Storage

Rule: Place all read-only data in flash/ROM using the `const` keyword.

5.159.1 Rationale

- ESP32 has **16MB flash** but only **512KB RAM**
- `const` data is stored in flash and accessed directly (no RAM copy)
- Saves precious SRAM for runtime variables

5.159.2 Examples

```

1  /* WRONG - Lookup table in RAM (wastes 256 bytes) */
2  static uint8_t sine_lookup[256] = {
3      0, 3, 6, 9, 12, /* ... */
4  };
5
6  /* CORRECT - Lookup table in flash (zero RAM usage) */
7  static const uint8_t sine_lookup[256] = {
8      0, 3, 6, 9, 12, /* ... */
9  };
10
11 /* WRONG - String literals without const */
12 static char* error_messages[] = {
13     "Sensor timeout",
14     "Invalid checksum",
15     "Bus error"
16 };
17
18 /* CORRECT - Strings in flash */
19 static const char* const error_messages[] = {
20     "Sensor timeout",
21     "Invalid checksum",
22     "Bus error"
23 };
24

```

```

25 /* CORRECT - Default configuration in flash */
26 static const star_pid_config_t default_pid_config = {
27     .kp = 1.0f,
28     .ki = 0.5f,
29     .kd = 0.1f,
30     .output_min = -100.0f,
31     .output_max = 100.0f,
32 };
33
34 /* Usage: Copy to RAM only when needed */
35 star_pid_config_t runtime_config;
36 memcpy(&runtime_config, &default_pid_config, sizeof(runtime_config));

```

5.159.3 Pointer Declarations

Be careful with pointer const placement:

```

1 /* Pointer to const data (data in flash, pointer in RAM) */
2 const char* error_msg = "Timeout";
3
4 /* Const pointer to data (pointer in flash, data in RAM) */
5 char* const buffer_ptr = buffer;
6
7 /* Const pointer to const data (both in flash) */
8 const char* const error_table[] = {"Error 1", "Error 2"};

```

5.160 Validate All Buffers

Rule: Always validate buffer sizes before `memcpy()`, DMA operations, or ring buffer manipulations.

5.160.1 Rationale

- Buffer overruns cause memory corruption
- Corruption may manifest much later, making debugging extremely difficult
- Real-time systems have no memory protection (unlike Linux)

5.160.2 Validation Patterns

```

1 /* WRONG - Unchecked memcpy */
2 void receive_packet(uint8_t* data, size_t len)
3 {
4     memcpy(rx_buffer, data, len); /* What if len > buffer size? */
5 }
6
7 /* CORRECT - Validate before copy */
8 #define RX_BUFFER_SIZE (512)
9 static uint8_t rx_buffer[RX_BUFFER_SIZE];
10
11 void receive_packet(const uint8_t* data, size_t len)
12 {

```

```

13  if (data == NULL || len > RX_BUFFER_SIZE) {
14      ESP_LOGE(s_TAG, "Invalid packet: len=%zu (max=%d)",
15              len, RX_BUFFER_SIZE);
16      return;
17  }
18  memcpy(rx_buffer, data, len);
19  }
20
21  /* CORRECT - Clamp to buffer size */
22  void receive_packet_safe(const uint8_t* data, size_t len,
23  size_t* bytes_copied)
24  {
25      size_t copy_len = (len < RX_BUFFER_SIZE) ? len : RX_BUFFER_SIZE;
26      memcpy(rx_buffer, data, copy_len);
27      *bytes_copied = copy_len;
28
29      if (len > RX_BUFFER_SIZE) {
30          ESP_LOGW(s_TAG, "Packet truncated: %zu bytes lost",
31                  len - RX_BUFFER_SIZE);
32      }
33  }

```

5.160.3 DMA Buffer Validation

```

1  /* DMA buffers must be validated AND aligned */
2  esp_err_t start_dma_transfer(const uint8_t* src, size_t len)
3  {
4      /* Validate buffer pointer */
5      if (src == NULL) {
6          return ESP_ERR_INVALID_ARG;
7      }
8
9      /* Validate length */
10     if (len == 0 || len > MAX_DMA_TRANSFER_SIZE) {
11         ESP_LOGE(s_TAG, "Invalid DMA length: %zu", len);
12         return ESP_ERR_INVALID_SIZE;
13     }
14
15     /* Check alignment (DMA requires 4-byte alignment) */
16     if (((uintptr_t)src & 0x3) != 0) {
17         ESP_LOGE(s_TAG, "DMA buffer not 4-byte aligned");
18         return ESP_ERR_INVALID_ARG;
19     }
20
21     /* Proceed with DMA */
22     return spi_device_transmit(spi_handle, &transaction);
23 }

```

5.160.4 Ring Buffer Validation

```

1  /* CORRECT - Ring buffer with overflow protection */

```

```

2 typedef struct {
3     uint8_t buffer[256];
4     size_t head;
5     size_t tail;
6     size_t count;
7 } ring_buffer_t;
8
9 esp_err_t ring_buffer_write(ring_buffer_t* rb, const uint8_t* data,
10 size_t len)
11 {
12     if (rb == NULL || data == NULL) {
13         return ESP_ERR_INVALID_ARG;
14     }
15
16     /* Check available space */
17     size_t available = sizeof(rb->buffer) - rb->count;
18     if (len > available) {
19         ESP_LOGW(s_TAG, "Ring buffer overflow: %zu bytes lost",
20 len - available);
21         return ESP_ERR_NO_MEM;
22     }
23
24     /* Safe to write */
25     for (size_t i = 0; i < len; i++) {
26         rb->buffer[rb->head] = data[i];
27         rb->head = (rb->head + 1) % sizeof(rb->buffer);
28         rb->count++;
29     }
30
31     return ESP_OK;
32 }

```

5.161 Summary

Table 22: Memory Management Rules Summary

Rule	Implementation
No dynamic allocation	Use static/global buffers instead of malloc/free
Avoid large stack arrays	Use static storage for arrays > 256 bytes
Use const	Place lookup tables, strings, defaults in flash
Validate all buffers	Check lengths before memcpy, DMA, ring buffers

Following these rules ensures stable, predictable memory usage and prevents an entire class of memory-related bugs that are extremely difficult to debug in embedded systems.

This section establishes best practices for real-time embedded systems development on ESP32. These guidelines are critical for motor control (1000Hz loops), sensor fusion, and safety-critical robotics applications.

5.162 Interrupt Service Routines (ISRs)

5.162.1 Minimize ISR Work

Rule: Keep ISRs as short as possible. Set flags or queue work to background tasks instead of performing complex operations.

```

1  /* WRONG - Too much work in ISR */
2  static void IRAM_ATTR gpio_isr_handler(void* arg)
3  {
4      /* Reading I2C sensor blocks for milliseconds! */
5      float temperature = read_i2c_sensor();
6      update_pid_controller(temperature);
7      adjust_motor_speed();
8      log_telemetry_to_sd_card(); /* File I/O in ISR! */
9  }
10
11 /* CORRECT - Minimal ISR, queue work to task */
12 static volatile bool s_encoder_event = false;
13 static QueueHandle_t s_encoder_queue;
14
15 static void IRAM_ATTR encoder_isr_handler(void* arg)
16 {
17     /* Only set flag and queue event */
18     s_encoder_event = true;
19
20     encoder_event_t event = {.timestamp_us = esp_timer_get_time()};
21     xQueueSendFromISR(s_encoder_queue, &event, NULL);
22
23     /* ISR done in microseconds, not milliseconds */
24 }
25
26 /* Background task processes events */
27 void encoder_task(void* arg)
28 {
29     encoder_event_t event;
30     while (1) {
31         if (xQueueReceive(s_encoder_queue, &event, portMAX_DELAY)) {
32             /* Heavy processing here, not in ISR */
33             process_encoder_event(&event);
34         }
35     }
36 }

```

5.162.2 ISR Constraints

- **No blocking operations:** No mutexes, delays, or I/O
- **Use IRAM_ATTR:** Places ISR in IRAM for faster execution
- **FromISR variants:** Use xQueueSendFromISR(), not xQueueSend()
- **Keep stack usage low:** ISR stack is typically 1KB or less
- **No floating-point:** Context save/restore is expensive

5.163 No Blocking Delays

Rule: Never use `vTaskDelay()` or busy-wait loops in time-critical code. Use event-driven design.

```
1  /* WRONG - Blocking delay in motor control */
2  void motor_control_loop(void)
3  {
4      while (1) {
5          read_encoder();
6          compute_pid();
7          update_motor();
8
9          vTaskDelay(pdMS_TO_TICKS(1)); /* Blocks, jitter from scheduler */
10     }
11 }
12
13 /* CORRECT - Timer-driven execution */
14 static void motor_control_timer_callback(void* arg)
15 {
16     /* Called exactly every 1ms by hardware timer */
17     read_encoder();
18     compute_pid();
19     update_motor();
20 }
21
22 void setup_motor_control(void)
23 {
24     esp_timer_create_args_t timer_args = {
25         .callback = motor_control_timer_callback,
26         .name = "motor_ctrl"
27     };
28
29     esp_timer_handle_t timer;
30     esp_timer_create(&timer_args, &timer);
31     esp_timer_start_periodic(timer, 1000); /* 1ms = 1000us */
32 }
33
34 /* CORRECT - Event-driven sensor reading */
35 void sensor_task(void* arg)
36 {
37     while (1) {
38         /* Wait for event, no polling */
39         ulTaskNotifyTake(pdTRUE, portMAX_DELAY);
40
41         /* Event occurred, process it */
42         process_sensor_data();
43     }
44 }
```

5.164 Real-Time and Safety

Table 23: Real-Time Best Practices

Practice	Implementation
Know timing constraints	Profile worst-case execution times (WCET)
Use watchdog timers	Automatic reset on firmware lockup
Handle sensor timeouts	Never assume sensors always respond
Fail safe	On error, disable actuators and enter safe state
CRC critical data	Validate packets and stored configuration

5.164.1 Timing Constraints

```

1  /* Profile execution time using GPIO toggle */
2  void motor_control_loop(void)
3  {
4      gpio_set_level(DEBUG_PIN, 1);  /* Pin HIGH */
5
6      /* Critical timing section */
7      read_encoder();
8      compute_pid();
9      update_motor();
10
11     gpio_set_level(DEBUG_PIN, 0);  /* Pin LOW */
12
13     /* Measure with oscilloscope:
14     * Pulse width = worst-case execution time (WCET) */
15 }
16
17 /* Or use ESP32 cycle counter */
18 uint32_t start_cycles = esp_cpu_get_cycle_count();
19
20 critical_function();
21
22 uint32_t end_cycles = esp_cpu_get_cycle_count();
23 uint32_t elapsed_us = (end_cycles - start_cycles) / (CPU_FREQ_MHZ);
24 ESP_LOGI(s_TAG, "Execution time: %lu us", elapsed_us);

```

5.164.2 Watchdog Timers

```

1  /* Initialize task watchdog */
2  #include "esp_task_wdt.h"
3
4  void motor_control_task(void* arg)
5  {
6      /* Subscribe to watchdog (5 second timeout) */
7      esp_task_wdt_add(NULL);
8
9      while (1) {
10         /* Kick watchdog to prove we're alive */

```



```

11     esp_task_wdt_reset();
12
13     /* Control loop */
14     motor_control_iteration();
15
16     vTaskDelay(pdMS_TO_TICKS(1));
17 }
18 }
19
20 /* If task hangs, watchdog resets ESP32 after 5 seconds */

```

5.164.3 Sensor Timeouts

```

1  /* WRONG - Assumes sensor always responds */
2  float read_temperature(void)
3  {
4      send_i2c_command(TEMP_SENSOR_ADDR, READ_TEMP_CMD);
5      return receive_i2c_data(TEMP_SENSOR_ADDR); /* Hangs if sensor fails */
6  }
7
8  /* CORRECT - Timeout and error handling */
9  esp_err_t read_temperature_safe(float* temp_celsius)
10 {
11     esp_err_t ret;
12     uint8_t data[2];
13
14     /* I2C read with 100ms timeout */
15     ret = star_bus_i2c_read(bus_manager, "temp_i2c", data, 2,
16     TEMP_READ_REG, NULL);
17     if (ret != ESP_OK) {
18         ESP_LOGW(s_TAG, "Temperature sensor timeout");
19         return ret;
20     }
21
22     *temp_celsius = (data[0] << 8 | data[1]) / 10.0f;
23     return ESP_OK;
24 }
25
26 /* Use sensor timeout for safety */
27 esp_err_t ret = read_temperature_safe(&motor_temp);
28 if (ret != ESP_OK) {
29     /* Sensor failed - enter safe state */
30     star_drv8243_stop(&motor, true);
31     system_enter_safe_mode();
32 }

```

5.164.4 Fail-Safe Design

```

1  /* Safe state: all motors stopped, systems halted */
2  void system_enter_safe_mode(void)
3  {

```

```

4  ESP_LOGE(s_TAG, "ENTERING SAFE MODE");
5
6  /* Stop all motors */
7  for (int i = 0; i < NUM_MOTORS; i++) {
8      star_drv8243_stop(&motors[i], true); /* Brake */
9  }
10
11 /* Disable all actuators */
12 gpio_set_level(RELAY_ENABLE_PIN, 0);
13
14 /* Set error LED */
15 gpio_set_level(ERROR_LED_PIN, 1);
16
17 /* Log error to flash */
18 log_critical_error_to_nvs("Safe mode triggered");
19
20 /* Infinite loop - require manual reset */
21 while (1) {
22     vTaskDelay(pdMS_TO_TICKS(100));
23 }
24
25
26 /* Trigger safe mode on critical errors */
27 if (battery_voltage_mv < BATTERY_CRITICAL_MV) {
28     system_enter_safe_mode();
29 }
30
31 if (motor_temp_celsius > THERMAL_SHUTDOWN_CELSIUS) {
32     system_enter_safe_mode();
33 }

```

5.164.5 CRC Validation

```

1  /* Validate critical configuration with CRC */
2  typedef struct {
3      star_pid_config_t pid;
4      uint32_t crc32; /* CRC of pid structure */
5  } stored_config_t;
6
7  esp_err_t save_config_to_nvs(const star_pid_config_t* config)
8  {
9      stored_config_t stored;
10     memcpy(&stored.pid, config, sizeof(star_pid_config_t));
11
12     /* Calculate CRC-32 */
13     stored.crc32 = esp_crc32_le(0, (uint8_t*)&stored.pid,
14     sizeof(star_pid_config_t));
15
16     /* Save to NVS */
17     return nvs_set_blob(nvs_handle, "pid_config", &stored,
18     sizeof(stored_config_t));
19 }
20

```

```

21 esp_err_t load_config_from_nvs(star_pid_config_t* config)
22 {
23     stored_config_t stored;
24     size_t len = sizeof(stored_config_t);
25
26     esp_err_t ret = nvs_get_blob(nvs_handle, "pid_config",
27     &stored, &len);
28     if (ret != ESP_OK) {
29         return ret;
30     }
31
32     /* Verify CRC */
33     uint32_t calculated_crc = esp_crc32_le(0, (uint8_t*)&stored.pid,
34     sizeof(star_pid_config_t));
35     if (calculated_crc != stored.crc32) {
36         ESP_LOGE(s_TAG, "Config CRC mismatch! Flash corrupted?");
37         return ESP_ERR_INVALID_CRC;
38     }
39
40     memcpy(config, &stored.pid, sizeof(star_pid_config_t));
41     return ESP_OK;
42 }

```

5.165 Hardware Abstraction

5.165.1 Use Abstract Bus Interfaces

```

1  /* CORRECT - Driver depends on bus abstraction, not ESP-IDF */
2  esp_err_t mpu6050_init(mpu6050_handle_t* handle,
3  star_bus_manager_t* bus_manager,
4  const char* bus_name)
5  {
6      handle->bus_manager = bus_manager;
7      handle->bus_name = bus_name;
8
9      /* Access I2C through bus manager */
10     uint8_t whoami;
11     esp_err_t ret = star_bus_i2c_read(bus_manager, bus_name,
12     &whoami, 1, MPU6050_WHO_AM_I, NULL);
13
14     /* Easy to mock for testing */
15     return ret;
16 }

```

5.165.2 Debounce Mechanical Inputs

```

1  /* Software debounce for button */
2  typedef struct {
3      gpio_num_t pin;
4      bool state;
5      uint32_t last_change_ms;
6      uint32_t debounce_ms;

```

```

7 } debounced_button_t;
8
9 bool button_read(debounced_button_t* btn)
10 {
11     bool raw_state = gpio_get_level(btn->pin);
12     uint32_t now_ms = xTaskGetTickCount() * portTICK_PERIOD_MS;
13
14     if (raw_state != btn->state) {
15         /* State change detected */
16         if (now_ms - btn->last_change_ms > btn->debounce_ms) {
17             /* Stable for debounce period */
18             btn->state = raw_state;
19             btn->last_change_ms = now_ms;
20         }
21     }
22
23     return btn->state;
24 }

```

5.165.3 Enable Brown-Out Detection

```

1 /* In sdkconfig or menuconfig:
2  * CONFIG_ESP32_BROWNOUT_DET=y
3  * CONFIG_ESP32_BROWNOUT_DET_LVL_SEL_7=y (3.08V threshold)
4  */
5
6 /* ESP32 automatically resets on voltage dip below threshold */
7 /* Prevents flash corruption and erratic behavior */

```

5.165.4 Flash Wear-Leveling

```

1 /* Use NVS (Non-Volatile Storage) for wear-leveling */
2 #include "nvs_flash.h"
3
4 esp_err_t init_nvs(void)
5 {
6     esp_err_t ret = nvs_flash_init();
7     if (ret == ESP_ERR_NVS_NO_FREE_PAGES ||
8         ret == ESP_ERR_NVS_NEW_VERSION_FOUND) {
9         /* Erase and retry */
10         ESP_ERROR_CHECK(nvs_flash_erase());
11         ret = nvs_flash_init();
12     }
13     return ret;
14 }
15
16 /* NVS automatically rotates writes across sectors */
17 nvs_handle_t nvs_handle;
18 nvs_open("storage", NVS_READWRITE, &nvs_handle);
19 nvs_set_u32(nvs_handle, "boot_count", boot_count);
20 nvs_commit(nvs_handle);

```

5.165.5 Always Use Pull-Up/Pull-Down

```

1  /* WRONG - Floating input */
2  gpio_config_t io_conf = {
3      .pin_bit_mask = (1ULL << GPIO_NUM_35),
4      .mode = GPIO_MODE_INPUT,
5      /* No pull-up or pull-down! Reads random values */
6  };
7
8  /* CORRECT - Enable pull-up */
9  gpio_config_t io_conf = {
10     .pin_bit_mask = (1ULL << GPIO_NUM_35),
11     .mode = GPIO_MODE_INPUT,
12     .pull_up_en = GPIO_PULLUP_ENABLE,
13     .pull_down_en = GPIO_PULLEDOWN_DISABLE,
14 };
15 gpio_config(&io_conf);
16
17 /* For active-low signals (like nFAULT), use pull-up */
18 /* For active-high signals, use pull-down */

```

5.166 Performance and Real-Time

5.166.1 Benchmark, Don't Guess

```

1  /* Use GPIO toggle to measure timing on oscilloscope */
2  #define BENCHMARK_START()  gpio_set_level(BENCHMARK_PIN, 1)
3  #define BENCHMARK_END()    gpio_set_level(BENCHMARK_PIN, 0)
4
5  void pid_control_loop(void)
6  {
7      BENCHMARK_START();
8
9      float velocity_rpm;
10     star_encoder_get_velocity_rpm(&encoder, 1.0f, 500, &velocity_rpm);
11
12     float output;
13     star_pid_compute(&pid, setpoint, velocity_rpm, 0.001f, &output);
14
15     star_motor_set_duty(&motor, output);
16
17     BENCHMARK_END();
18     /* Oscilloscope shows pulse width = execution time */
19 }

```

5.166.2 Avoid Blocking Operations

```

1  /* WRONG - Blocking I2C read in control loop */
2  void control_loop(void)
3  {
4      while (1) {

```

```

5     uint8_t data[6];
6     i2c_master_read_from_device(I2C_NUM_0, SENSOR_ADDR,
7     data, 6, 100 / portTICK_PERIOD_MS);
8     /* Blocks for up to 100ms! */
9
10    process_data(data);
11  }
12 }
13
14 /* CORRECT - Non-blocking with DMA */
15 void setup_nonblocking_i2c(void)
16 {
17     /* Configure I2C interrupt + DMA */
18     /* When data ready, ISR triggers */
19     /* Background task processes results */
20 }

```

5.167 Power Management

5.167.1 Use Sleep Modes

```

1 /* Light sleep between sensor reads */
2 void sensor_polling_task(void* arg)
3 {
4     while (1) {
5         read_sensors();
6         process_data();
7
8         /* Sleep for 100ms instead of busy-wait */
9         esp_sleep_enable_timer_wakeup(100000); /* 100ms in us */
10        esp_light_sleep_start();
11    }
12 }

```

5.167.2 Use DMA to Reduce CPU Load

```

1 /* DMA-enabled SPI transfer (CPU-free) */
2 spi_device_transmit(spi_handle, &transaction);
3 /* DMA handles transfer, CPU does other work */

```

5.168 Testing and Debugging

5.168.1 Use Mock Drivers

```

1 /* Create mock bus for unit testing */
2 star_bus_manager_t mock_bus_manager;
3 star_bus_manager_init(&mock_bus_manager, "mock", NULL, NULL);
4
5 /* Test driver without hardware */
6 mpu6050_handle_t sensor;
7 mpu6050_init(&sensor, &mock_bus_manager, "mock_i2c");

```

5.168.2 Meaningful Logging

```
1  /* Include timestamp, module, and error codes */
2  ESP_LOGE(s_TAG, "[%lu] Motor fault detected: current=%lu mA (limit=%d
   mA)",
3  esp_timer_get_time() / 1000, /* Timestamp in ms */
4  current_ma,
5  MAX_MOTOR_CURRENT_MA);
```

5.169 Reliability and Fault Tolerance

5.169.1 Reset Peripherals if Stuck

```
1  /* I2C bus recovery */
2  esp_err_t recover_i2c_bus(i2c_port_t port)
3  {
4      ESP_LOGW(s_TAG, "Recovering I2C bus %d", port);
5
6      /* Delete driver */
7      i2c_driver_delete(port);
8
9      /* Wait */
10     vTaskDelay(pdMS_TO_TICKS(10));
11
12     /* Re-initialize */
13     i2c_config_t conf = { /* ... */ };
14     i2c_param_config(port, &conf);
15     return i2c_driver_install(port, conf.mode, 0, 0, 0);
16 }
```

5.170 Summary

Following these real-time and embedded best practices ensures:

- **Predictable timing** for motor control and sensor fusion
- **Robust error handling** for fault tolerance
- **Safe operation** in robotics applications
- **Maintainable code** through proper abstractions
- **Testability** with mock drivers and measurement tools

These practices are the foundation of reliable embedded systems for robotics.

This section establishes configuration requirements for Protocol Buffer messages used in ESP32 firmware. The STAR project uses **nanopb** (Nano Protocol Buffers) for embedded systems, which requires static memory allocation.

5.171 nanopb Overview

nanopb is a small, portable Protocol Buffers implementation designed for embedded systems:

- **Static Allocation:** All buffers allocated at compile time (no malloc)
- **Small Code Size:** Minimal flash/RAM footprint
- **C Language:** Compatible with C99 (not C++ required)
- **Field Size Constraints:** Maximum sizes configured in `.options` files

5.172 Why Static Allocation?

ESP32 firmware follows strict "no dynamic allocation" rules (see Section 25). `nanopb` field sizes must be explicitly configured to enable compile-time buffer allocation.

5.173 Field Size Configuration

Field sizes are configured in `.options` files located alongside `.proto` files:

```
1 # File: star-proto/proto/star.options
2 star.v1.RequestHeader.request_id max_size:64
3 star.v1.ResponseHeader.error_message max_size:128
4 star.v1.FirmwareChunk.data max_size:1024
```

5.174 Standard Field Size Constraints

Table 24: `nanopb` Field Size Constraints for ESP32

Field Type	Constraint	Example
String (UUID)	<code>max_size:64</code>	<code>request_id</code>
String (error)	<code>max_size:128</code>	<code>error_message</code>
String (version)	<code>max_size:32</code>	<code>firmware_version</code>
Repeated (cells)	<code>max_count:16</code>	<code>cell_mv[]</code>
Repeated (motors)	<code>max_count:4</code>	<code>motor_status[]</code>
Bytes (firmware)	<code>max_size:1024</code>	<code>chunk_data</code>

5.175 Examples

5.175.1 String Fields

```
1 // File: robot_command.proto
2 message RequestHeader {
3     string request_id = 1; // UUID (36 chars + null terminator)
4 }
5
6 // File: robot_command.options
7 RequestHeader.request_id max_size:64
```

Generated C code:


```

1 typedef struct _RequestHeader {
2     char request_id[64]; /* Static allocation */
3 } RequestHeader;

```

5.175.2 Repeated Fields

```

1 // File: battery_state.proto
2 message BatteryState {
3     repeated uint32 cell_mv = 1; // Cell voltages in millivolts
4 }
5
6 // File: battery_state.options
7 BatteryState.cell_mv max_count:16

```

Generated C code:

```

1 typedef struct _BatteryState {
2     uint32_t cell_mv[16]; /* Static array */
3     pb_size_t cell_mv_count; /* Actual element count */
4 } BatteryState;

```

5.175.3 Bytes Fields

```

1 // File: firmware_update.proto
2 message FirmwareChunk {
3     bytes data = 1; // Binary firmware data
4 }
5
6 // File: firmware_update.options
7 FirmwareChunk.data max_size:1024

```

Generated C code:

```

1 typedef struct _FirmwareChunk {
2     pb_bytes_array_t data; /* Contains uint8_t bytes[1024] */
3 } FirmwareChunk;

```

5.176 Sizing Guidelines

5.176.1 String Sizes

Table 25: Recommended String Field Sizes

Purpose	Size	Example
UUID (RFC 4122)	64 bytes	request_id, session_id
Error messages	128 bytes	error_message
Version strings	32 bytes	firmware_version
Short names	32 bytes	component_name
Log messages	256 bytes	log_text

5.176.2 Repeated Field Counts

Table 26: Recommended Repeated Field Counts

Purpose	Count	Example
Battery cells (3S LiPo)	3	cell_mv[]
Motors (quadruped)	4	motor_status[]
Motors (hexapod)	8	motor_status[]
Sensor array	8–16	temperature_sensors[]
Telemetry samples	32–64	velocity_samples[]

5.176.3 Bytes Field Sizes

Table 27: Recommended Bytes Field Sizes

Purpose	Size	Example
Firmware chunk	1024 bytes	chunk_data
Image thumbnail	4096 bytes	thumbnail
Small binary data	256 bytes	calibration_blob
Cryptographic key	64 bytes	aes_key

5.177 Memory Impact

5.177.1 Buffer Size Calculation

Each message allocates memory for the largest possible size:

```

1  /* Example: BatteryState with 3 cells (3S LiPo) */
2  typedef struct {
3      uint32_t cell_mv[3];          /* 3 * 4 = 12 bytes */
4      pb_size_t cell_mv_count;      /* 4 bytes */
5      uint32_t current_ma;          /* 4 bytes */
6      uint8_t soc_percent;          /* 1 byte */
7      /* Total: ~21 bytes + padding */
8  } BatteryState;
9
10 /* Even if only 1 cell used, full array is allocated */

```

5.177.2 RAM Usage Example

```

1  /* Static allocation of message buffers */
2  static RequestHeader s_request;    /* 64 bytes */
3  static ResponseHeader s_response; /* 128 bytes */
4  static BatteryState s_battery_state; /* 24 bytes */
5  static FirmwareChunk s_firmware_chunk; /* 1024 bytes */
6
7  /* Total: ~1240 bytes of static RAM */

```

5.178 Best Practices

5.178.1 Right-Size Your Buffers

- **Don't over-allocate:** max_size:1024 for a 20-byte string wastes 1004 bytes
- **Don't under-allocate:** Truncated data causes silent bugs
- **Measure actual usage:** Profile message sizes in production

5.178.2 Use Fixed-Size Types

```

1  /* CORRECT - Fixed-size integer types */
2  message MotorCommand {
3      int32 speed_percent = 1;      // Always 4 bytes
4      uint64 timestamp_us = 2;     // Always 8 bytes
5  }
6
7  /* AVOID - Variable-length encoding */
8  message MotorCommand {
9      sint32 speed_percent = 1;     // 1-5 bytes (varint encoding)
10 }

```

5.178.3 Validate Message Sizes

```

1  /* Check encoded message size before transmission */
2  uint8_t buffer[256];
3  pb_ostream_t stream = pb_ostream_from_buffer(buffer, sizeof(buffer));
4
5  bool status = pb_encode(&stream, RequestHeader_fields, &request);
6  if (!status) {
7      ESP_LOGE(s_TAG, "Encoding failed: %s", PB_GET_ERROR(&stream));
8      return ESP_FAIL;
9  }
10
11 size_t message_length = stream.bytes_written;
12 ESP_LOGI(s_TAG, "Encoded message: %zu bytes", message_length);
13
14 /* Verify it fits in SPI transaction buffer */
15 if (message_length > SPI_MAX_TRANSFER_SIZE) {
16     ESP_LOGE(s_TAG, "Message too large for SPI: %zu > %d",
17         message_length, SPI_MAX_TRANSFER_SIZE);
18     return ESP_ERR_INVALID_SIZE;
19 }

```

5.179 Integration with Code Generation

5.179.1 buf.gen.yaml Configuration

```

1  # File: star-proto/buf.gen.yaml
2  version: v2
3  plugins:

```

```

4 - local: nanopb_generator
5 out: gen/nanopb
6 opt:
7 - --extension=.pb

```

5.179.2 Options File Naming

```

1 # Options file must match .proto file name
2 proto/
3 robot_command.proto      -> robot_command.options
4 battery_state.proto      -> battery_state.options
5 firmware_update.proto    -> firmware_update.options

```

5.180 Common Pitfalls

5.180.1 Missing Options File

```

1 /* ERROR: No .options file for string field */
2 // robot_command.proto
3 message Command {
4     string name = 1; // No max_size specified
5 }
6
7 /* Generated code uses default (usually too small or error) */

```

Fix: Always create .options file for messages with strings, bytes, or repeated fields.

5.180.2 Mismatched Field Names

```

1 # WRONG - Field name mismatch
2 # robot_command.options
3 Command.command_name max_size:32 # Field is 'name', not 'command_name'
4
5 # CORRECT
6 Command.name max_size:32

```

5.180.3 Forgetting Array Count

```

1 /* Common mistake: Forgetting to set count field */
2 BatteryState battery = {0};
3 battery.cell_mv[0] = 3700;
4 battery.cell_mv[1] = 3720;
5 battery.cell_mv[2] = 3710;
6 /* BUG: battery.cell_mv_count not set! Encoder sends 0 cells */
7
8 /* CORRECT */
9 battery.cell_mv_count = 3; /* Set actual element count (3S LiPo) */

```

5.181 Summary

- **Always create .options files** for strings, bytes, repeated fields
- **Right-size buffers** based on actual usage (measure, don't guess)
- **Use fixed-size types** to avoid varint encoding overhead
- **Validate encoded sizes** before transmission
- **Set array count fields** for repeated fields

Properly configured nanopb constraints enable reliable, predictable Protocol Buffer communication on resource-constrained ESP32 hardware.

5.182 NASA Power of 10: Rules for Safety-Critical Code

This section documents the NASA/JPL Power of 10 coding rules developed by Gerard J. Holzmann in 2006 for safety-critical embedded systems. These rules prioritize verification and predictability over coding flexibility.

STAR Compliance Philosophy: The STAR project follows the Power of 10 rules, with one **intentional architectural deviation** for Dependency Inversion Principle (DIP) — a conscious trade-off for testability and modularity.

5.182.1 Compliance Status Legend

Symbol	Meaning
COMPLIANT	STAR follows this rule
INTENTIONAL DEVIATION	Architectural choice (DIP pattern)

5.182.2 Rule 1: Simplify Control Flow

Rule: Restrict all code to very simple control flow constructs — no `goto`, `setjmp/longjmp`, or recursion (direct or indirect).

Rationale: Creates predictable program structure for human understanding and static analysis. Recursion causes unbounded stack usage — dangerous in embedded systems.

STAR Compliance: **COMPLIANT**

```

1  /* COMPLIANT - No goto, no recursion */
2  esp_err_t star_motor_init(star_motor_handle_t* handle,
3                           const star_motor_config_t* config)
4  {
5      if (handle == NULL || config == NULL) {
6          return ESP_ERR_INVALID_ARG;
7      }
8
9      /* All control flow uses if/while/for */
10     esp_err_t ret = configure_timer(handle, config);
11     if (ret != ESP_OK) {
12         return ret;
13     }
14 
```

```

15     return configure_gpio(handle, config);
16 }

```

5.182.3 Rule 2: Fixed Loop Upper-Bounds

Rule: All loops must have a fixed upper bound that static analysis tools can trivially prove.

Rationale: Prevents runaway code, guarantees termination, enables real-time deadline verification.

STAR Compliance: **COMPLIANT**

```

1  /* COMPLIANT - Fixed iteration count */
2  #define MAX_RETRIES (3)
3
4  for (uint8_t i = 0; i < MAX_RETRIES; i++) {
5      if (sensor_read() == ESP_OK) {
6          break;
7      }
8      vTaskDelay(pdMS_TO_TICKS(100));
9  }
10
11 /* COMPLIANT - Array with known size */
12 #define BQ7850_MAX_CELLS (16)
13
14 for (uint8_t cell = 0; cell < BQ7850_MAX_CELLS; cell++) {
15     read_cell_voltage(cell, &voltages[cell]);
16 }

```

5.182.4 Rule 3: No Dynamic Memory After Initialization

Rule: Prohibit malloc/free after initialization phase.

Rationale: Dynamic allocation causes unpredictable performance, memory leaks, fragmentation, and corruption.

STAR Compliance: **COMPLIANT**

Best Practice: Use pre-allocated static pools instead of runtime memory requests.

```

1  /* COMPLIANT - Static allocation with fixed limits */
2  #define MAX_ITEMS (8)
3  #define MAX_DESC_LEN (64)
4
5  typedef struct {
6      char items[MAX_ITEMS][MAX_DESC_LEN]; /* Static 2D array */
7      uint8_t item_count;
8  } static_pool_t;
9
10 /* Usage */
11 if (pool->item_count >= MAX_ITEMS) {
12     return ESP_ERR_NO_MEM;
13 }
14 strncpy(pool->items[pool->item_count], desc, MAX_DESC_LEN - 1);
15 pool->items[pool->item_count][MAX_DESC_LEN - 1] = '\0';
16 pool->item_count++;

```

5.182.5 Rule 4: Keep Functions Short

Rule: Functions should not exceed ~60 lines (one printed page, one statement per line).

Rationale: Represents single verifiable logical units; improves comprehension, review, and testability.

STAR Compliance: **MOSTLY COMPLIANT**

```

1  /* COMPLIANT - Single responsibility, short function */
2  esp_err_t star_drv8243_read_current(star_drv8243_handle_t* handle,
3                                     float* current_ma)
4  {
5      if (handle == NULL || current_ma == NULL || !handle->initialized) {
6          return ESP_ERR_INVALID_ARG;
7      }
8
9      int raw_value = 0;
10     esp_err_t ret = read_adc_channel(handle, &raw_value);
11     if (ret != ESP_OK) {
12         return ret;
13     }
14
15     *current_ma = (float)raw_value * handle->ki_propi / 1000.0f;
16     return ESP_OK;
17 }
```

5.182.6 Rule 5: Use Assertions Generously

Rule: Minimum two assertions per function; assertions must be side-effect-free.

Rationale: Acts as executable documentation verifying pre-conditions, post-conditions, and invariants.

STAR Compliance: **COMPLIANT**

Note: STAR uses explicit parameter validation instead of assertions for production code (assertions disabled in release builds).

```

1  /* COMPLIANT - Explicit validation (better than assert for production) */
2  esp_err_t star_pid_compute(star_pid_handle_t* handle, float setpoint,
3                             float measurement, float dt, float* output)
4  {
5      /* Pre-condition checks (effectively runtime assertions) */
6      if (handle == NULL) {
7          return ESP_ERR_INVALID_ARG;
8      }
9      if (!handle->initialized) {
10         return ESP_ERR_INVALID_STATE;
11     }
12     if (output == NULL) {
13         return ESP_ERR_INVALID_ARG;
14     }
15     if (dt <= 0.0f || dt > 1.0f) { /* Sanity check time delta */
16         return ESP_ERR_INVALID_ARG;
17     }
18
19     /* Function logic */
```

```

20 float error = setpoint - measurement;
21 handle->integral += error * dt;
22
23 /* Post-condition: integral within bounds */
24 if (handle->integral > handle->integral_max) {
25     handle->integral = handle->integral_max;
26 } else if (handle->integral < handle->integral_min) {
27     handle->integral = handle->integral_min;
28 }
29
30 *output = handle->kp * error + handle->ki * handle->integral;
31 return ESP_OK;
32 }

```

5.182.7 Rule 6: Declare Data at Smallest Scope

Rule: Minimize variable scope to reduce accidental corruption risks.

Rationale: Limits chances of accidental reference or corruption from other code.

STAR Compliance: **COMPLIANT**

```

1 /* COMPLIANT - Variables declared at minimal scope */
2 void process_sensors(void)
3 {
4     /* Loop variable declared in for statement */
5     for (uint8_t i = 0; i < SENSOR_COUNT; i++) {
6         /* Temporary variable for single sensor */
7         float temp_c = 0.0f;
8         if (read_sensor(i, &temp_c) == ESP_OK) {
9             /* Process immediately, temp_c out of scope after block */
10            update_telemetry(i, temp_c);
11        }
12    }
13 }

```

5.182.8 Rule 7: Check All Return Values and Parameters

Rule: Validate all function returns and input parameters; treat warnings as errors.

Rationale: Forces explicit failure-condition handling rather than ignoring error codes.

STAR Compliance: **COMPLIANT**

```

1 /* COMPLIANT - All returns checked, all parameters validated */
2 esp_err_t star_bus_manager_add_bus(star_bus_manager_t* manager,
3                                     star_bus_config_t* config)
4 {
5     /* Parameter validation */
6     if (manager == NULL || config == NULL) {
7         return ESP_ERR_INVALID_ARG;
8     }
9
10    /* Check return value and propagate errors */
11    esp_err_t ret = star_bus_config_init(config);
12    if (ret != ESP_OK) {
13        ESP_LOGE(TAG, "Bus initialization failed: %s", esp_err_to_name(ret));

```



```

14     return ret;
15 }
16
17 /* Mutex operation checked */
18 if (xSemaphoreTake(manager->mutex, pdMS_TO_TICKS(1000)) != pdTRUE) {
19     ESP_LOGE(TAG, "Failed to acquire mutex");
20     return ESP_ERR_TIMEOUT;
21 }
22
23 /* Critical section */
24 add_to_list(manager, config);
25 xSemaphoreGive(manager->mutex);
26
27 return ESP_OK;
28 }

```

5.182.9 Rule 8: Limit Preprocessor Use

Rule: Restrict to header inclusion and simple macros; forbid token pasting, recursive macros, and incomplete syntactic units.

Rationale: Preprocessor complexity obscures operations, fooling developers and static analysis tools.

STAR Compliance: **COMPLIANT**

```

1  /* COMPLIANT - Simple constant macros */
2  #define MAX_RETRY_COUNT (3)
3  #define MOTOR_PWM_FREQ_HZ (20000)
4  #define TIMEOUT_MS (1000)
5
6  /* COMPLIANT - Simple inline functions (better than complex macros) */
7  static inline float voltage_mv_to_v(uint16_t mv)
8  {
9      return (float)mv / 1000.0f;
10 }
11
12 /* AVOID - Complex token-pasting macros */
13 // #define CREATE_HANDLER(name) handler_##name##_init() /* Too complex */

```

5.182.10 Rule 9: Restrict Pointer Use

Rule: Maximum one dereferencing level (*ptr OK, **ptr forbidden); no function pointers.

Rationale: Multiple indirection and function pointers make static data-flow analysis impossible.

STAR Compliance: **INTENTIONAL DEVIATION**

Intentional Architectural Deviation: STAR uses function pointers for **Dependency Inversion Principle (DIP)** — an industry-standard pattern for testability and modularity. This is a conscious trade-off for better architecture.

DIP Pattern (Intentional Function Pointer Use):

```

1  /* INTENTIONAL DEVIATION - DIP interface with function pointers */
2  typedef struct star_error_interface {
3      esp_err_t (*record_error)(void* ctx, esp_err_t error, const char*
        message,

```

```

4         const char* file, int line, const char* func);
5     bool (*can_retry)(void* ctx);
6     esp_err_t (*reset_state)(void* ctx);
7     void* ctx; /* Implementation context */
8 } star_error_interface_t;
9
10 /* Why we keep this:
11  * 1. Enables mock implementations for unit testing
12  * 2. Allows swapping error handlers without recompilation
13  * 3. Follows SOLID principles (Dependency Inversion)
14  * 4. Industry-standard pattern (used in Linux kernel, RTOS, etc.)
15  */

```

5.182.11 Rule 10: Compile with Maximum Warnings

Rule: Enable all compiler warnings at highest pedantry; achieve zero-warning builds with static analyzers.

Rationale: Modern compilers catch bugs before runtime. Zero warnings, no excuses.

STAR Compliance: **COMPLIANT**

```

1 # platformio.ini - Compiler flags
2 build_flags =
3     -Wall           # Enable all warnings
4     -Wextra         # Extra warnings
5     -Werror         # Treat warnings as errors
6     -Wno-unused-parameter # Allow unused params in interface
7                     implementations
8     -Wno-missing-field-initializers

```

CI/CD Enforcement: Build fails on any compiler warning.

5.182.12 Summary: STAR Compliance Matrix

Table 28: NASA Power of 10 Compliance Status

Rule	Description	STAR Status
1	No goto/recursion	COMPLIANT
2	Fixed loop bounds	COMPLIANT
3	No dynamic memory	COMPLIANT
4	Short functions (<60 lines)	COMPLIANT
5	Use assertions	COMPLIANT
6	Minimal variable scope	COMPLIANT
7	Check all returns	COMPLIANT
8	Limit preprocessor	COMPLIANT
9	Restrict pointers	INTENTIONAL (DIP)
10	Maximum warnings	COMPLIANT

5.182.13 References

- Holzmann, Gerard J. “The Power of 10: Rules for Developing Safety-Critical Code.” IEEE Computer, 39(6), pp. 95-97, June 2006.

- JPL Institutional Coding Standard for the C Programming Language (JPL DOCID D-60411)
- MISRA C:2012 Guidelines for the Use of the C Language in Critical Systems

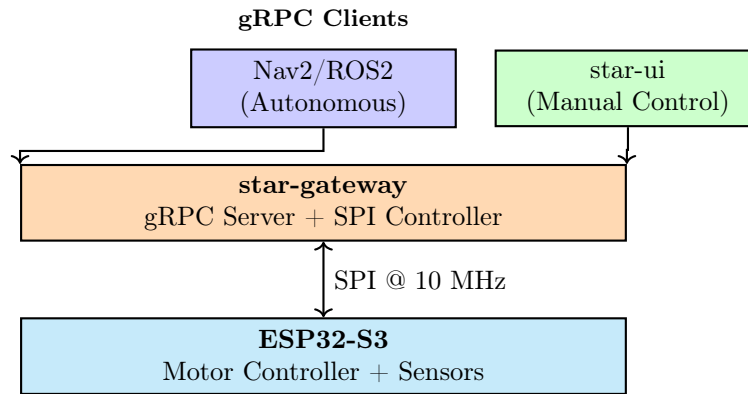
6 Gateway Architecture

This section documents the `star-gateway` Go service that bridges the Raspberry Pi 5 control stack with the ESP32-S3 motor controller.

6.1 Overview

The gateway service runs on the Raspberry Pi 5 and serves as the communication bridge between:

- **High-level side:** gRPC clients (Nav2/ROS2 for autonomous navigation, UI for manual control)
- **Low-level side:** ESP32-S3 motor controller via SPI at 10 MHz



6.2 Protocol Stack Implementation

The gateway implements Layers 1–4 of the communication protocol (see Section ??).

Table 29: Gateway Protocol Stack Implementation

Layer	Name	Go Package	Status
5	Application	<code>internal/service/</code>	Placeholder
4	Serialization	<code>star-proto/gen/go/</code>	Generated
3	ARQ	<code>internal/arq/</code>	Placeholder
2	Framing	<code>internal/frame/</code>	Constants defined
1	Transport	<code>internal/transport/</code>	Placeholder

6.3 Directory Structure

```

star-gateway/
cmd/
  star-gateway/
    main.go          # Entry point
internal/
  transport/         # Layer 1: SPI transport
    spi.go           # Transport interface
    spi_test.go
  frame/             # Layer 2: Frame protocol
    frame.go         # Constants and types
  
```

```

    encoder.go          # Frame encoder
    decoder.go          # Frame decoder
    frame_test.go
  arq/                  # Layer 3: ARQ protocol
    arq.go              # Stop-and-Wait ARQ
    arq_test.go
  service/              # Layer 5: gRPC services
    motor_control.go
    telemetry.go
    battery.go
    configuration.go
    firmware.go
go.mod
CLAUDE.md

```

6.4 Frame Package

The `internal/frame` package defines the wire protocol constants and types.

6.4.1 Frame Constants

Table 30: Frame Protocol Constants

Constant	Value	Description
SyncWord	0x55AA	Frame sync marker (alternating bits: [0x55, 0xAA])
SyncSize	2 bytes	Sync word size
HeaderSize	6 bytes	SEQ(2) + LEN(2) + TYPE(1) + FLAGS(1)
CRCSize	4 bytes	CRC-32 checksum
MaxPayloadSize	1024 bytes	Maximum protobuf payload
MaxFrameSize	1036 bytes	Total frame size
MinFrameSize	12 bytes	Empty payload frame

6.4.2 Frame Types

- `FrameTypeCommand` – Command from controller to peripheral
- `FrameTypeResponse` – Response from peripheral to controller
- `FrameTypeAck` – Acknowledgment of successful receipt
- `FrameTypeNack` – Negative acknowledgment (CRC error, etc.)

6.5 Transport Package

The `internal/transport` package provides the SPI transport layer.

6.5.1 SPI Configuration

Table 31: SPI Configuration Constants

Parameter	Value	Description
Device	/dev/spidev0.0	SPI device path
Speed	10 MHz	SPI clock frequency
Mode	0	CPOL=0, CPHA=0
Bits per word	8	Word size
Timeout	100 ms	Default operation timeout

6.5.2 Transport Interface

```

type Transport interface {
    Send(data []byte) (int, error)
    Receive(maxLen int) ([]byte, error)
    Transfer(txData []byte) ([]byte, error)
    Close() error
}

```

6.6 ARQ Package

The `internal/arq` package implements Stop-and-Wait ARQ for reliable delivery.

6.6.1 ARQ Parameters

Table 32: ARQ Protocol Parameters

Parameter	Value	Description
Max Retries	3	Maximum transmission attempts
Timeout	10 ms	ACK wait timeout
Sequence Max	0xFFFF	16-bit wraparound

6.6.2 ARQ State Machine

- **IDLE** – Ready to send or receive
- **WAITING_ACK** – Waiting for acknowledgment
- **RETRANSMITTING** – Retransmitting after timeout/NACK
- **ERROR** – Max retries exceeded

6.7 gRPC Services

The gateway exposes five gRPC services (see Section ?? for detailed specifications):

Table 33: Gateway gRPC Services

Service	Purpose
MotorControlService	Differential drive control, encoder streaming
TelemetryService	IMU, GPS, system status
BatteryManagementService	BQ7850 BMS monitoring
ConfigurationService	Runtime parameters, NVS persistence
FirmwareUpdateService	OTA updates, rollback

6.8 Build and Test

```
# Build the gateway binary
cd star-gateway
go build ./cmd/star-gateway
```

```
# Run all tests
go test ./...
```

```
# Run with verbose output
go test -v ./...
```

```
# Format code
go fmt ./...
```

```
# Static analysis
go vet ./...
```

6.9 Dependencies

The gateway depends on:

- github.com/Locked-Inc/star-proto/gen/go – Generated protobuf and gRPC code
- google.golang.org/grpc – gRPC framework
- google.golang.org/protobuf – Protocol Buffers runtime

The `go.work` file at the repository root enables workspace mode for seamless cross-module development:

```
go 1.21

use (
    ./star-proto/gen/go
    ./star-proto/tests/go
    ./star-gateway
)
```